High-Frequency Radio Observations of Galaxy Cluster Minihalos

by

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Abstract

Diffuse radio sources have been detected within the cool cores of a growing number of relaxed galaxy clusters. These faint 'minihalos' are a result of synchrotron-emitting, highly relativistic electrons. Typically exhibiting spatial extents over several hundred kiloparsec, minihalos are far beyond the expected scale from a single source of relativistic electrons. The processes of *in situ* turbulent reacceleration and hadronic production have been proposed to explain this phenomenon. Spectral steepening is expected at high radio frequencies for the reacceleration scenario. However, the majority of previous studies have been performed at lower radio frequencies (144 MHz to 1.4 GHz), where this is difficult to detect.

To investigate potential steepening, Very Large Array radio interferometer observations at 10 GHz of 12 clusters hosting minihalos are calibrated and imaged. This represents the first step towards a high-frequency survey of minihalos. Source-subtracted measurements of minihalo flux are performed for four clusters, enabling comparison with prior observations at lower frequencies. Steepening was not observed for three clusters, RXJ1720+2638 ($\alpha = 1.1 \pm 0.05$), RXCJ1115+0129 ($\alpha = 1.03 \pm 0.18$), and RXJ2129+0005 ($\alpha = 1.18 \pm 0.09$), suggesting a hadronic origin for these minihalos. One cluster, A478 ($\alpha = 2.02 \pm 0.10$), exhibits tentative evidence of steepening. However, more research is required to ensure this is not from instrumental effects. The remainder of the sample requires more extensive analysis. Challenges encountered during the calibration and imaging are discussed, and recommendations for future analysis of high-frequency minihalo studies are presented.

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Contents

Chapter 1

Introduction

When looking into the sky on a clear night, those without an interest in astrophysics might imagine many of the points in the sky as galaxies just like our own, floating alone in the deep void of space. Although true to an extent, it doesn't paint an accurate picture of the cosmos. In reality, many of these are associated with galaxy clusters, massive structures of tens to thousands of individual galaxies that are gravitationally bound into a single system (somewhat analogous to the planets in our solar system, but on a vastly different scale).[1] Galaxy clusters are the largest gravitationally bound structures in the Universe, with total masses of order 10^{14}–10^{15} $M_\odot$ and diameters larger than 1.5 million parsec.[2]

One might naturally expect that the majority of the cluster mass originates from the innumerable galaxies held within, however, only a few percent of the total mass comes from this baryonic matter[3] (such as stars, planets, etc). The remainder of the baryonic mass comes from a hot diffuse gas (10^7–10^8 K) spread throughout the cluster called the intracluster medium (ICM). This dilute gas ($\sim 10^3$ particles m^{-3}) is held together by the gravitational pull of the cluster, with the highest density in the core and most sparse at the edges (Meekins et al., 1971; Serlemitsos et al., 1977). Even then, this amount of visible baryonic matter only comprises 15% of the gravitational mass calculated from the dynamics of the cluster, a huge discrepancy that became a key piece of evidence for the existence of dark matter. Indeed, the rough consensus from decades of study on galaxy clusters has $\sim$3% of the mass budget taken by galaxies, $\sim$12% from the ICM, and a staggering $\sim$85% from unobservable dark matter (Blumenthal et al., 1984; Vikhlinin et al., 2006; Pratt et al., 2019).

Between these clusters, long filaments of galaxies span across space, with their own warm-hot intergalactic medium (WHIM) mimicking the role of the ICM. The WHIM is cooler and more diffuse than the ICM, with densities $\leq 10^2$ particles m^{-3} and temperatures 10^5–10^7 K.

[1] Smaller clusters, with only tens of galaxies, are often known as groups.

[2] $M_\odot$ is the solar mass and defined as 1 $M_\odot \approx 10^{30}$ kg, while a parsec is defined as 1 pc $\approx 3 \times 10^{16}$ m.

[3] Astronomers use the term 'baryonic' to broadly refer to all matter that is not 'dark' matter, rather than the proper particle physics definition.

These filaments are far larger than clusters, though not gravitationally bound, stretching up to hundreds of Mpc to create a cosmic web throughout the Universe with galaxy clusters located at intersections between filaments (Böhringer et al., 2025). This cosmic web is the largest structure in the Universe, with observable galaxies and clusters tracing a vast network of dark matter (Bond et al., 1996; Pimbblet, 2005). Detecting and understanding these filaments observationally is the subject of ongoing work (e.g. Tudorache et al., 2025). Clusters are known to form in part due to accretion of the WHIM at intersections of the cosmic web, as well as by mergers between smaller clusters (Press et al., 1974; Voit, 2005; Kravtsov et al., 2012).

An important distinction should be made when analysing clusters, between 'disturbed' clusters undergoing merger events or still in a turbulent state from a recent merger, and 'relaxed' clusters which have settled into an equilibrium state. Relaxed clusters favour theoretical studies due to their greater spherical symmetries, fewer substructures and, therefore, simpler modelling (Mantz et al., 2016). Many of these relaxed clusters also feature a 'cool core', where the dense central region of the cluster exhibits a lower temperature than the outer regions. The high density of the ICM in the core causes a large outflux of X-ray emission. The resulting cooling time may then be lower than the age of the Universe (known as Hubble time), causing the core temperature to drop and an inwards flow of gas to develop (Peterson et al., 2006; Hudson et al., 2010). Although cool cores are observed, there is little evidence for the predicted inward cooling flow, suggesting the existence of a counteracting heating mechanism. The most widely supported theory involves active galactic nuclei (AGN), supermassive black holes at the centre of some galaxies. AGNs consume massive amounts of matter and produce radio synchrotron emission that forms jets and lobes, which have been observed interacting with the ICM to distribute energy in 'radio-mode' feedback (Fabian, 2012). The exact method of this heating is still disputed and undergoing research.

Within the high mass subset of relaxed cool core clusters ($M_{500} > 6\times10^{14}$ $M_\odot$),[4] recent works have shown that 80% contain radio minihalos (Giacintucci et al., 2017). Minihalos are diffuse radio sources with currently unknown origins, observed solely in cool-core clusters. This radio emission is localised within the cool core, with radii from 50 kpc to $0.2R_{500}$.[5] The emission cannot be directly formed by radio sources such as AGN, as the size of these halos is well beyond the expected spatial extent for particles produced by a single source (e.g. Bagchi et al., 2002; Giacintucci et al., 2014b). Understanding the origins and physics behind minihalos is an ongoing process, with two viable production methods proposed, but a lack of conclusive observational evidence to differentiate between them. These mechanisms are detailed in Chapter 2.

This thesis presents new high-frequency observations for a sample of twelve minihalos,

[4]The value M_{500} refers to the total mass within a radius of R_{500}. This is defined as the radius at which a cluster's average enclosed density is 500 times the critical density of the Universe at the cluster redshift. The critical density is a cosmological quantity that has varied over the lifetime of the Universe.

[5]The radius $0.2R_{500}$ is a standard metric to separate the core and the outer ICM. Within this radius processes such as cooling and AGN feedback become significant to the behaviour of the ICM.

alongside a systematic and reproducible analysis framework. These results provide new constraints on high-frequency spectral properties of minihalo emission and help assess the relative likelihood of the proposed production mechanisms.

1.1 Observations of Galaxy Clusters

Since the phenomena we study are far beyond the reach of direct physical measurement, a natural question is how astronomers are able to investigate these processes and structures. This is achieved through the electromagnetic radiation and gravitational effects that can be detected through observation. Different processes within the cluster, and more specifically within the ICM, produce emission across the electromagnetic spectrum. These processes have characteristic spectral and spatial signatures that can be identified. Therefore, telescopes are designed to observe different frequency bands in order to distinguish the various components of the emission. The question then becomes: what do observations at different frequency bands inform us about the structure and processes occurring within a cluster?

1.1.1 Optical

The vast majority of optical light from clusters comes from stars within the component galaxies. While incredibly informative about the stars and galaxies themselves, in the context of minihalos, optical light has limited use. Optical observations are useful in determining the component galaxies of a cluster, alongside investigating the radial velocities of galaxies. Since the radial velocity is dependent on the gravitational potential of the cluster, the distribution allows us to estimate the cluster mass (Voit, 2005). This analysis is difficult for more distant clusters and disturbed clusters, as the dynamical state will affect the radial velocities and skew the resulting mass.

1.1.2 X-ray

The hot gas of the intracluster medium forms an optically thin plasma, which has a degree of ionisation that depends on the temperature and density. We observe an X-ray spectrum from this gas with both continuum and line characteristics, and photon energies between 0.1 keV and 100 keV.[6] The energy of the photons is dependent on the temperature of the ICM, resulting in most emission being in the narrower range of 1 to 10 keV (Schneider, 2014).

Line emission occurs due to electron transitions between orbitals within an atom. The temperature and density of the ICM changes the measured rate of this line emission, allowing the probing of these features. However, this becomes limited at higher temperatures as the ionisation

[6] X-ray emission is historically characterised by kilo-electron volts (energy) rather than frequency or wavelength.

fraction increases. The distribution of line emissions can also be an indicator of how abundant elements such as C, N, Ne, Mg, Si, S, Ar, Ca, and Fe are. This is particularly evident at lower temperatures ($\lesssim 2$ keV) where the total X-ray emission can be dominated by the line component (Voit, 2005; Schneider, 2014).

Continuum radiation is due to bremsstrahlung, or free-free radiation, where electrons are deflected via Coulomb interaction with nearby nuclei. This deflection is an acceleration which induces a change in kinetic energy that is emitted as a photon. The spectral energy distribution of these emissions is related to the kinetic energy of the particles and, therefore, the temperature T of the ICM. The scale of emissions is similarly impacted by the electron number density n_e, which traces the gas density of the ICM. This gives rise to the following relationship to total continuum emissivity ϵ_c (Schneider, 2014).

$$\epsilon_c \propto T^{\frac{1}{2}} n_e^2 \tag{1.1}$$

X-ray spectroscopy is then a powerful tool to measure the temperature and density of the ICM and, therefore, a cluster's dynamical state.

In general, telescopes trade off between high spatial or spectral resolutions, or in other words, between detecting exactly where a photon originated from or an accurate measurement of the photon energy. This becomes somewhat of a constraint when observing galaxy clusters as both a reasonable spatial and spectral resolution is required for these large objects. One common instrument is the *Chandra X-ray Observatory*[7], a space telescope launched in 1999. Many studies of minihalos have utilised *Chandra* observations to map the structure of the cool core and investigate potential relationships to the radio emission (e.g. Giacintucci et al., 2024). However, due to the ageing detectors and contaminant build up on *Chandra*, the sensitivity of the telescope is decreasing. This is particularly poor at low energies (< 2 keV), where sensitivity is below 10% of where it was two decades ago (Grant et al., 2024). A new space telescope named the *X-ray Imaging and Spectroscopy Mission* (XRISM)[8] was launched in late 2023 and was designed as a successor to *Chandra*. Over the last year researchers have begun to utilise the greater resolution to more deeply probe the dynamics of the ICM in some clusters (Sarkar et al., 2025; XRISM Collaboration et al., 2025). Although not the focus of this thesis, X-ray observations provide key ancillary data in investigating the processes involved.

1.1.3 Sunyaev–Zeldovich Effect

The Sunyaev–Zeldovich (SZ) effect is a spectral distortion in the cosmic microwave background (CMB) arising from inverse-Compton scattering of CMB photons by high-energy electrons within the ICM (Zeldovich et al., 1969). In this process low-energy CMB photons gain energy through

[7] https://www.nasa.gov/mission/chandra-x-ray-observatory/
[8] https://heasarc.gsfc.nasa.gov/docs/xrism/

interactions with the hot thermal electron population, producing a characteristic frequency-dependent distortion of the CMB spectrum. This distortion can be detected at millimetre and centimetre wavelengths.

The SZ surface brightness is independent of cluster redshift. This is not the case for the majority of observed emission, where distant clusters are highly dimmed. As such, the SZ effect is a powerful probe for distant galaxy clusters.

The amplitude of the thermal SZ effect is proportional to the integrated electron pressure along the line of sight through the ICM, expressed via the Compton y-parameter (see Figure 1.1). When combined with X-ray measurements of the electron gas temperature and density profiles, and with assumptions regarding the cluster's dynamical state (e.g. hydrostatic equilibrium)[9], SZ observations can be used to estimate the cluster mass. Consequently, the SZ effect is a valuable method of identifying galaxy clusters and an independent means of estimating their masses (e.g. Planck Collaboration et al., 2016).

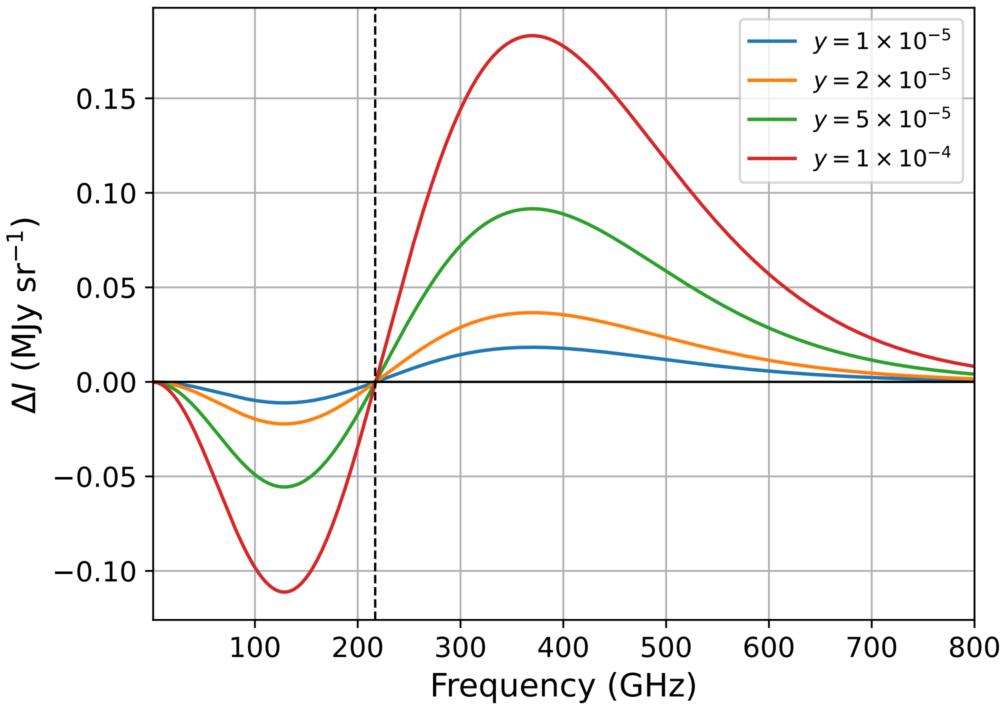


Figure 1.1: Thermal SZ intensity spectrum illustrating characteristic frequency dependence. Curves correspond to representative Compton-y values, with larger y typically associated with more massive clusters. The dotted black line denotes the null frequency at 217 GHz.

1.1.4 Radio

Radio observations cover several orders of magnitude in wavelength, with corresponding frequencies from 10 MHz to over 100 GHz. This massive range comes with several difficulties

[9]Hydrostatic equilibrium assumes that the gravitational force is balanced by the pressure of the ICM, a standard approximation for relaxed clusters.

for astronomers, with the primary one being the angular resolution of the telescope. Angular resolution θ defines the smallest area that can be resolved given an aperture of diameter D and the observing wavelength λ.[10] Equation 1.2 gives the formula for angular resolution in arcseconds, or $\frac{1}{3600}$th of a degree.[11]

$$\theta = 2.5 \times 10^5 \frac{\lambda}{D} \tag{1.2}$$

Radio waves range from 3 cm to 3 km, orders of magnitude larger than optical ($\sim 1\mu$m) and X-ray (0.01–10 nm) wavelengths. To resolve the same object at radio frequencies as at X-ray or optical, it is clear that vastly larger telescopes would be necessary. In order to observe over this range, several types of telescopes have been developed including single dish telescopes, interferometer arrays, and very long baseline interferometer arrays (VLBI).

The size of a single dish antenna is limited by both practical construction and from scientific standpoint. The largest dish is the *Five-hundred-meter Aperture Spherical Telescope* (FAST)[12] located in Guizhou China, though of the 500 metre diameter, only a 300 metre area is usable at a time. Furthermore, as an immovable dish, it is limited in its view of the sky, with a maximum angle of 40 degrees from zenith. Several smaller 100 metre diameter dishes exist such as the *Green Bank Telescope* in West Virginia United States, and the *Effelsburg Telescope*[13] near Bonn, Germany. These dishes also have a limited view of the sky and variable amounts of steerability due to their size, alongside worsening resolution at decreasing frequencies. These restrictions resulted in the development of interferometry, which uses an array of single dish telescopes in tandem to simulate a larger dish. The *Karl G. Jansky Very Large Array* (VLA)[14] in New Mexico United States, is comprised of 28 fully steerable 25-metre radio telescopes mounted on railway tracks in a Y-shaped array, observing between 73 MHz and 50 GHz. The *MeerKAT*[15] telescope array in the Meerkat National Park of South Africa contains 64 13.5-metre telescopes and operates between 580 MHz and 3500 GHz. These arrays can simulate observations by dishes several kilometres in diameter, while simultaneously being able to track objects across the entire observable sky. VLBI takes this concept even further by using radio telescopes across different continents, synchronised by atomic clocks, to simulate dishes comparable to the size of the Earth. This thesis uses observations taken by the VLA, with the underlying physics of these telescopes discussed in more detail in Chapter 2.

Radio emission detected by these telescopes from clusters falls into the category of 'synchrotron' emission. This synchrotron emission is produced by particles travelling at relativistic velocities through a magnetic field, emitting photons at radio energies characterised by their velocity and the magnetic field strength (Feretti et al., 2012; Schneider, 2014). This is due to the

[10] The angular resolution of a human eye (with 20/20 vision) is 1 arcminute. *See note 11*
[11] Arcseconds ("), arcminutes ($1' = 60$"), and degrees are common units in astrophysics.
[12] https://fast.bao.ac.cn/
[13] https://www.mpifr-bonn.mpg.de/en/effelsberg
[14] https://www.vla.nrao.edu/
[15] https://www.sarao.ac.za/science/meerkat/

Lorentz force ($\boldsymbol{F} = \boldsymbol{v} \times \boldsymbol{B}$) perpendicular to their direction of motion, which induces a helical path through the field and, therefore, constant acceleration. This is analogous to bremsstrahlung radiation, but occurs due to a magnetic interaction instead of electric, which is why it is uncommonly called 'magnetobremsstrahlung'. A more detailed discussion of the types of radio emission is given in Section 1.2. A review of the physical processes that give rise to this emission, and the corresponding spectral and spatial markers, is given in Chapter 2.

1.1.5 Gamma

The last region of interest for minihalo research is in detecting gamma radiation, or gamma rays. This is the high energy limit for astronomy, where telescopes are designed to detect photons with energies in the MeV and GeV range.[16] These prove difficult to not only detect, due to the majority of rays being absorbed in the atmosphere, but also to determine their origin (Schneider, 2014). The *Fermi Gamma-ray Space Telescope*[17] was built specifically for this research and was launched in 2008 carrying the Large Area Telescope (LAT), with an energy range between 30 MeV and 300 GeV.

There are various sources for gamma rays, such as Gamma Ray Bursts (GRB), which are extremely short and intense flashes. Many of these have been spatially located to coincide with bright AGNs called blazars, and rapidly spinning magnetised neutron stars called pulsars. Another source is from gamma ray background radiation, which is expected via neutral pion decays resulting from proton-proton collisions. Attempts to detect gamma ray emissions connected to a cluster ICM have been ongoing as they allow us to put limits on effectiveness of proposed mechanisms to create radio halos (e.g. Reimer et al., 2003; Ackermann et al., 2010; Zandanel et al., 2014a). Though progress has been made for giant radio halos, with firm constraints found for the Coma cluster (Brunetti et al., 2012), minihalos have a smaller spatial extent and are much fainter, with no conclusive gamma ray studies. The role of gamma rays in determining the processes creating minihalos is detailed in Chapter 2.

1.2 Diffuse Radio Emission

It has long been known that some clusters exhibit diffuse radio synchrotron emission that cannot be directly associated with radio galaxies within the cluster. This synchrotron emission can take many spatial arrangements, with the primary categories being radio halos, radio relics, and revived AGN sources. Radio halos are extended sources that trace the ICM distribution and operate at the scales matching the full ICM extent, called giant radio halos, or contained within a cluster cool core, named minihalos. Relics instead reflect particles that have been reaccelerated

[16] Like X-rays, gamma rays are measured by their energy rather than frequency.
[17] `https://fermi.gsfc.nasa.gov/`

to relativistic energies by shock waves travelling through the ICM, such as the violent energies released during merger events. Relics tend to be elongated and irregularly shaped, located at the cluster periphery instead of the centre. Revived AGN sources are radio plasma that has been emitted from an AGN and re-energised similarly to the radio relics. From this point, this thesis will focus on radio emission in the context of halos and, in particular, minihalos. Several types of diffuse radio emission are shown in Figure 1.2.

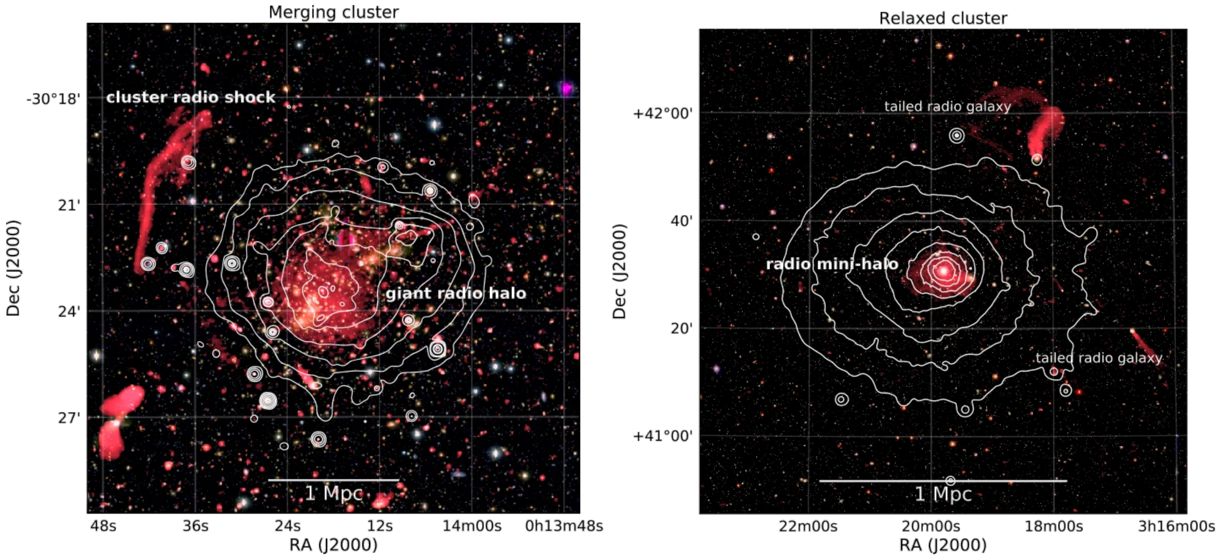


Figure 1.2: *Left panel:* Merging galaxy cluster A2744, VLA 1-4 GHz image (red) and *Chandra* X-ray contours (white) overlaid on *Subaru* BRz optical image. X-ray surface brightness contours are drawn proportional to $[1, 4, 16, 64, \ldots]$. The luminous giant radio halo and a cluster radio shock (relic) are labelled. *Right panel:* Relaxed cool core Perseus cluster, VLA 230-470 MHz image (red) and *XMM-Newton* X-ray contours (white) in the 0.4-1.3 keV band are overlaid on an optical image. X-ray contours have the same spacing as the left image. The Perseus cluster hosts a radio mini-halo as well as two head-tail radio galaxies. This figure is reproduced, in an unmodified form, from Figure 3 of van Weeren et al. (2019), licensed under CC BY 4.0.[18]

1.2.1 Production

The origin of radio halos is a key uncertainty for these phenomena. Radio-emitting relativistic particles progressively lose energy, resulting in a characteristic lifetime for observable radio emission. This therefore creates a limited diffusion length-scale in which we should be able to observe them. Within a weak magnetic field ($\sim 10^{-7}$ G) these distances are about 10 pc, which even when accounting for bulk motion of the ICM and more favourable diffusion environments, are well below the kpc and Mpc scales observed in many clusters (Bagchi et al., 2002). Two primary theories have been proposed to explain this phenomenon, one being turbulence in the ICM causing 'reacceleration' of once-relativistic particles (e.g. Brunetti et al., 2001; Gitti et al., 2002; ZuHone et al., 2013; Zandanel et al., 2014b). The other is inelastic collisions between

[18] https://creativecommons.org/licenses/by/4.0/

relativistic cosmic ray protons and thermal protons, termed ‘hadronic’ or secondary production (e.g. Dennison, 1980; Dolag et al., 2000; Keshet et al., 2010; ZuHone et al., 2015; Ignesti et al., 2020).

Both of these mechanisms produce similar observable features, which makes disentangling the true source much more challenging. Investigation into spatial extents and spectral features of giant radio halos has given the reacceleration theory more credence than hadronic production (Brunetti et al., 2012), however, the debate is still ongoing for minihalos. Due to their faint nature and smaller size, minihalo observations require high sensitivity and resolution in order to accurately capture an image. A detailed explanation of these production mechanisms is provided in Section 2.3.

1.2.2 Minihalo detections

Much effort has been devoted towards detection and validation of minihalos in order to build a comprehensive sample. Over the last two decades the number of candidate and confirmed minihalos has increased substantially from 10 in 2011 (Feretti et al., 2012), to 22 in 2017 (Giacintucci et al., 2014b), and 40 in 2024 (Giacintucci et al., 2024).

One of the major issues with detection is in separating out the much brighter sources embedded within and nearby the minihalo emission. Alongside the minihalo in the core is the brightest cluster galaxy (BCG), generally the most radio-loud AGN within the cluster. Minihalos typically have a surface brightness of $1 \sim 5$ μJy arcsec^{-2}, while the BCG can have a flux measured in excess of tens to hundreds of mJy.[19] The BCG must then be masked or subtracted away from an observation to reveal the minihalo, but this requires the observation to have good resolution of point sources as well as the larger spatial minihalos. This requirement makes current general sky surveys insufficient for detecting minihalos, with some studies suggesting that next generation radio telescopes could reveal thousands more minihalos due to improved capabilities (Gitti et al., 2018).

1.3 This Thesis

1.3.1 Aims

The primary goal of this thesis is to calibrate, image, and analyse new radio observations of 12 minihalos at 10 GHz. Due to the time constraints of a Master’s thesis, this aim is limited to the initial calibration and imaging of each cluster, alongside a deeper pilot investigation of one. These observations were performed by the VLA in C- and D-configuration over two periods in

[19] The Jansky (Jy) is a unit measurement of spectral flux density common in radio astronomy. It measures the rate of energy received per square meter per hertz, or 10^{-26} W m^{-2} Hz^{-1} in SI units.

early 2024 and early 2025. This data constitutes the first broad survey of minihalos at high radio frequencies.

This data will complement previous surveys at lower frequencies taken by the *Low-Frequency Array* (LOFAR) at 140 MHz and MeerKAT at 1.3 GHz (e.g. Riseley et al., 2022). Processing this data will allow measurements of the minihalo flux to investigate if steepening of the spectrum occurs at high frequencies, which will assist in differentiating between the reacceleration and hadronic scenarios.

1.3.2 Work

In pursuing these aims, I downloaded the raw interferometric data from the National Radio Astronomy Observatory (NRAO) Archive[20] and documented the calibration and imaging process. I then performed various analyses on the final images for each minihalo to evaluate the limits and usefulness of 10 GHz observations for characterising the production mechanisms of minihalos.

This work was done in a source-controlled repository[21] so that the results are reproducible and documented. Due to the size of the datasets, (uncalibrated $\sim$ 1.4 TB compressed, calibrated $\sim$ 2.5 TB, imaging $\sim$ 0.5 TB), these are stored on the Storage fOr Learning And Research (SOLAR) large scale storage provided by Te Herenga Waka—Victoria University of Wellington.

The repository is split into two primary sections, *Data Processing* and *Image Processing*. The first section follows the calibration of the raw data and separation from observation runs into cluster-specific datasets, split by observation ID. The following image processing includes the cleaning and creation of the images from the trimmed datasets and the associated analysis and is split by cluster.

1.3.3 Cosmology

Several physical values depend on the constants chosen for the cosmological model of the Universe. This work assumes a flat ΛCDM model with Hubble's constant $H_0 = 70$ km s^{-1} Mpc^{-1}, fractional density of dark energy $\Omega_\Lambda = 0.7$, and fractional density of matter $\Omega_m = 0.3$.

1.3.4 Layout

The introduction has given a brief overview on several concepts necessary for understanding what minihalos are. The remainder of the thesis is organised as follows.

Chapter 2 provides a deeper explanation of synchrotron radiation, the hadronic and turbulent reacceleration processes, and a summary of previous research into minihalos.

[20] `https://data.nrao.edu/portal/`

[21] `https://github.com/MoonlitMoo/Masters-2025`

Chapter 3 introduces the observational data and explains the calibration and imaging process used.

Chapter 4 explains the tools and techniques used in the analysis of the final images.

Chapter 5 details the results found for each cluster, with the context of previous studies.

Chapter 6 frames the results and their implications with regards to the production scenarios. Limitations and uncertainties for the results are discussed, alongside future work arising from this thesis.

Chapter 7 summarises the content and findings of this thesis.

Chapter 2

Radio Halos

2.1 Synchrotron emission

Since the first discoveries of diffuse radio emission, much research has been devoted towards understanding the mechanisms behind the production of synchrotron-emitting electrons. These electrons are also known as 'cosmic ray' electrons (CRe) or non-thermal electrons to differentiate them from the thermal electrons of the ICM. These CRe have an individual energy $E = \gamma m_e c^2$ dependent on their Lorentz factor γ.

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \tag{2.1}$$

In the case of radio halos, we can expect $\gamma \gtrsim 10^4$ which corresponds to velocities within 99.9999% the speed of light. Throughout their radiative lifetime CRe produce radio synchrotron emission with power dependent on the magnetic field B and their energy E. This is broad continuum radiation with the synchrotron power loss described by the following equation, where σ_T is the Thomson cross-section, m_e the electron mass, c the speed of light, and $U_B = B^2/(8\pi)$ is the energy density of the magnetic field (Sarazin, 1999).

$$\left[-\frac{dE}{dt}\right]_{\text{syn}} = \frac{4}{3}\frac{\sigma_T}{m_e c}\gamma^2 U_B \tag{2.2}$$

CRe also interact with low-energy photons in the ICM via inverse Compton (IC) scattering,[1] of which the dominant photon population originates from the Cosmic Microwave Background (CMB). The scattered photon has a boosted frequency ν_p that is proportional to its original frequency $\nu_{p,0}$ as,

$$\nu_p = \frac{3}{4}\gamma^2 \nu_{p,0} \tag{2.3}$$

[1]Standard Compton scattering has ultrarelativistic photons interacting with thermal electrons, the opposite to what we are considering.

For the CRe under consideration this results in a boosting factor $\geq 10^6$. CMB photons have a temperature of T = 2.73 K, to which the Planck function peaks at $\nu \approx 1.6 \times 10^{11}$ Hz. This interaction causes emission at $\nu \gtrsim 10^{17}$ Hz or energies in the range 1–50 keV, corresponding to high energy X-rays (Sarazin, 1999; Govoni et al., 2004). The energy losses due to IC scattering with CMB photons can be expressed with respect to the CMB energy density $U_{\rm CMB} = B^2_{\rm CMB}/(8\pi)$, and an equivalent magnetic field strength $B_{\rm CMB} = 3.25(1+z)^2$ [μG]. This magnetic field is not real, but emulates the equivalent energy losses as if electrons were interacting with a real magnetic field within the cluster.

$$\left[-\frac{dE}{dt}\right]_{\rm IC} = \frac{4}{3}\frac{\sigma_T}{m_e c}\gamma^2 U_{\rm CMB} \tag{2.4}$$

Further losses due to Coulomb and bremsstrahlung interactions are possible, but scale with γ as follows (Sarazin, 1999).

$$\left[-\frac{dE}{dt}\right]_{\rm Coul} \propto \ln\gamma, \qquad \left[-\frac{dE}{dt}\right]_{\rm brem} \propto \gamma\ln(\gamma)$$

These mechanisms are naturally highly dominated by the synchrotron and IC losses, which scale $\propto \gamma^2$, at high Lorentz factors and so Coulomb and bremsstrahlung losses can be neglected. The combined synchrotron and IC losses can then be written as a single loss equation where the magnetic fields are measured in μG.

$$-\frac{dE}{dt} = \frac{1}{6\pi}\frac{\sigma_T}{m_e c}\gamma^2\left(B^2 + B^2_{\rm CMB}\right) \tag{2.5}$$

The standard assumptions regarding the population of CRe within the ICM are that it is homogeneous, isotropic, and follows a power-law energy distribution $n(E) = n_0 E^{-\delta}$. The total synchrotron emissivity ϵ_s at a given frequency ν is dependent on the CRe number density n_0, the spectral index of the CRe distribution δ, and the strength of the ICM magnetic field.

$$\epsilon_s(\nu) \propto n_0 B^{(\delta+1)/2}\nu^{-(\delta-1)/2} \tag{2.6}$$

This can be rewritten in terms of $\alpha = (\delta-1)/2$, the spectral index of the synchrotron emission.

$$\epsilon_s(\nu) \propto n_0 B^{\alpha+1}\nu^{-\alpha} \tag{2.7}$$

This relation shows a degeneracy between the magnetic field strength and the CRe number density, hence, the synchrotron radiation alone is insufficient to directly measure the magnetic field. Estimates of the magnetic field are generally found by minimising the total energy. This is the combination of the energy of the relativistic particles $(1+k)U_e$, where k is the ratio of energy of relativistic protons U_p to the electrons U_e, and the magnetic field U_B.

$$U_{\rm tot} = (1+k)U_e + U_B \tag{2.8}$$

This occurs when the total energy of the relativistic particles is close to equilibrium with the total magnetic field energy $U_B = \frac{3}{4}(1+k)U_e$ and, therefore, $U_{\min} = \frac{7}{3}U_B$, often referred to as the equipartition value (Govoni et al., 2004).[2] This allows the calculation of the minimum energy density $u_{\min}$ which can be used to derive the associated equipartition field B_{eq}.

$$B_{\mathrm{eq}} = \left(\frac{24\pi}{7}u_{\min}\right)^{\frac{1}{2}} \tag{2.9}$$

Since the IC emission is independent of the magnetic field, comparisons between these fluxes would allow direct measurement of the magnetic field strength (Govoni et al., 2004; Wik et al., 2014). Unfortunately, detection of the X-ray IC component is difficult as low X-ray energies are dominated by the thermal ICM emission, while upper energies lack spatial resolution in current telescope instruments. An interesting topic in its own right, IC emission and numeric estimations of cluster magnetic fields are outside the scope of this thesis.

2.2 Lifetime of synchrotron emission

The CRe interact with the electromagnetic fields and particles in the ICM losing energy over time. Therefore, they eventually are no longer energetic enough to produce observable radio emission. Considering losses due to synchrotron and IC processes, the lifetime of a CRe τ can be given as the following function of the magnetic field in the ICM B, the equivalent magnetic field of the CMB B_{CMB} (both fields are measured in μG), the cluster redshift z, and the observed frequency of the particle ν (van Weeren et al., 2019).

$$\tau = 32\frac{B^{\frac{1}{2}}}{B^2 + B_{\mathrm{CMB}}^2}\left[(1+z)\nu\right]^{-\frac{1}{2}} \quad [\mathrm{Gyr}] \tag{2.10}$$

As the majority of observed minihalos have redshifts in the range $[0.1, 0.3]$[3] and magnetic fields within the ICM are estimated at $\lesssim \mu$G, the maximum expected lifetime of a CRe is about 10^8 years (Sarazin, 1999; Feretti et al., 2012; Gitti et al., 2018).

A basic check for the spatial extent for these particles, in absentia of external forces $d \approx c\tau$, finds a distance of $\sim 10^{16}$ m or 1 pc from the source. This is orders of magnitude below the extents of observed diffuse radio emission, so we then consider the effect of the bulk motion and magnetic fields of the ICM on the CRe. By assuming there is a strong enough turbulence to promote diffusion of CRe, we can use the Bohm approximation $D = \frac{k_B T}{eB}$, which models the diffusion of plasma across a magnetic field. Bagchi et al. (2002) finds that the expected

[2]This is not the true equipartition field, rather it is the minimum energy field, though the two fields are quite similar in magnitude.

[3]A handful of minihalos are notably outside this range, with the Phoenix cluster at $z = 0.596$ (van Weeren et al., 2014), ACTJ0022-0036 at $z = 0.8$ (Knowles et al., 2019), and a recent candidate SpARCS1049+56 at $z = 1.709$ (HLavacek-Larrondo et al., 2025)

length-scale L_{diff} of this diffusion (at $z = 0.3$) is that of a random walk across its lifetime with a diffusion coefficient κ and length-scale,

$$\kappa = 3.31 \times 10^{21} \left(\frac{E}{\text{GeV}}\right)\left(\frac{B}{10^{-1}\mu\text{G}}\right) \text{ m s}^{-1} \tag{2.11}$$

$$L_{\text{diff}} = \sqrt{\kappa\tau} = 6.7 \left(\frac{B}{10^{-1}\mu\text{G}}\right)^{-\frac{1}{2}} \text{ pc} \tag{2.12}$$

Once again, we find the expected diffusion and halo produced by a point source is measured in the 10s of parsecs, far below the kiloparsec and megaparsec halos observed. It then seems a reasonable conclusion that particles are instead reaccelerated or produced *in situ* by a process that is extended over the kiloparsec scale and long-lived across the particle lifetime.

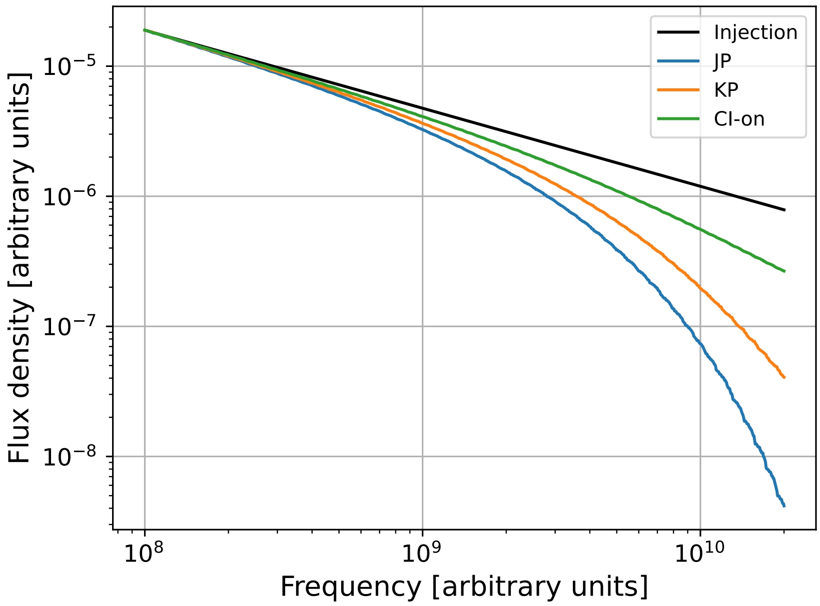


Figure 2.1: A representation of radio spectral shapes with $\alpha_{\text{inj}} = -0.6$. The injection spectrum depicts the shape prior to energy losses. The curves have been normalised to the injection spectrum at low frequency for better comparison. Generated with the Python module `synchrofit`.[4]

The spectral shape of synchrotron emission is also affected by the lifetime of the particles, as higher energy particles have a shorter lifetime ($\tau \propto \nu^{-\frac{1}{2}}$), causing the population of CRe to change over time from the initial power-law spectrum. Several models have been developed to describe the resulting spectra, depending on the initial injection spectrum α_{inj}, break frequency, and any continuous injection after the initial state. The Jaffe-Perola (JP; Jaffe et al. 1973) model assumes all CRe are created in a single burst, before ageing with continuous scattering of electron pitch angles. This predicts that the pitch angle (between the electron velocity vector and local magnetic field vector) will be isotropic due to scattering. The Kardashev-Pacholczyk

[4] `https://github.com/synchrofit/synchrofit`

(KP; Kardashev 1962, Pacholczyk 1970) model is similar to the JP model as it uses the same initial conditions, but evolves without pitch angle scattering. Continuous injection (CI) models describe sources that exist for longer than a single burst of acceleration, such as until current time (CI-on; Kardashev 1962), or for a fixed amount of time $\tau_{\rm on}$ (CI-off; Komissarov et al. 1994). An example of how these models differ in their shape is provided in Figure 2.1. The initial burst models show extreme steepening towards high frequency. In contrast, the CI model offsets the losses at high energy due to newly injected particles, but still exhibits a lower level of steepening due to the presence of aged particles alongside newer ones.

The basic observables of synchrotron radiation can then be summarised in the following points:

- The synchrotron emissivity is dependent on the number density of the CR electrons and the strength of the magnetic field (Equation 2.7). Assuming a constant magnetic field strength, the emissivity follows a power-law $\epsilon_s(\nu) \propto \nu^{-\alpha}$ which is related to the CRe energy spectral index δ by $\alpha = (\delta - 1)/2$. Typical minihalo spectral indices are steep with $\alpha > 0.7$.[5]

- As the emitting particles lose energy the particle energy distribution changes and, therefore, the emitted radio spectrum changes. This appears as a steepening or curved spectrum at high frequencies, rather than a single power-law. The break frequency at which this occurs is related to the time since acceleration. This means that CRe accelerating events that have stopped will show an aged spectrum with a characteristic break, such as for radio relics. Ongoing events that are continually injecting new particles will not feature the same level of steepening.

2.3 Production mechanisms

With this understanding of the spectrum of CRe, we now consider the methods that can produce them. This can be broken down into primary models involving first and second order Fermi acceleration, and secondary models where the electrons are a byproduct or resulting from decay.

2.3.1 Turbulent reacceleration

This process takes a pre-existing lower energy CRe population of $\gamma \sim 100$ and boosts it to the factors observed $\gamma \gtrsim 1000$. This occurs via interactions with plasma waves and shocks within the ICM which can be caused by several factors, such as violent cluster mergers and gas sloshing within the core (Brunetti et al., 2001; Fujita et al., 2004; Mazzotta et al., 2008).

Shocks within the ICM can cause first order Fermi acceleration (Fermi-I), also known as diffuse shock acceleration (DSA). As particles cross back and forth over the shock front, they can

[5] A typical radio galaxy emits synchrotron radiation with a spectral index ~ 0.7.

be reflected by magnetic instabilities upstream and downstream from the shock. Each crossing of the front increases the particle energy, raising trapped particles to relativistic levels before they escape. The energy E gained via Fermi-I acceleration from a shock with velocity $\mathrm{v}_{\mathrm{wave}}$ scales compared to the speed of light c as,

$$\Delta E \propto \frac{\mathrm{v}_{\mathrm{wave}}}{c} E \tag{2.13}$$

Shocks are generally created due to merger activity in disturbed clusters and, therefore, are not observed in relaxed clusters where minihalos are found. As such, it is not considered a major mechanism at play in the context of minihalos.

Gas sloshing within the cool core of relaxed clusters has been shown as a natural consequence of the larger motions of the ICM (Fujita et al., 2004). This turbulent environment naturally allows for low relativistic 'seed' electrons to undergo second order Fermi (Fermi-II) acceleration. This type of acceleration is stochastic as particles are scattered by the magnetic instabilities from the randomly moving plasma waves. Particles meeting these waves head-on are reflected with an increased energy, while particles in a head-tail collision are reflected with a lower energy. The chance for a head-on collision is higher than that of a head-tail collision allowing for a net gain. The mean energy gained is dependent on the speed of the waves and energy of the seed particle,

$$\Delta E \propto \left(\frac{\mathrm{v}_{\mathrm{wave}}}{c}\right)^2 E \tag{2.14}$$

The random nature of Fermi-II acceleration makes for an inefficient process compared to the aligned shock fronts of Fermi-I. Since $\mathrm{v}_{\mathrm{wave}} \ll c$, Fermi-II scales far more slowly than Fermi-I $(\mathrm{v}_{\mathrm{wave}}/c)^2 \ll \mathrm{v}_{\mathrm{wave}}/c$.

Brunetti et al. (2007) showed that the reacceleration of relativistic particles moving through magnetohydrodynamic (MHD) turbulence is primarily achieved by two Fermi-II type mechanisms.

The first mechanism is the damping of fast magnetosonic waves due to resonant transit time damping (TTD). The compressible and low-frequency plasma waves can resonate with the naturally gyrating particles at the zeroth order, $\omega = \mathrm{v}_{\parallel} k_{\parallel}$, where ω is the frequency of the wave, $k_{\parallel}$ is the wavenumber, and $\mathrm{v}_{\parallel}$ is the speed of the particle along the axis of the magnetic field. The random acceleration is then captured as a random walk in momentum space, giving rise to a momentum-diffusion coefficient $D_{pp,\mathrm{TTD}}$. ZuHone et al. (2013) simulated this process numerically in the case of minihalos and simplified this diffusion coefficient to be proportional to the average wavenumber $\langle k \rangle$, the turbulent velocity of the particle v_t,[6] and fraction of turbulent energy in compressible motion R_c.

$$D_{pp,\mathrm{TTD}} \propto \langle k \rangle \mathrm{v}_t^2 R_c p^2 \tag{2.15}$$

[6]This is essentially the instantaneous velocity of the particle due to turbulence.

The second mechanism is through the non-resonant compression and expansion of the ICM. Adiabatic compression increases the particle momentum, while expansion reduces it. The particle is more likely to experience compression than expansion, leading to a net gain in energy. Here ZuHone et al. (2013) took the fast diffusion limit, where particles escape the eddies of compression before they re-expand. This non-resonant momentum-diffusion coefficient $D_{pp,\mathrm{C}}$ scales as,

$$D_{pp,\mathrm{C}} \propto \frac{V_l^2}{D} p^2 \tag{2.16}$$

where V_l is the compressive velocity at scale l, and $D \propto c l_{\mathrm{mfp}}$ is the spatial diffusion rate for a mean free path of l_{mfp}.

The reacceleration efficiency is then given via $\chi = \frac{4D_{pp}}{p^2}$, where D_{pp} is the sum of the prior diffusion coefficients. The total rate of momentum increase for each particle can then be clearly written as the sum of two Fermi-II processes when compared to Equation 2.14.[7]

$$\frac{dp}{dt} = \chi p \tag{2.17}$$

$$\frac{dp}{dt} \propto \langle k \rangle R_c v_t^2 p + \frac{1}{D} V_l^2 p \tag{2.18}$$

One remaining loose thread in this explanation is the source of the seed electrons themselves. A possible explanation is that the seed population is provided from radio AGN and spread throughout the cluster core by the gas sloshing (Brunetti et al., 2001). The CRe from the central BCG lose energy and disappear out of our detectable range by the physics described earlier, however, they don't stop existing after this timescale. These 'relic' CRe are lower energy but still relativistic with $\gamma \sim 100$, radiating at radio frequencies that we are unable to detect. In this state they still continue to move and disperse through the core, creating the seeds for the large-scale minihalos that we observe. The central BCG may not be the only source as other AGN within the cluster, galactic winds, and supernovae may all contribute relic CRe to be later reaccelerated. ZuHone et al. (2013) found that gas sloshing should be sufficient to redistribute these particles to create a minihalo. Hadronic interactions as described in a later section could also contribute seed electrons.

2.3.2 Observables of turbulent reacceleration

With this understanding we can now explore the spectral and spatial features that we expect to observe from emission resulting from the reacceleration scenario. Turbulence is an inherently transient and intermittent phenomenon. This spatial variance means that when sampling a given region of space, we may instead be observing several 3D components superimposed on our 2D image. Areas with higher turbulent velocity inspire greater magnetic fields and greater levels of

[7] The factors of c have been absorbed into the constants, with the main focus being the squared scaling with velocity and linear with momentum

emission (ZuHone et al., 2013). The lack of a steady state and variable components means that we might expect variation in the flux brightness and spectral index across different locations of the minihalo (Riseley et al., 2022). However, due to the integration over the line of sight, these artefacts could be averaged and less visible in the 2D observation.

The particles must also trace the distribution of turbulence within the ICM, leading to a spatial bounding of the minihalo within the gas sloshing of the core. The boundary of the cluster core is often associated with cold fronts visible in X-ray data, allowing comparison between X-ray and radio images. Cold fronts denote the boundary between cooler denser gas clouds moving through a hotter and more diffuse surrounding gas. These are notably different to shock fronts, where the boundary separates a hot and dense downstream region that the shock has passed through, from the cooler and less dense upstream region the shock is moving towards (Markevitch et al., 2007). ZuHone et al. (2013) have shown that synchrotron emission arising from simulated turbulent reacceleration is indeed bounded by cold fronts (Figure 2.2), with a sharp steepening at the core radius. They also found that depending on the projection through the ICM, it may be difficult to locate the cold fronts in X-ray images due to the aforementioned line of sight integration.

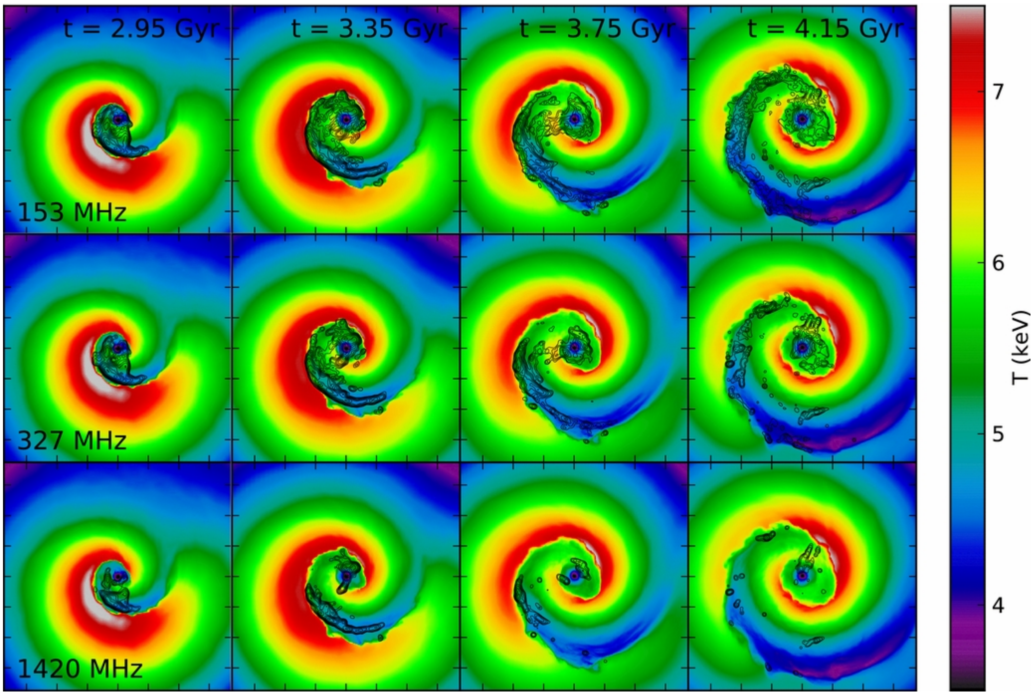


Figure 2.2: 2D projected gas temperature maps with overlaid radio contours at several simulation timesteps for frequencies 153, 327, and 1420 MHz. Radio emission contours begin at $(1.0, 0.5, 0.125) \times 10^{-3}$ mJy arcsec^{-2} and increase by a factor of 2. Image dimensions are 750×750 kpc, with tick marks every 100 kpc. This figure is reproduced, in unmodified form, from Figure 10 of ZuHone et al. (2013), © AAS. Reproduced with permission.

A spectral feature of the reacceleration model arises from the balancing of reacceleration and losses, creating a cutoff energy at which the population of CRe declines more sharply. This is indeed the logic behind the CI model visualised in Figure 2.1, with a steepening in the spectral index beyond the break frequency. ZuHone et al. (2013) noted that as there is a range of magnetic field strengths within the core, this break frequency will differ between individual CRe, causing a smearing of the break and a more gradual steepening towards high frequency (see Figure 2.3). The simulations found typical spectral indices in the range $\alpha = 1.5$–2 at low frequencies before the spectral break, varying between time steps.

2.3.3 Hadronic Production

For the secondary production we instead consider the protons within the ICM, rather than the electrons. We again separate them into a relativistic 'cosmic ray' proton (CRp) population and a thermal population and examine the resultant interactions. CRp have significantly longer radiative lifetimes than CRe, with $\tau_p \sim 10^{10}$ yrs, allowing them to travel up to 1 Mpc (Dennison, 1980). The spread out CRp (p_{CR}) then collide inelastically with thermal protons (p_{th}) to produce charged and neutral pions π,

$$p_{CR} + p_{th} \longrightarrow \pi^0 + \pi^+ + \pi^- \tag{2.19}$$

The charged pions then decay into (anti)muons $\mu^\pm$ and muon (anti)neutrinos $\nu_\mu(\tilde{\nu}_\mu)$, while the neutral pions decay into gamma rays γ,

$$\pi^\pm \longrightarrow \mu^\pm + \nu_\mu(\tilde{\nu}_\mu) \tag{2.20}$$

$$\pi^0 \longrightarrow 2\gamma \tag{2.21}$$

The muons then decay into highly relativistic (positrons) electrons $e^\pm$ and electron (anti)neutrinos $\nu_e(\tilde{\nu}_e)$.

$$\mu^\pm \longrightarrow e^\pm + \nu_e(\tilde{\nu}_e) + \tilde{\nu}_\mu(\nu_\mu) \tag{2.22}$$

From this chain reaction we end up with two observables, CRes and gamma rays. Importantly, it is these secondary CRe products that are predicted to produce the minihalo emission. The emission in this model then traces the population of CRp producing the CRe, rather than the turbulence of the ICM. Due to our assumption of the long radiative lifetimes, we expect a continuous injection of CRe from the hadronic process.

ZuHone et al. (2015) investigated the hadronic method via simulation and assumed a power-law momentum distribution for the CRp, $n_p(p_p) = C_p p_p^{-\alpha_p}$ (where C_p is a normalisation constant), and estimated the CRp momentum distribution spectral index to be in the range $\alpha_p \sim 2.0$–2.5. They then derived a distribution for the produced CRe momentum p based upon the CRp momentum distribution to be,

$$Q(p)dp = \frac{4}{3} 16^{1-\alpha_p} c n_{th} C_p \sigma_{pp}(16p) p^{-\alpha_p} dp \tag{2.23}$$

with n_{th} the density of thermal protons, $\sigma_{pp}(16p)$ the proton-proton inelastic collision cross-section at 16 times the CRe momentum.[8] Ignoring the numerical terms we can also write,

$$Q(p)dp \propto n_{th}\sigma_{pp}p^{-\alpha_p}dp \tag{2.24}$$

to clearly see that the efficiency of the hadronic method scales with the density of thermal protons, collision cross-section, and as a power-law in momentum. By approximating σ_{pp} as constant[9] we are able to find a steady state for the number density of CRe by equating the synchrotron and IC losses $\frac{dp}{dt} \propto -p^2$. This occurs when $\frac{d}{dp}(-p^2 n_e) + Q(p) = 0$, which following through by integrating for all particles $p \to \infty$,

$$\begin{aligned}
-p^2 n_e &= -\int_p^\infty Q(p) \\
p^2 n_e &\propto \int_p^\infty p^{-\alpha_p} \\
n_e &\propto \frac{p^{-\alpha_p+1}}{p^2} \\
n_e &\propto p^{-(\alpha_p+1)}
\end{aligned}$$

This gives us a CRe density power-law with index $\delta = \alpha_p + 1$ and therefore a synchrotron spectral index of $\alpha = \frac{\alpha_p}{2}$. Using the expected values for α_p we should then expect $\alpha \sim 1$–1.25 which is similar to that of observed minihalos. However, this assumes steady state injection, which is a somewhat idealised case, for the intermittent and turbulent core of the cluster.

2.3.4 Observables of hadronic production

This hadronic process predicts a constant spectral index extending into high frequencies, lacking the steepening observed in the reacceleration model. This also implies the spectral index will be constant throughout the spatial extent of the minihalo, rather than varying due to localised high amounts of turbulence. Keshet et al. (2010) showed that changes in the magnetic field strength relative to B_{CMB} can induce changes in the emissivity. For regions of the magnetic field where $B < B_{CMB}$ they predicted a radio suppression $\propto (B/B_{CMB})^2$. Assuming that the magnetic field inside a cluster has a radial steepening, there then exists a radius r_b where the magnetic field drops below B_{CMB} and a radial break may be observed in the radio profile. As such, the extent of the minihalo would be more constrained via the strength of the magnetic field, than the cold fronts of the cool core.

The simulation of hadronic production by ZuHone et al. (2015) found that the majority of particles have spectral indices close to that of steady state, at approximately $\alpha = 1.1$. As B

[8] This is an approximation of the CRp momentum in the proton-proton collision, $p_p = 16p$.

[9] There is a weak dependence on the CRp and thermal proton energies, which ZuHone et al. (2015) account for, resulting in a minor flattening effect on the synchrotron spectral index.

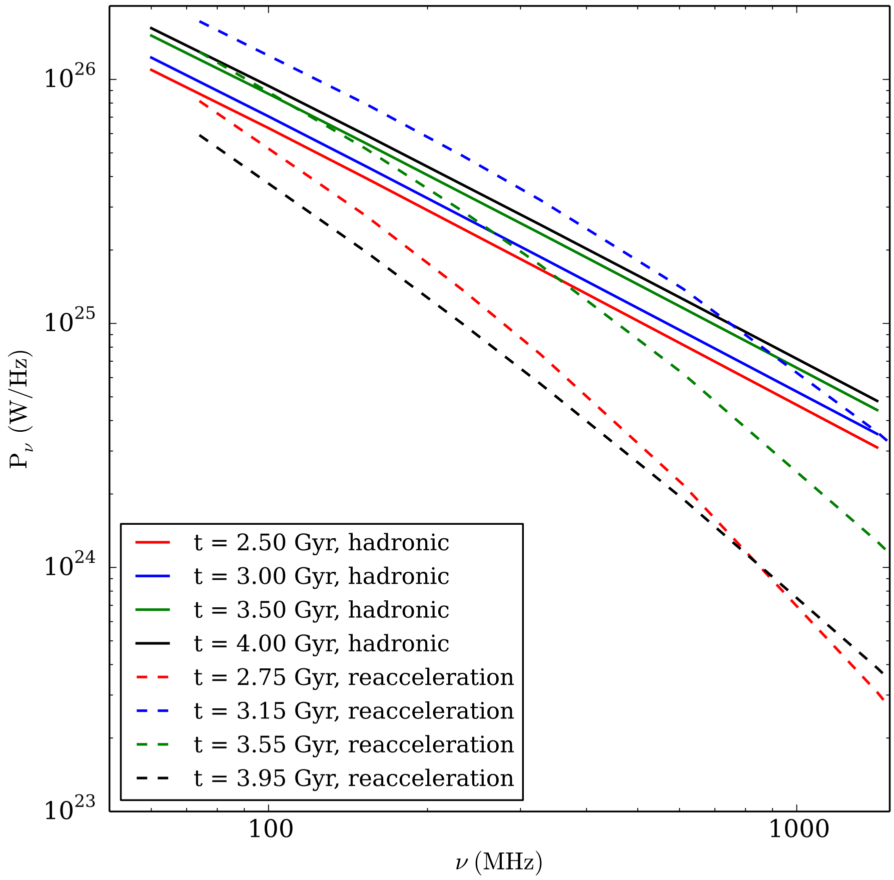


Figure 2.3: The total simulated radio spectrum within 300 kpc radius of the cluster centre, comparing the hadronic and reacceleration model at several timesteps. This figure is reproduced, in unmodified form, from Figure 8 of ZuHone et al. (2015), © AAS. Reproduced with permission.

decreases towards $B \ll B_{CMB}$ the spread of spectral indices converges tightly. The opposite happens at $B \gg B_{CMB}$ with broadening and a small skew towards steeper indices. They attributed the broadening to sampling regions with rapidly decreasing magnetic fields (flatter index) and rapidly increasing magnetic fields (steeper index). They also noted that the spatial bounding of the hadronic emission is less constraining than that of the reacceleration model. A drop in the emission across the cold front occurs due to a drop in the magnetic field strength and falls $\propto B^2$, but they simultaneously observed emission extending outside of the cool core.

Importantly, ZuHone et al. (2015) did not find evidence of spectral steepening at high frequencies, showing that a single power-law relationship holds to beyond 10 GHz. They compared the spectral distribution of the hadronic and reacceleration models and clearly show the difference in spectral shape (Figure 2.3). The observed spectral index for the hadronic model was also flatter ($\alpha \approx 1.1$), compared to that of the reacceleration model ($\alpha \approx 1.5$–2).

A final clear distinction from the reacceleration model is the emission of gamma rays with the hadronic model. This implies there will be a diffuse gamma emission aligned spatially with the minihalo. This would be a ‘smoking gun’ proof for the hadronic model. However, as mentioned earlier, current instruments do not have the spatial or spectral resolution to accurately detect diffuse gamma emission at minihalo scales.

2.4 Previous minihalo studies

Minihalos are the most poorly understood class of diffuse radio emission, with few in-depth spectral studies to look at. For many minihalos we only have a couple of observations between 144 MHz and 1.4 GHz taken by the *Giant Metrewave Radio Telescope* (GMRT) and VLA (Giacintucci et al., 2014b; Giacintucci et al., 2019). This lack of data means that for many minihalos the spectral index is calculated from a few points, if it can be at all. Furthermore, the lack of high frequency data > 1.4 GHz makes detecting steepening from the reacceleration theory far more challenging, with both models resembling a power-law within this range (Figure 2.3).

The most in-depth spectral study at a range of frequencies has been performed on RXJ1720+2638 (e.g. Mazzotta et al., 2008; Giacintucci et al., 2014a; Perrott et al., 2021). This revealed a tail to the minihalo, with a steeper spectrum than the central region. Early analysis showed a potential for steepening at 5 GHz, however, later observations at 13-18 GHz were consistent with a power-law index $\alpha = 1.0$ (Perrott et al., 2021). These spectral features lend impetus towards the hadronic scenario for the core, while also suggesting that reacceleration might play a role within the tail. The tail of the minihalo also appears to be constrained by the cold fronts of the core. More detailed analysis for this cluster is given in Section 5.1, which includes images produced as part of this thesis.

The recent MeerKAT-meets-LOFAR series of papers (Riseley et al., 2022; Riseley et al., 2023; Riseley et al., 2024) aims to perform a more uniform sample of minihalo observations to investigate the method of CRe production. These observations are performed at 145 MHz to 1.7 GHz with the aforementioned LOFAR and MeerKAT telescopes and used alongside archival *Chandra* X-ray data. These detailed analyses of MS1455+2232, A1413, and A2142 have revealed that the minihalo morphology generally contains multiple components, a single steep-spectrum power-law, and spatially homogeneous spectral index. They found that both scenarios can explain several observables, but not all, and suggested that both mechanisms might be at play. MS1455+2232 and A1413 are discussed in more detail, alongside images produced during this thesis, in sections 5.7 and 5.8 respectively.

Another point of interest is the possible connection to giant radio halos. Although minihalos are found exclusively within cool-core clusters and giant radio halos are found almost entirely within disturbed clusters, they may have a physical relation to each other. The general consensus is that giant halos are formed by reacceleration due to merger-induced turbulence, evidenced by a lack of gamma emission at the larger cluster scale (Brunetti et al., 2012; Brunetti et al., 2017). These two phenomena may have a transitional state where a minihalo switches into a giant halo due to a merger, and the inverse as the cluster relaxes (Brunetti et al., 2014). An alternative scenario could be that mini-halos are produced via the hadronic process, which shifts towards turbulent reacceleration at larger scales (Zandanel et al., 2014b).

The cluster A2142 shows some evidence towards these theories, with a large-scale 2 Mpc halo comprised of two parts (Venturi et al., 2017). This cluster is undergoing a minor merger and has only small disturbances in the ICM. The brighter inner core component has a spectral index $\alpha_i = 1.33 \pm 0.08$, which appears to be produced similarly to that of a minihalo. The outer component is fainter and somewhat steeper with $\alpha_o \approx 1.5$, suggested to be from large-scale turbulence within the ICM. Clusters may also show evidence of a minihalo and giant halo existing simultaneously, with RXJ1720+2638 and PSZ139.61+24.20 both exhibiting large-scale steep spectrum emission outside the core region (Savini et al., 2018; Savini et al., 2019). Physical relations to giant radio halos are useful in our understanding of how mergers change the behaviour of a cluster, however, are outside the scope of this thesis.

2.5 Interferometry

To better understand the reasoning behind the calibration and imaging processes, it is useful to first cover the basics of interferometry. Interferometer arrays consist of individual antennas that measure the amplitude and phase of incoming electromagnetic waves. These signals are then combined with each other to produce an interference pattern. The simplest example for this is an array of two antennas. Since each antenna has a small difference in distance from the source, there is a geometric time delay τ_g between each receiving a signal. By combining the two individual signals, the interference pattern is created due to the phase shift caused by the time delay. Given a cosinusoidal wave travelling at the speed of light c, from a source in the far field at $\boldsymbol{s}$, and the baseline separation between the two antennas $\boldsymbol{b}$, the geometric delay is $\tau_g = \boldsymbol{b} \cdot \boldsymbol{s}/c$ (see Figure 2.4). The measured interference pattern is then the product of the individual fields $V_1 = E\cos(\omega t)$ and $V_2 = E\cos(\omega(t - \tau_g))$, which can be written in the form $\frac{E^2}{2}\left[\cos(\omega\tau_g) + \cos(2\omega t - \omega\tau_g)\right]$ (Perley, 2018). The second term varies much faster than the first and can be averaged out, and the prefactor can be shortened to $P = \frac{E^2}{2}$. This gives the final cosine response as,

$$R_C = P\cos(\omega\tau_g) \tag{2.25}$$

or written in terms of the signal wavelength $\lambda = 2\pi c/\omega$ and the baseline,

$$R_C = P\cos\left(2\pi\frac{\boldsymbol{b}\cdot\boldsymbol{s}}{\lambda}\right) \tag{2.26}$$

The fringes of the interference pattern then sample the sky, with longer baselines corresponding to small spatial features, and small baselines mapping large spatial features. For the sky

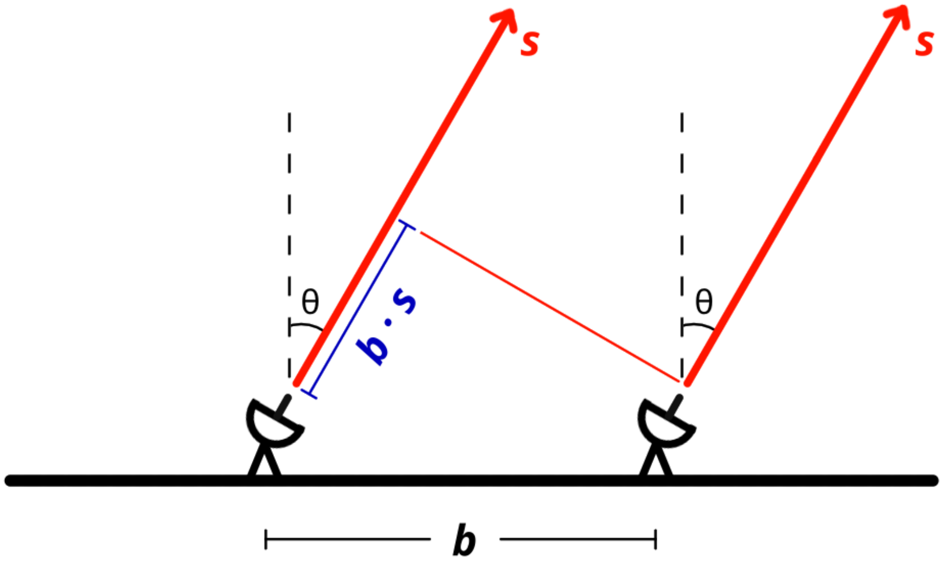


Figure 2.4: Geometry of a two antenna interferometer. The antenna is separated by baseline $\boldsymbol{b}$ and observe a source $\boldsymbol{s}$ at angle θ from zenith. The difference in path length is shown in blue $\boldsymbol{b} \cdot \boldsymbol{s}$, resulting in the geometric delay τ_g.

brightness map $I_\nu(\boldsymbol{s})$ at the frequency ν, the interferometer response is the spatial integral[10] of the sky modified by the cosinusoidal fringe pattern.

$$R_C = \iint I_\nu(\boldsymbol{s}) \cos\left(2\pi\nu \frac{\boldsymbol{b} \cdot \boldsymbol{s}}{c}\right) d\Omega \tag{2.27}$$

This response function is even due to the cosine term, resulting in any odd functions described by the intensity map being averaged out. By modifying the phase of one of the antennas by 90 degrees, $V_2 = E \sin(\omega(t - \tau_g))$, and applying the same processes as before, an odd response function is created $R_S = P \sin(\omega\tau_g)$. The two responses R_C and R_S can then be used to define the complex visibility function in terms of $\boldsymbol{u} = \boldsymbol{b}/\lambda$ (Perley, 2018).

$$V(\boldsymbol{b}) = R_C - iR_S = \iint I_\nu(\boldsymbol{s}) e^{-2\pi i \boldsymbol{u} \cdot \boldsymbol{s}} \, d\Omega \tag{2.28}$$

If the baselines lie on a flat plane, the complex visibility is then a 2D Fourier transform of the sky brightness at the observing frequency, allowing an inverse transform to be performed to reconstruct the sky given complete sampling. The baseline vectors $\boldsymbol{u} = (u, v, w)$ are defined in units of wavelength, with the flat uv-plane defined as a special case having $w = 0$. In 3D coordinates, the unit vector $\hat{\boldsymbol{u}}$ is aligned East-West, $\hat{\boldsymbol{v}}$ is North-South, and $\hat{\boldsymbol{w}}$ points towards the source (or phase centre). If the antenna array is not coplanar to the source, baselines may vary in w, which invalidates the 2D Fourier transform. To solve this, many imaging packages project the visibilities onto the uv-plane (called w-projection) (Perley, 2018).

[10]The spatial integral is in units of solid angle Ω, as the usual axes of the sky are given in arcseconds, or arcminutes.

An individual antenna pair only samples a single point in $\boldsymbol{u}$, so to maximise coverage of the uv-plane, interferometer arrays use N antennas in tandem to achieve $N(N-1)/2$ unique pairs simultaneously. Similarly, the rotation of the Earth allows these baselines to move with respect to the source, sweeping through the uv-plane in an ellipse to increase coverage (see Figure 2.5). As source declination approaches 0°, the ellipse flattens due to the source alignment with the Earth's rotation, reducing effective coverage. The final coverage of the uv-plane then depends on the arrangement and number of antennas, the target location in the sky, and the length of time that the source is being observed.

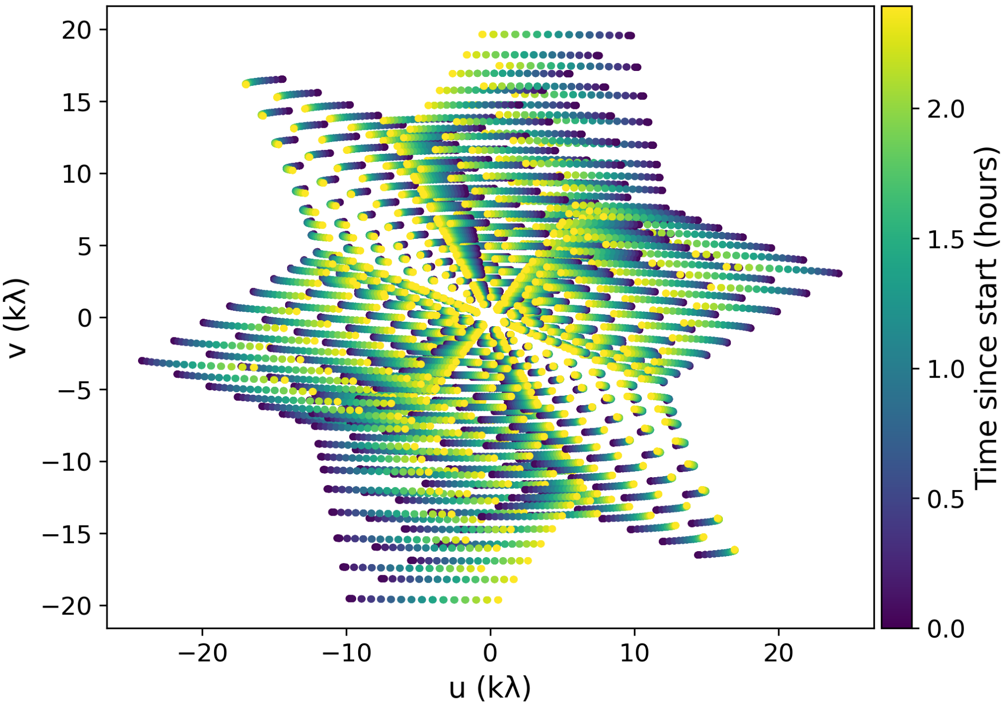


Figure 2.5: Example uv-coverage of a VLA D-configuration observation. The colour scale indicates time, and the coverage increases over the observation due to Earth rotation.

The incomplete coverage of the uv-plane results in a synthesised beam with radial spikes and periodics in image space, called the point spread function (PSF). The inverse transform of the sampled complex visibilities is then called the 'dirty' image, having the true sky convolved with the PSF. Several methods have been developed to deconvolve the PSF, with the most common method being the CLEAN algorithm (Högbom, 1974; Clark, 1980). This follows an iterative major-minor cycle to build a model of the sky brightness. Minor cycles operate in the image space, identifying source components in the residual image and adding them to the sky model. The major cycle transforms the model image into visibilities and subtracts them from the observed visibilities. These residuals are then Fourier transformed to update the residual image. With tuned parameterisation, the CLEAN algorithm can successfully deconvolve the PSF and return an accurate image of the sky brightness.

In practice, noise and calibration errors add further complexity to the observed complex visibilities. These can be delay errors, antenna-dependent gain and phase variations, atmospheric effects, and instrumental noise. This can be summarised via the Measurement Equation, which defines the observed visibilities $\boldsymbol{V}_{ij}$ between antennas i and j in terms of the ideal visibilities $\boldsymbol{V}_{ij}^{\mathrm{IDEAL}}$ and an accumulation of corrupting effects $\boldsymbol{J}_{ij}$ for that baseline (Hamaker et al., 1996; Sault et al., 1996).

$$\boldsymbol{V}_{ij} = \boldsymbol{J}_{ij}\boldsymbol{V}_{ij}^{\mathrm{IDEAL}} \tag{2.29}$$

The term $\boldsymbol{J}_{ij}$ can be factored into a sequence of individual effects applied to the incoming wave, with different physical origins and importances. Antennas measure the polarisation of the incoming wave, forming four correlations in a circular basis: LL, RR, LR, RL.[11] As a result, the matrices of the Measurement Equation are 2×2 to describe the polarisation response. In the context of minihalos for this thesis, the cross-polarisations are not of scientific interest and are used during the calibration process for flagging. The specific calibration and imaging procedures applied to the data analysed in this thesis are presented in the following chapter.

[11]The VLA uses a circular basis, though other antennas may measure in a linear basis: XX, YY, XY, YX.

Chapter 3

Calibration and Imaging

The observations to be calibrated are VLA X-band (8–12 GHz) data, taken in C- and D-configuration, of 12 galaxy clusters hosting minihalos. VLA antennas may be arranged in four standard configurations, termed A through D. A-configuration has the longest baselines and best angular resolution, at the expense of lower brightness sensitivity to extended emission. Baseline length decreases towards D-configuration, which has the poorest angular resolution, but greatest brightness sensitivity to extended emission. D-configuration was chosen to maximise the extended brightness sensitivity to detect the faint and extended minihalo emission. C-configuration observations were added to allow accurate imaging of compact AGN sources embedded within the minihalos.

The C-configuration observations were performed between January and February 2024, under project code 24A-411 (PI: Yvette Perrott). The D-configuration observations were taken the following year during February and March 2025, under project code 25A-157 (PI: Yvette Perrott).

The frequency range for X-band spans 7.977–12.025 GHz, with a total bandwidth of 4.048 GHz. This is split into 32 spectral windows with individual bandwidths of 128 MHz. Each spectral window has a spectral resolution of 2 MHz, resulting in a total of 64 channels per window. The baselines for C-configuration cover approximately 1–135 kλ, while the more compact D-configuration has the range 1–34 kλ.[1] For reference, an overview of the targeted clusters is given in Table 3.1, while details of the radio observations of each cluster are given in Table 3.2.

The *Common Astronomy Software Applications* (CASA)[2] was used to calibrate and image these observations. This software has been developed by an international collaboration of scientists and software developers and provides Python-based tools for the processing, analysis, and manipulation of observations from the VLA and the Atacama Large Millimeter/submillimeter

[1] As $\boldsymbol{u} = \boldsymbol{b}/\lambda$, baseline lengths are often expressed in units of the observing wavelength.

[2] https://casa.nrao.edu/index.shtml

Table 3.1: List of observed clusters, sorted by redshift.

Cluster	R.A.$_{J2000}$ (h, m, s)	Decl$_{J2000}$ (deg, arcmin, arcsec)	z	Scale (kpc arcsec^{-1})	Reference
2A0335+096	03 38 40.5	+09 58 12.4	0.036	0.716	(1)
A2626	23 36 30.5	+21 08 47.6	0.056	1.087	(2)
A1795	13 48 52.5	+26 35 34.0	0.062	1.196	(1)
A478	04 13 25.3	+10 27 55.0	0.088	1.646	(3)
A1413	11 55 18.0	+23 24 19.0	0.139	2.455	(1)
A2204	16 32 47.0	+05 34 35.0	0.152	2.646	(1)
RXJ1720+2638	17 20 10.0	+26 37 32.0	0.161	2.775	(2)
RXJ2129+0005	21 29 39.9	+00 05 21.3	0.234	3.727	(1)
MS1455+2232[a]	14 57 15.1	+22 20 34.0	0.258	4.007	(2)
Z3146	10 23 39.6	+04 11 11.0	0.289	4.347	(2)
RXCJ1115+0129[b]	11 15 52.0	+01 29 54.0	0.340	4.856	(2)
ACTJ0022-0036	00 22 13.0	−00 36 33.0	0.805	7.545	(4)

Notes. Right ascension and Declination values are taken from the observation phase centres. Values for kpc arcsec^{-1} are calculated using the Cosmology Calculator I (Wright, 2006) using the previously stated cosmology. Redshifts are taken from the corresponding reference: (1) Sanders et al. (2011), (2) Wen et al. (2015), (3) Liu et al. (2024), (4) Menanteau et al. (2013).
[a] Also known as ZwCl 1454.8+2233.
[b] Also known as ZwCl 1113.8+0144.

Array (ALMA). Part of this software suite is the VLA pipeline, which is able to perform calibration and automated flagging of raw observations. The pipeline allows a streamlined process with reduced manual intervention, while producing a final weblog detailing the applied steps for reproducibility. Flagging of data refers to the exclusion of problematic points, often due to the presence of radio frequency interference (RFI). For these observations, pipeline flagging did not sufficiently mitigate the RFI for accurate calibration, necessitating the development of a modified version. A description of the calibration process and pipeline steps is provided in Section 3.2. The following section details the RFI present in the data and the measures taken to mitigate it.

A concise overview of the full reduction and imaging workflow used throughout this chapter is shown in Figure 3.1.

3.1 Radio Frequency Interference

Picking up spurious radio sources, broadly termed RFI, is an unavoidable consequence of ground-based antennas. These sources can be attributed to the internal mechanisms of the telescope, as well as external sources such as ground-based and satellite communications. With the increasing number of satellites in orbit, RFI is a growing problem, as only a few narrow frequency ranges are protected for radio astronomy. RFI is particularly strong near the declination of 0°, since this aligns with the orbit of geostationary satellites, known as the Clarke Belt. RFI can be categorised

Table 3.2: Summary of the VLA X-band observations and calibration sources used.

Cluster	Cfg.	Date	Time (min)	Primary Calibrator	Secondary Calibrator
2A0335+096	C	2024-01-27	45.3	3C48	J0321+1221
	D	2025-02-28	68.5		
	D	2025-03-15	68.5		
A2626	C	2024-02-10	41.5	3C48	J2253+1608
	C	2024-02-11	41.5		
	D	2025-02-28	41.5		
	D	2025-02-28	33.2		
A1795	C	2024-02-04	71.5	3C286	3C286[a]
	D	2025-03-19	35.6		J1330+2509
	D	2025-03-27	35.9		
A478	C	2024-01-27	44.5	3C48	J0424+0204
	D	2025-02-27	124.5		
	D	2025-03-11	124.5		
A1413	C	2024-02-04	36.9	3C48	J1150+2417
	D	2025-02-28	129.2		
	D	2025-02-28	129.2		
A2204	C	2024-02-03	64.0	3C286	J1616+0459
	D	2025-02-28	36.6		
	D	2025-03-27	36.6		
RXJ1720+2638	C	2024-02-03	62.2	3C286	J1719+1745
	D	2025-02-28	35.6		
	D	2025-03-27	35.6		
RXJ2129+0005	C	2024-02-10	138.1	3C48	J2136+0041
	C	2024-02-11	138.1		
	D	2025-02-28	64.5		
	D	2025-02-28	64.5		
MS1455+2232	C	2024-02-03	61.1	3C286	J1436+2321
	D	2025-03-19	36.2		
	D	2025-03-27	35.6		
Z3146	C	2024-02-04	73.1	3C48	J1041+0610
	D	2025-02-27	36.5		
	D	2025-03-16	36.5		
RXCJ1115+0129	C	2024-02-04	73.1	3C286	J1058+0133
	D	2025-02-27	36.5		
	D	2025-03-16	36.5		
ACTJ0022-0036	C	2024-02-10	109.5	3C48	J0022+0014
	C	2024-02-11	109.5		
	D	2025-02-28	27.4		
	D	2025-02-28	36.5		

Notes. The date and length of each observation is provided. The primary calibrator was used for setting the flux scale and bandpass calibration. The secondary calibrator was used for complex gain and phase calibrations.
[a] The C-configuration observation for A1795 reuses the primary calibrator for the secondary calibration.

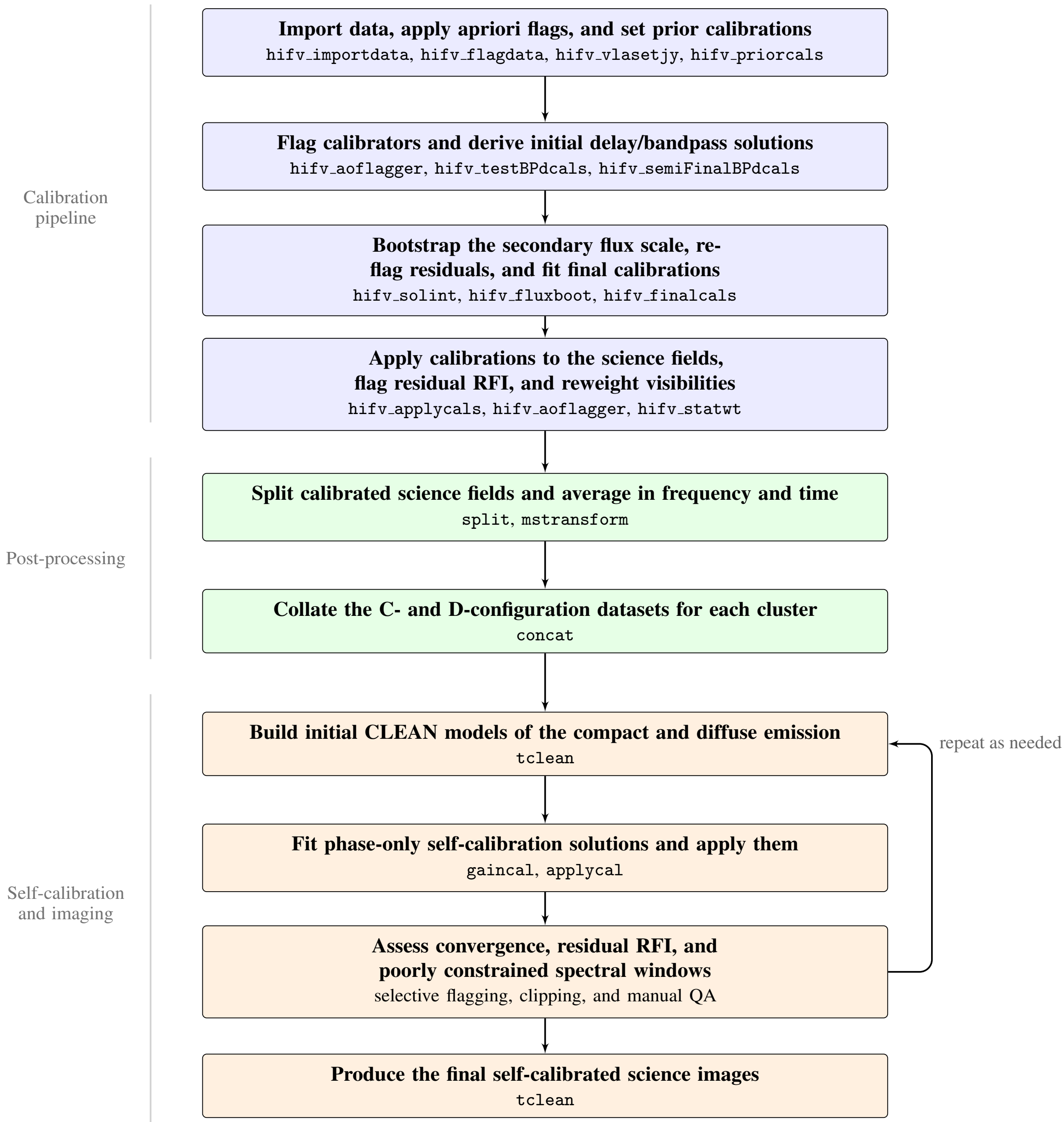


Figure 3.1: Overview of the full calibration and imaging workflow used in this chapter, from the modified VLA pipeline to the final self-calibrated science images.

as intermittent or continuous in time, and as broadband or narrowband in frequency (e.g. Figure 3.2). Detecting and automatically removing these data can be difficult due to the wide range of forms RFI can take. NRAO monitors RFI across the VLA frequency bands and provides descriptions of known sources to aid in cleaning the data. For observations at X-band, satellite and weather radars are an intermittent source between 9.3–9.9 GHz, continuous ground-based microwave links contaminate 10.7–11.5 GHz, and strong RFI exists between 11.7–12 GHz due to emission associated with the Clarke Belt.[3]

For the faint sources of minihalos ($\sim \mu$Jy), RFI signals can be orders of magnitude brighter (10–100s Jy). This smothers the signal and can bias calibration results. Sources of RFI may

[3] https://science.nrao.edu/facilities/vla/observing/RFI/X-Band

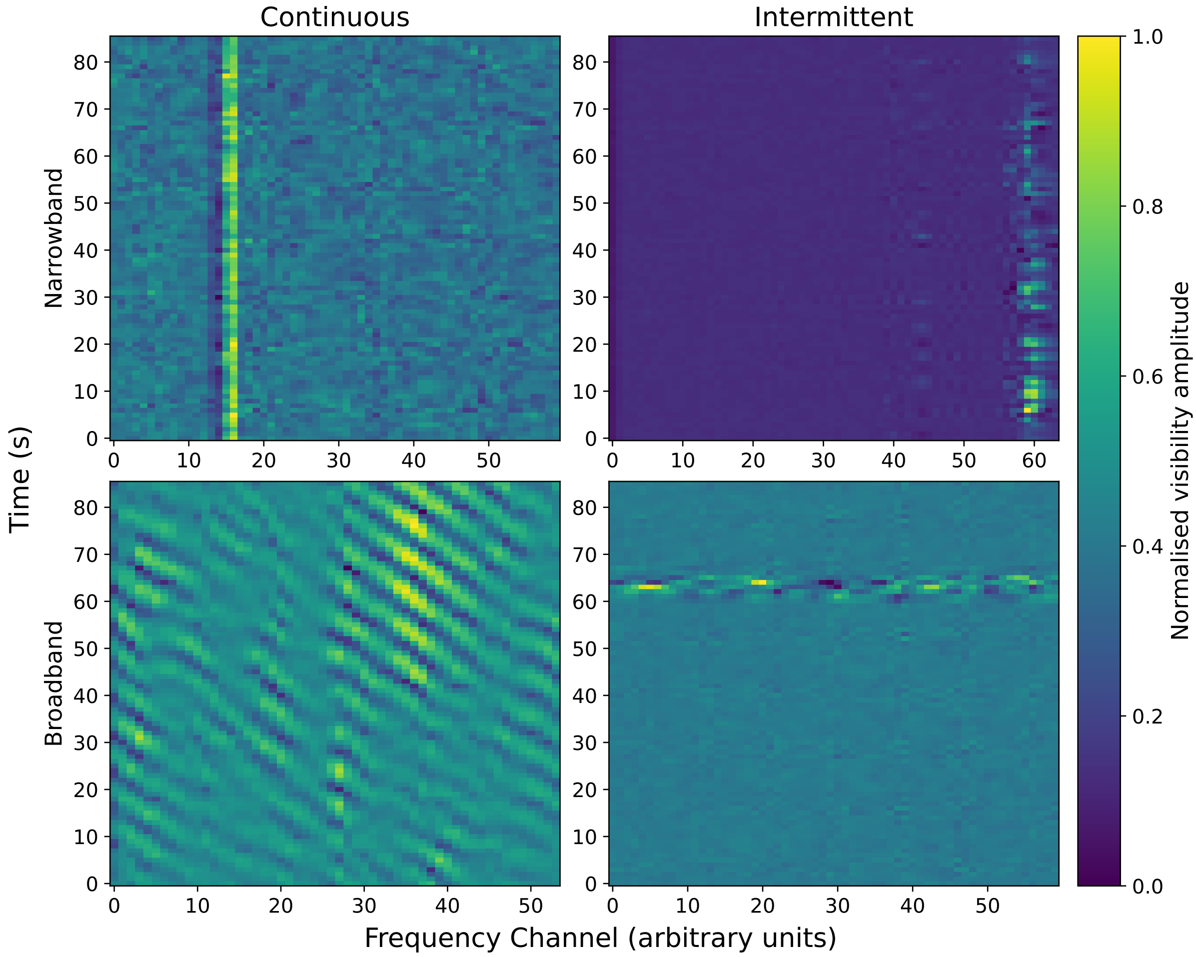


Figure 3.2: Examples of RFI in time–frequency space for a representative baseline. Columns show continuous and intermittent behaviour, while rows show narrowband (top: representative ~128 MHz sub-bands within 9.3–9.9 GHz, likely associated with satellite and weather radar emissions) and broadband RFI (bottom: representative ~128 MHz sub-bands within 10.7–12 GHz, likely associated with ground-based links and geostationary satellites).

have different delays in reaching antenna compared to the science target, which affects delay and phase calibrations and increases noise. This degrades the angular and brightness sensitivities. When imaging, RFI may appear as bright wave-like artefacts, arising from the Fourier transform of these high-amplitude visibilities.

The standard pipeline flagging uses a combination of the `tfcrop` and `rflag` algorithms to detect and remove RFI (Greisen, 2011).[4] `tfcrop` uses the visibility amplitudes to fit a linear time series, and a polynomial model for the frequency bandshape. Using this model, it then flags outlier points beyond a given relative threshold. This is adept at flagging short-duration spikes of RFI in both narrowband and broadband ranges, as well as time-persistent RFI in the narrowband for observations of bright sources such as a calibrator. `rflag` uses a sliding window to calculate the local visibility rms, and flags points beyond a relative threshold. This operates on individual

[4]Although the AIPS cookbook is readily available online, these algorithms are referenced to Appendix E.5 which is much harder to locate.

baselines over single frequencies (channels) with the sliding window over the time axis, followed by single time steps with the sliding window across channels. rflag is best at identifying weak and sparse RFI that may be affecting a subsection of antennas. The pipeline also extends the flags for each baseline, such that any channel or time step that is > 50% flagged will be entirely flagged. Flags are also propagated between polarisations, with all four correlations flagged if any individual polarisation is flagged.

The software suite AOFLAGGER (Offringa et al., 2010; Offringa et al., 2012; Offringa et al., 2023),[5] showed great improvements compared to the results from the pipeline flagging. This software was originally designed for cleaning LOFAR data but has since expanded for use with several other telescopes. The cleaning process is implemented via a flexible Lua script, allowing customisation in algorithms for an individual telescope and observation setup. The default AOFLAGGER VLA script performs rms-based time step and channel flagging, high-pass filtering, and morphological filtering. The morphological analysis is performed via the SumThreshold algorithm, which uses several decreasing thresholds to evaluate a sequence of samples. Points exceeding the highest threshold are marked for flagging, after which next lower threshold is evaluated, and the process is repeated. This works well for broadband and peaked RFI. This script has two primary parameters, base_threshold and transient_threshold_factor which define the thresholds to use for flagging. Lower thresholds correspond to increased sensitivity and, therefore, more aggressive flagging. For frequent but intermittent RFI bursts, transient_threshold_factor can be used to tune detection separately to the main threshold. The script was altered to preserve any pre-existing flags applied by the pipeline.

Based on the NRAO description of RFI in the X-band, the script was configured to do three passes with separate parameters for each. The first two passes flag the 9.3–9.5 GHz (weather radar) and 10.7–12 GHz (ground communications) ranges, respectively, followed by a final full frequency range pass. This allows for tuning of aggression based upon the RFI present in each observation. The higher angular resolution C-configuration observations exhibited a much lower level of RFI compared to the D-configuration, where many datasets were sufficiently contaminated by the ground communications that these spectral windows were completely removed. A summary of the flagging for datasets used in final imaging is provided in Section 3.6. Scripts were tuned for each source and observation independently using a manual and iterative testing process.

This process was incorporated into a copy of the VLA Pipeline version 6.6.1 as a custom step hifv_aoflagger with three arguments: the source target to be flagged flag_target, the Lua script aoflagger_file, and the data to operate over (raw or residuals). If the residual data column is selected, this first subtracts the model from the corrected data to estimate the residual spectrum. Given an accurate source model, these residuals are expected to be closer to a

[5] https://aoflagger.readthedocs.io/en/latest/index.html

Gaussian noise distribution, improving rms-based RFI detection. This provides flexibility per observed source with customisable parameters per source and region of RFI. The implementation of this custom pipeline step is available in a separate repository.[6]

3.2 Calibration pipeline

All observations were calibrated using a single Python file, `casa_pipescript.py`. An example pipeline script is given in Appendix A. The outcome of this calibration process are visibilities with correctly scaled amplitudes and phases to the true sky. The raw measurements contain several frequency and time dependencies, effects that can arise from the atmosphere or instrument for example. The visibilities also require an absolute amplitude scale describing the instrumental response to true sky brightness.

The absolute visibility scale is set by comparison to a compact source called a primary calibrator. These sources are well-studied, have a stable brightness over long time periods, and are extremely bright. Scaling the visibilities to match the source model allows the raw visibility amplitudes to be measured in physical units.

The following calibrations take varying forms, but the primary ones are the bandpass and gain calibrations. The bandpass calibration is a frequency-dependent but time-averaged solution, while the gain solutions are frequency-averaged and time-dependent. Each calibration has an amplitude and phase component, and when combined, correct the visibilities for both frequency-varying and time-varying distortions.

The following subsections detail the calibration steps in more detail and are written with reference to the Pipeline Tasks Reference Manual and CASA documentation.[7,8].

3.2.1 Apriori Flagging

Some flagging is able to be determined prior to the analysis and arise from the observing configuration and instrument. These deterministic (apriori) flags are applied with the `hifv_flagdata` task. An overview of these categories are:

- Antenna shadowing occurs when the target is close to the horizon, as antennas may block the field of view of antennas behind them. CASA automatically models the positioning of antennas against the source in order to remove shadowed antennas.

[6] https://github.com/MoonlitMoo/casa-pipeline/tree/main/hifv

[7] https://almascience.eso.org/processing/documents-and-tools/cycle12/reference-manual-2025

[8] https://casadocs.readthedocs.io/en/stable/index.html

- Unwanted intents are scans in the MS that are not science targets or necessary for calibration. These are primarily system configuration related or unspecified scans (e.g. Listing A.1).
- Auto-correlation is the correlation of the antenna to itself, these zero distance baselines are removed.
- Edge channel flagging removes the first and last channel of each spectral window, as the interferometer is least sensitive at these boundaries.
- Clipping removes any visibility amplitudes equal to zero, likely arising from an instrument fault.
- 'Quacking' flags a short period at the start and end of each scan. Since the antenna physically moves (slews) between scans, this eliminates data measured while the antennas may still be settling.
- Baseband flagging removes ten channels at the start and end of each baseband, analogous to the spectral window flagging. For these observations, there are two basebands covering 8–10 GHz and 10–12 GHz.

3.2.2 Prior calibrations

Similarly, there are several deterministic calibrations that are monitored during the observation. These are defined by the instrument configuration. A summary of those applied during `hifv_priorcals` are:

- Gain curves account for elevation-dependent gains, where the angle of the antenna causes gravitational deformation of the dish surface.
- Changes in elevation also alter the atmospheric path-length, which is not entirely transparent to radio emission. The atmospheric opacity correction adjusts for this attenuation, while other atmospheric effects (re-emitted radio emission, delay errors due to density fluctuations) are corrected by later fitted calibrations.
- The geographical coordinates of antennas may have small time-dependent errors over the course of the observation, which result in delay and baseline errors when correlated. This calibration references an online database of antenna position corrections to apply adjustments.
- Requantiser gains are amplitude offsets introduced when visibilities are re-quantised to a lower bit size to reduce data rates. These effects are antenna and time-dependent and estimated during the observation.

3.2.3 Primary calibrator flux scale

The visibilities measured by each antenna have a gain and phase scale that differs depending on the antenna pointing. These relative amplitudes and phases must then be scaled to an absolute flux scale to obtain comparable measurements. Primary calibrators are unresolved point sources with high flux densities (typically $\gg 10$ Jy at $\lambda = 1$ m) and stable over long timescales (years). CASA implements the Perley–Butler flux density scale, spanning the frequency range 50 MHz to 50 GHz (Perley et al., 2017). This scale contains polynomial flux models of several primary calibrators, allowing the raw visibilities to be scaled by minimising residuals to these models. This assumes an accurate model and low time-variability of the primary calibrator between the modelling and observation. The Perley–Butler scale quotes accuracy to 3–5% over the full frequency range. NRAO recommends observations of 3C286, 3C48, 3C147, or 3C138 to set the absolute flux scale. For the observed clusters, the nearest primary calibrators were either 3C286 or 3C48 (see Table 3.2). Scans of the primary calibrators were performed at the start or end of each observing block.

Due to the small number of primary calibrators, the likelihood of a science target lying in close proximity to one is low. This necessitates the use of secondary calibrators to account for residual pointing-dependent gain and phase errors. The calibrators are also unresolved compact sources, but may exhibit significant time variability on the timescale of months to years. At 5 GHz, observations have shown a median variability of 4.3% over 60–260 days, increasing to 5.3% over the same time period at 33 GHz (Kurinsky et al., 2013). Because of this variability, precise emission models for these sources are not available. The absolute flux density scale obtained from the primary calibrator is instead used to scale the measured flux density of the secondary sources. These closer sources then allow for more accurate gain and phase solutions for the science targets.

3.2.4 Calibrator flagging

Before using the primary calibrator observation to fit calibrations, it is flagged for RFI using the custom step `hifv_aoflagger` discussed previously. This step is the first non-*a priori* flagging applied to this source and operates on the uncalibrated visibilities. A comparison of the post-flagging visibilities is given in Figure 3.3. This may share a script and parameter set with other calibrators or science targets, depending on the similarity of the RFI present. These parameters were tested prior to running the full calibration pipeline to optimise the mitigation of RFI signals and preservation of source data. A second round of flagging is performed prior to the `hifv_finalcals` step, described in a following section. The second flagging step operates on the residual data, which are obtained by subtracting the model from the corrected data. By operating on the residual data, variation due to calibration effects are suppressed, allowing RFI to be more easily detected. These two passes improve the RFI detection as the quality of the

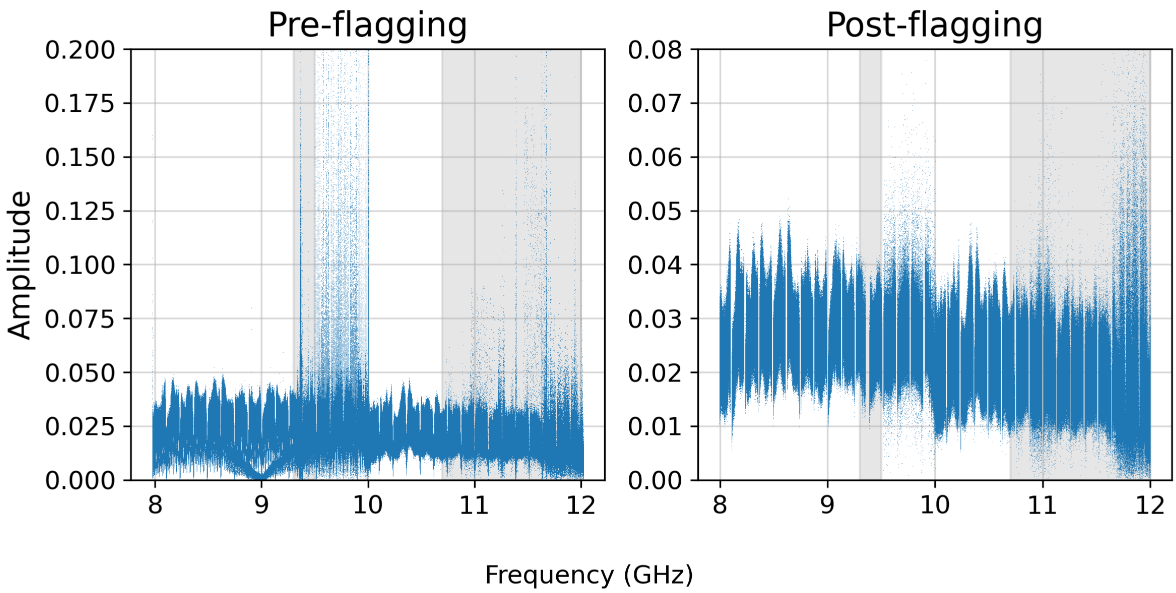


Figure 3.3: Frequency versus amplitude of the raw primary calibrator visibilities before (left) and after (right) *a priori* and AOFLAGGER flagging. Note the different y-axis scales between panels. Known high-RFI regions are shaded in grey. Data is shown for all baselines. Residual shape in the post-flagging data is primarily due to the absence of the bandpass correction (e.g. Figure 3.4).

visibilities increase, resulting in cleaner gain and phase solutions during the final calibration stage.

Similarly to the primary calibrators, the secondary calibrators are flagged prior to fitting any calibrations. A first flagging run is performed prior to setting the flux density scale on the raw visibilities. A second pass is performed prior to creating the final calibrations on the corrected visibilities.

3.2.5 Delay and Bandpass calibrations

Errors in antenna position and time measurements cause systematic errors in the measured delay between antennas, which produces errors in the correlated phase. These effects are antenna-dependent and can be corrected using observations of a strong calibrator. By assuming one antenna as a reference with zero delay, the relative delays of all other antennas can be calibrated. This provides one solution per antenna, per spectral window, and per parallel correlation (LL, RR). Following the delay calibration, residual frequency-dependent errors in the phase and gain remain. This is correlated with the spectral window sub-bands and the basebands, resulting in a total bandpass response. These errors are also corrected using observations of a strong calibrator. These calibrations are iteratively constructed over the `hifv_testBPdcals`, `hifv_semiFinalBPdcals`, and `hifv_finalcals` stages.

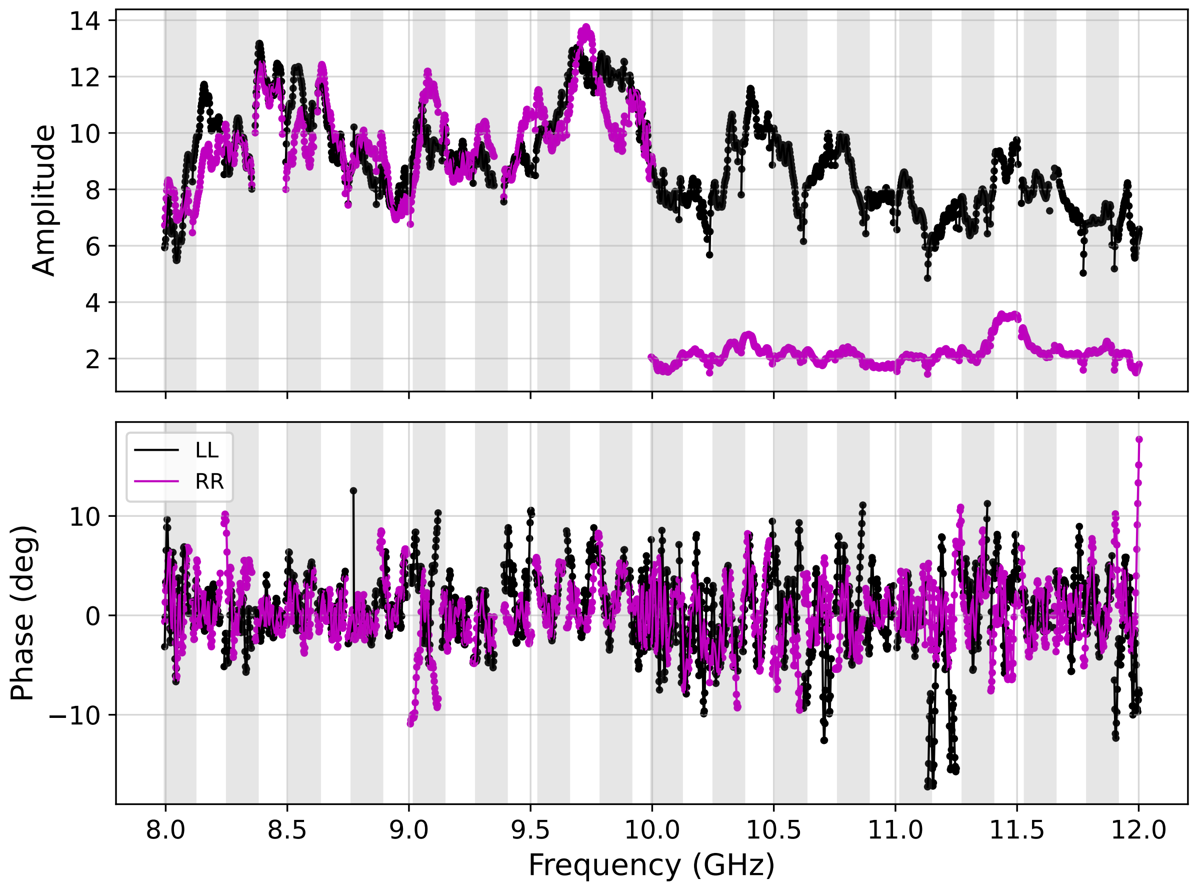


Figure 3.4: Example bandpass calibration. Amplitude (top) and phase (bottom) time-independent solutions are shown for the parallel-hand (LL, RR) correlations of a single antenna. Spectral windows are indicated by alternating grey shaded regions to highlight discontinuities at their boundaries. The pronounced discontinuity in the RR correlation at 10 GHz marks the baseband transition between 8–10 and 10–12 GHz.

3.2.6 Secondary calibrator flux scale

The absolute flux density scale must be transferred from the primary calibrator to the secondary calibrator. This is performed by fitting and scaling a polynomial model to the delay and bandpass-corrected secondary calibrator with the `hifv_fluxboot` task. To accurately transfer the known flux scale to the unknown source flux, this model must be well-constrained. Therefore, to reduce time-variable noise contributing to the model, a short time interval complex gain solution is also used. This is defined by a minimal timescale found by `hifv_solint` task. The minimum possible interval is the integration timescale, however, a high flagging fraction may make this infeasible. Outliers from these solutions are then excluded from fitting the polynomial model. This scaled model then represents the predicted true spectrum of the secondary calibrator and added to the dataset. It is important to note that any missing spectral windows or antennas in the primary calibrator solutions are propagated to the secondary calibrator and subsequent calibrations.

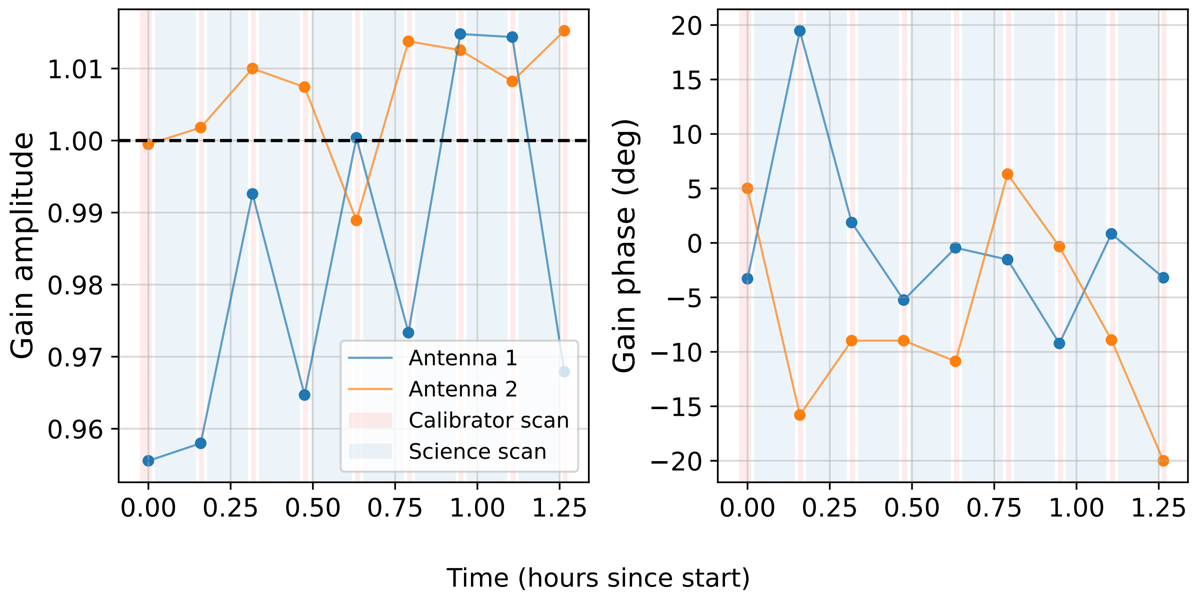


Figure 3.5: Example complex gain calibration. Amplitude (left) and phase (right) solutions for a single polarisation and spectral window are shown for two representative antennas over the course of an observation. Red shaded regions mark secondary calibrator scans, interleaved between blue shaded science target scans.

3.2.7 Final calibration

With flux density models established for the calibrators and flagging performed on the corrected visibilities, the final calibrations are fitted using the `hifv_finalcals` task. This step first refits the delay and bandpass calibrations using the primary calibrator. An average phase solution is then computed over all scans to normalise the time-dependent phase errors. The short-timescale phase corrections are then computed for all calibrators and used to derive the final complex gain and phase solutions. An example of these time-dependent solutions are shown in Figure 3.5.

The final calibration list then consists of the prior, delay, bandpass, average phase, complex gain, and phase calibrations. These calibrations are applied to the raw visibilities to obtain the final corrected data. This includes applying the calibrations from the secondary calibrators to the science targets. These solutions contain the pointing-dependent corrections to the scaling derived from the primary calibration. Once again, any missing spectral windows or antennas in the secondary calibrator solutions are propagated to the science targets to maintain consistency.

3.2.8 Science target processing

The science targets then undergo a single flagging run performed on the residual visibilities. Ideally, this would occur without the model subtraction, but this was not implemented in the pipeline step at the time. The default model column of a Measurement Set is a 1 Jy point source located at the phase centre. Subtracting this is equivalent to offsetting the visibilities by a constant, for which rms-based detection is insensitive. As the science targets are primarily comprised of faint and diffuse emission, the visibilities are visually similar to noise, and deviations above the rms can be attributed to RFI. The benefits gained in using the residual data, rather than the raw visibilities, for flagging were considered to outweigh the risk of erroneous flagging.

The visibilities are then weighted using the `hifv_statwt` task. This performs a sliding window rms measurement, weighted per baseline based on the resulting exposure time and the number of unflagged points. This improves the sensitivity of the final images by down-weighting baselines that have noisier data and, therefore, a lower signal-to-noise ratio.

3.2.9 Post-processing outputs

The calibrated science fields were split from the full dataset, keeping only the parallel correlations and averaging over frequency and time bins. Averaging over frequency alters the *uv* distribution due to the relation $\boldsymbol{u} = \boldsymbol{b}/\lambda$, which results in radial smearing.

A typical maximum limit for frequency bin size $\Delta\nu$ can be found in terms of the antenna diameter D, the maximum baseline B, observing frequency ν, and sensitivity x (in the range 1–4, where lower values are stricter limits). For high frequency VLA data, the sensitivity was set to 4.

$$\Delta\nu = x\frac{\nu D}{4B} \tag{3.1}$$

For VLA dishes of diameter 25 m, maximum baselines $\sim 1.25\times10^5$ m, and a minimum observing frequency of 8 GHz, this gives $\Delta\nu \approx 57$ MHz. For channels of width 2 MHz, this corresponds channel averaging by a factor of 28.5. Erring on the side of caution, frequency averaging was decided over 16 channels.

Time averaging can also cause smearing due to the Earth's rotation, with an analogous time bin limit Δt in terms of antenna diameter D, maximum baseline B, sensitivity x (in the range 1–3.7), and the angular frequency of Earth's rotation ω. The maximum value for the sensitivity $x = 3.7$ was chosen to match the spectral averaging.

$$\Delta t = x\frac{D}{4\omega B} \tag{3.2}$$

With $\omega \approx 7.27\times10^{-5}$ rad s^{-1}, the maximal time bin is $\Delta t \approx 91$ s. The time averaging was chosen at 60s based upon this limit. Since these values are independent of a specific observation, the same averaging was applied to all datasets.

These metrics are based on the C-configuration data. The D-configuration observations have shorter maximum baselines and, therefore, could be averaged over a larger frequency bin. For simplicity of imaging, the frequency bins were kept the same between antenna configurations. Although this averaging reduces the precision of the data, it also reduces the total size by a factor $\sim$60. By choosing these reasonable limits, the impact on the final image is minimal while vastly improving the associated computation time.

3.3 Calibration exceptions

Two observing runs, ID `25A-157.sb47856643.eb48084985.60734.6548690162` and `25A-157.sb47896788.eb48078564.60733.06076180555`, required manual intervention during calibration. In both datasets this was due to a secondary calibrator exhibiting noisy and inconsistent amplitude and phase solutions that affected spectral windows 22–31. Following the final calibration, this appeared as an antenna-dependent spread of amplitudes and phases over these spectral windows. This was caused by the pipeline selecting a long solution interval despite the presence of short-timescale phase variations, likely due to high RFI in these spectral windows inflating the minimum timescale estimated during `hifv_solint`. To resolve this, a short-timescale phase calibration was fitted while excluding spectral windows contaminated with RFI to produce a single time-dependent solution per antenna and correlation. Due to the lack of frequency dependence, a residual phase and amplitude gradient across the spectral windows remained following this correction. This was solved by iteratively fitting a linear bandpass solution to this gradient, excluding outlying points until convergence. These phase and bandpass solutions were then applied in place of the final complex gain and phase solutions to the affected secondary calibrator and science field. This solution was sufficient to correct the phase issues in both datasets.

The observation `25A-157.sb47856643.eb48084985.60734.6548690162` also contained a mislabelled secondary calibrator scan. This scan was initially labelled with the intent for bandpass calibration. As such, it was handled alongside the primary calibrator data during the pipeline, significantly skewing the resulting bandpass calibration. This was fixed using the CASA `defintent` task to set the intent to the correct values.

3.4 Imaging

The collated calibrated datasets were then imaged using CASA's `tclean` task. This implements the CLEAN algorithm discussed in Section 2.5. There are several important imaging parameters that are independent of the source properties, but depend on the instrument configuration and scientific objective. For the calibrated observations, test imaging revealed a synthesised elliptical beam with a Full Width at Half Maximum (FWHM), defined as the beam width at half the peak sensitivity, of approximately 2.7×2.2 arcsec in C-configuration. To achieve sufficient spatial sampling, 3–5 pixels across the beam are recommended. Accordingly, a cell size of 0.5×0.5 arcsec was chosen for all imaging. The total image size was set to $2^8 \times 3^2 = 2304$ pixels per side to achieve an angular diameter of 19.2×19.2 arcminutes.[9] To prevent radial clipping due to the primary beam truncation, the fractional limit `pblimit` was set to -1×10^{-6}. This angular size

[9]Imaging is most efficient when the image size is even and factorisable into combinations of 2, 3, and 5 only.

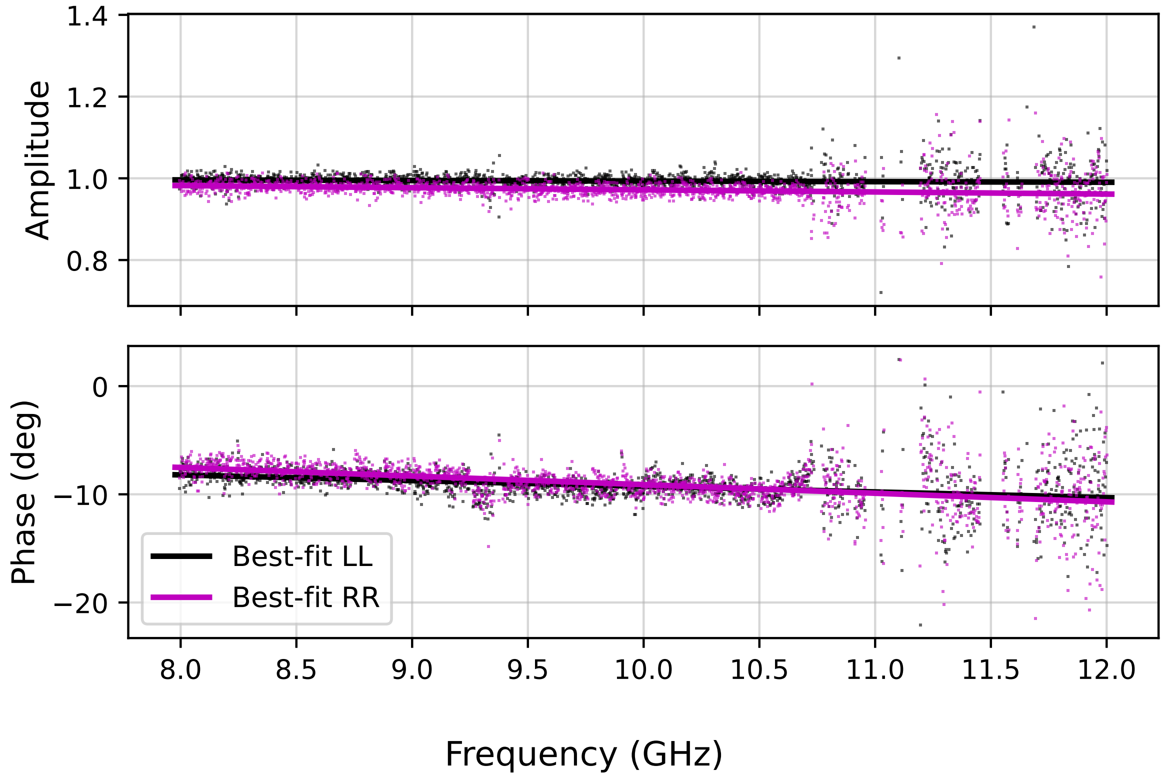


Figure 3.6: Example of a custom linear bandpass fit to the residual gain errors. Amplitude (top) and phase (bottom) solutions are found for the parallel-hand (LL, RR) correlations of a single antenna and scan. The fitted solutions remain stable despite increased RFI contamination at the highest frequencies.

captures the full minihalo region for all clusters and was sufficiently wide to image any nearby sources causing PSF artefacts in the central region.

These observations require continuum (8–12 GHz) imaging at several angular scales due to the relative sizes of AGN and minihalos. As such, the spectral definition mode was set to multi-frequency synthesis `mfs`, the deconvolver (minor cycle algorithm) to multi-term (i.e. multi-scale when provided scales) multi-frequency synthesis `mtmfs` (Rau et al., 2011), and the `gridder` was left as standard due to negligible widefield effects. As the total intensity is of interest, Stokes I imaging was used, which averages the two parallel polarisations $I = \frac{RR+LL}{2}$ for an intensity measurement. The gain, or fraction of source flux subtracted from the residual during CLEAN cycles, was left at the default of 0.1. These parameters, summarised in Table 3.3, define the general spatial and spectral sampling and deconvolution methods necessary for imaging minihalos.

The imaging can be further tuned by changing the weighting of visibilities. Natural weighting applies the baseline visibility weights determined during the calibration `statwt` task. This results in a weighting grid across the uv-plane that follows the sampling density which maximises the image sensitivity, but negatively impacts the PSF. This arises in brighter and more widely spread sidelobes convolved into the image, influencing the cleaning process. In contrast, uniform

Table 3.3: General imaging parameters for `tclean` used for all imaging.

Parameter	Value
`imsize`	2304×2304 pixels
`cell`	0.5×0.5 arcsec
`pblimit`	-1×10^{-6}
`specmode`	`mfs`
`deconvolver`	`mtmfs`
`gridder`	standard
`stokes`	I
`gain`	0.1

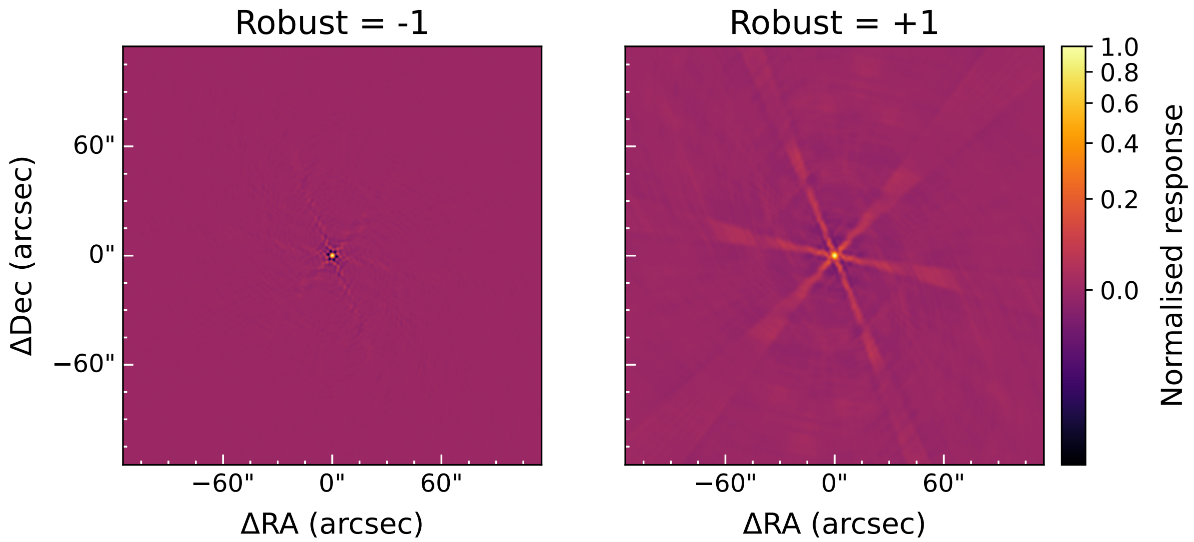


Figure 3.7: Example point spread functions for imaging with Briggs weighting `robust` $= -1$ (more uniformly weighted; left) and `robust` $= 1$ (more naturally weighted; right). Both images are normalised to a peak value of 1 and shown on a common colour scale to illustrate the difference in sidelobe prominence.

weighting normalises the base weights by the total weight of all data at each *uv* coordinate. This results in a close to uniform weighting across the *uv*-plane which suppresses PSF sidelobes at the cost of decreased sensitivity to extended emission. Briggs weighting creates smoothly varying scale between the natural and uniform schemes using a `robust` parameter r. For $r = -2$ Briggs weighting matches uniform, while $r = 2$ corresponds to natural weighting. Briggs weighting was chosen as the weighting scheme for this flexibility, with the robust parameter altered depending on the intention of the image. Positive robust values of 0.5 and 1 were used to image the minihalo by maximising the synthesised beam sensitivity and, therefore, maximising the detection of the faint surface brightness emission. Negative robusts of -0.5 and -1 were used when imaging AGN, which improves the modelling of these compact objects. Figure 3.7 illustrates the difference in the PSF between positive and negative robusts, with more prominent sidelobes visible in the positive robust version.

To model emission at different angular scales with the `mtmfs` deconvolver, a list of CLEAN scales can be provided. These scales correspond to possible pixel diameters of Gaussian components added to the model. This allows a more complex image to be modelled and is especially important to model the extended emission of minihalos. Point sources are a recommended scale to use, with the next smallest corresponding to the beam size, which is approximately four pixels for these images. In general the used scales corresponded to zero (point source) and powers of two beginning at 4, [0, 4, 8, 16, ...], with the maximum scale dependent on the observed emission. Rarely, the intermediate scale of 2 was found to improve the quality of modelling. The selection of different scales can also be adjusted using the `smallscalebias`, which biases the scale selection towards larger (-1) or smaller scales (1).

The final cleaning parameter is the `threshold`, which defines the minimum residual level to clean to. This was altered depending on the rms σ of the residual image and the lower limit was set in the range 3–5σ for automated cleaning ($\sim 10\ \mu$Jy), and close to σ ($\sim 3\ \mu$Jy) for manual cleaning. For limits close to σ, spurious noise peaks may be incorporated into the model which can negatively impact the resulting image. To prevent this, masking is usually used to define regions of true emission that are then cleaned.

Masking can be performed interactively, however, automasking is implemented for `tclean`. This has several rms-based thresholds (with the default CASA value), including `noisethreshold` (5) to prevent masking emission near the current residual rms, `sidelobethreshold` (1.25) to prevent adding bright sidelobes into the mask, `negativethreshold` (0) to mask negative features, and `lownoisethreshold` (2) to grow the current mask border using binary dilation. Automatic masking was generally well-behaved with these parameters, however, the `sidelobethreshold` was increased to values of 2.5 and 5 for some clusters that featured a PSF with bright and frequent sidelobes.

3.5 Self-calibration

The angular resolution and noise levels of the resulting images are then limited by residual antenna-based phase and gain errors. These arise from the interpolation from solutions at the secondary calibrator sky position to the science target position. The surface-brightness model created during cleaning can be used to fit further complex gain and phase solutions to improve the calibration model. This is called self-calibration as it is an iterative and self-consistent calibration method. Several rounds of phase-only calibration were performed for each cluster, which improved the spatial resolution of the emission. Figure 3.8 illustrates this improvement, with the diffuse emission exhibiting a greater spatial extent and a smoother morphology following self-calibration. Amplitude calibrations were not used as this requires a high signal-to-noise ratio not available for the faint minihalo emission.

Self-calibration was performed on the joint image produced by all datasets for a cluster. This

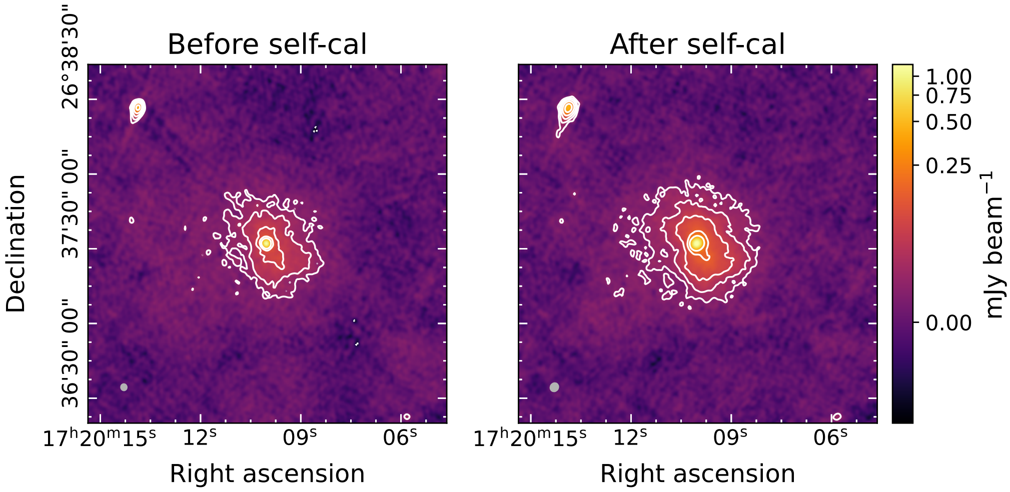


Figure 3.8: Before (left) and after (right) self-calibration images of RXJ1720+2638. Both images were produced with identical imaging parameters (`robust` = 1). The restoring beam in the left panel is 2.64" × 2.41" at position angle 7.3° (shown in grey, bottom left) and the rms is 3.28 μJy beam^{-1}; the right panel has a beam of 3.52" × 3.17" at position angle -34.2° with rms 3.18 μJy beam^{-1}. White contours increase by factors of two, starting from their respective 3σ. The negative -3σ level is shown as a white dotted contour.

combined higher angular resolution modelling of compact sources from the C-configuration data, with the more sensitive extended flux detection of D-configuration visibilities. The combined model was then used to fit phase solutions over the scan interval per antenna, per spectral window, and per correlation. These calibrated data was imaged to create a new model and the process repeated until the phase solutions converged. This was done in several stages for each cluster, with initial images using a negative robust $r = -0.5$, small scales $[0, 2]$, and high thresholds 0.1–0.05 mJy to create accurate models of bright and compact sources. Once a desirable model was achieved, the robust was swapped to positive $r = 0.5$, larger scales were added $[0, 4, 8]$, and the threshold was lowered towards 3σ. This process assisted in preventing prematurely deep cleaning, which can create model artefacts such as negative point sources and virtual sidelobe emission. The parameters discussed in the prior section were adjusted on a per-cluster basis in order to obtain physical and accurate models for self-calibration. The notes and parameters for each cluster are available in the associated repository under Image-Processing/<cluster_name>/full_image.

Between self-calibration rounds, the datasets were assessed for highly flagged spectral windows, remaining RFI in visibility amplitudes, and inconsistent phase solutions. Spectral windows where less than ∼25% of data remained were flagged as they decreased the signal-to-noise ratio. Spectral windows where phase solutions did not converge were flagged as this indicated that solutions were dominated by noise. The visibilities were also clipped to a maximum value to remove remaining spikes, with the threshold in the range 0.1–0.5 mJy based on the observed visibilities.

Time variability of AGN was noticed during calibration of several clusters. Due to the year offset between the C- and D-configuration observations, several bright AGN exhibited significantly different peak flux densities ($\approx$20–40%). When combined into a single brightness model, these inconsistencies produced deep negative features surrounding the variable sources. Two correction methods were adopted depending on the resulting image quality, being individual modelling or flux scaling.

In the first approach, each dataset was imaged independently to obtain sky models unaffected by variability. These were then merged with the joint model by replacing a small region around each variable source with the corresponding region from the individual model. The resulting modified joint models were applied to their corresponding datasets for continued self-calibration. This preserved the advantages of joint imaging across the majority of the image while correcting the variable components. This approach was most effective for localised sources where the splice boundaries did not introduce spatial discontinuities in the model.

For variable sources embedded within larger scale emission, such as the minihalo, flux scaling was used instead. Here, the individual models for each dataset were used to fit the peak brightness of the variable source. These values were then used to scale the joint image model flux within the small region surrounding the source. This approach mitigated large discontinuities while preserving the underlying extended emission, although existing negative pixels remained. The use of these methods is noted in the Results chapter where it was required.

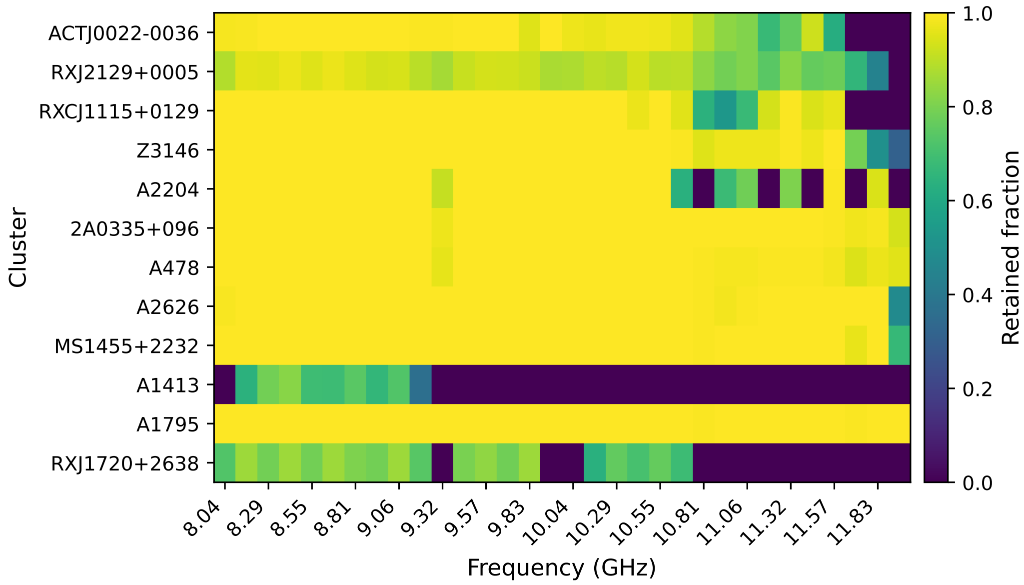
3.6 Final imaging datasets

The calibrated datasets exhibit varying levels of flagging, with D-configuration observations most affected. This is worst at frequencies > 10.7 GHz due to ground-based communications and geostationary satellites creating broadband interference, and at $\sim$ 9.3 GHz from narrowband weather radars. Excluding the outliers of A1413 and RXJ1720+2638, this flagging is more evident at lower declinations, nearer to the Clarke Belt. An average exposure time of 22% was lost for C-configuration observations, while an average of 32% was lost for D-configuration. Per cluster losses are summarised in Table 3.4.

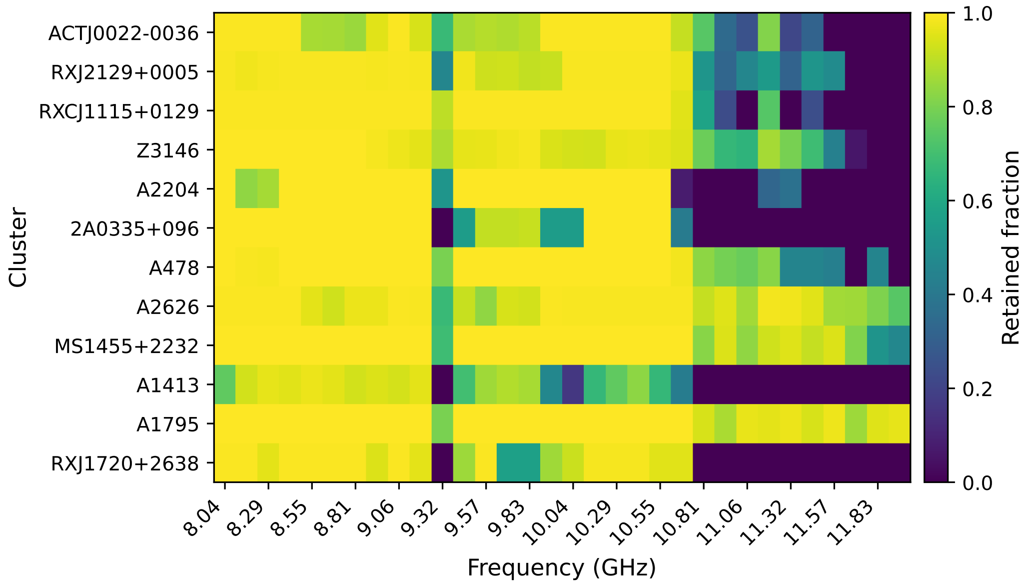
The effective exposure time T_{eff} is calculated as the sum of exposure times per integration time T_i, weighted by the unflagged fractional bandwidth $f_{i,j}$ of each polarisation correlation j (normalised by the number of correlations per integration time n_i), and the maximum number of baselines $N_b = N(N-1)/2$ for N antenna.

$$T_{\mathrm{eff}} = \frac{1}{N_b}\left(\sum_i T_i \sum_j \left[\frac{f_{i,j}}{n_i}\right]\right) \tag{3.3}$$

The single C-configuration observation of A1413 is a notable outlier, with the majority of spectral windows completely flagged. This occurred during self-calibration, rather than the pipeline, and may represent the removal of residual RFI. This observation was taken in the same block as A1795, Z3146, and RXCJ1115+0129, which do not exhibit the same level of flagging. The lack of C-configuration data for A1413 means that point source estimates and source subtraction should be interpreted with caution, as this likely results in a worse signal-to-noise and uncertainty. Furthermore, the effective frequency of the image is skewed towards a lower value as the majority of the highest frequencies are completely flagged.



(a) C-configuration



(b) D-configuration

Figure 3.9: Summary of retained data per cluster following the calibration pipeline and self-calibration. The results are separated for C-configuration and D-configuration and ordered in increasing declination. The fraction of retained data is shown as a function of frequency. Binning is done per spectral window and averaged with multiple observations weighted by exposure time.

Table 3.4: Effective exposure time following calibration pipeline and self-calibration. The total raw observation time, effective time, and percent lost is given per cluster and configuration.

Cluster	Config	Total Time (min)	Effective Time (min)	Reduced (%)
2A0335+096	C	45.3	43.4	4.1
	D	137.0	68.8	49.8
A1413	C	36.9	6.8	81.5
	D	258.4	116.3	55.0
A1795	C	71.5	70.8	1.0
	D	71.5	72.3	0.0
A2204	C	64.0	49.3	23.0
	D	73.2	46.0	37.2
A2626	C	83.0	67.8	18.3
	D	74.7	57.1	23.6
A478	C	44.5	41.9	5.9
	D	249.0	174.8	29.8
ACTJ0022-0036	C	219.0	171.7	21.6
	D	63.9	39.1	38.8
MS1455+2232	C	61.1	57.5	5.9
	D	71.8	67.3	6.3
RXCJ1115+0129	C	73.1	61.4	16.0
	D	73.0	49.0	32.9
RXJ1720+2638	C	62.2	26.5	57.3
	D	71.2	36.0	49.4
RXJ2129+0005	C	276.2	207.7	24.8
	D	129.0	79.7	38.2
Z3146	C	73.1	68.2	6.7
	D	73.0	60.4	17.2

Chapter 4

Analysis Methods

The primary goal of this thesis is to extract reliable flux density measurements of the minihalo emission. These values are then able to be compared with previous works to estimate the spectral index of the emission. With the calibrated datasets, several further steps in the image domain are necessary to isolate the diffuse emission from point sources, define the spatial region of the emission, and estimate the associated uncertainties. This process is summarised in Figure 4.1. This chapter details the methods used to reach a total flux measurement for the minihalo, with the following chapter presenting the results per cluster.

4.1 Removal of contaminating sources

Contamination of the minihalo emission can occur in several ways. This includes emission from AGN embedded within the minihalo and positive or negative radial artefacts from the convolution of the PSF with poorly modelled sources. These bias the total minihalo flux, necessitating the removal of such contaminating sources. Using a sky model of these sources, the Fourier transformed model can be subtracted from the observed visibilities. The resulting source-subtracted images then depict the uncontaminated minihalo emission.

Sources with variable fluxes between datasets were primarily attributed to the associated time-variability between the observation dates. This variability resulted in an unphysical joint model and artefacts in the joint imaging. Independent models of the source were created for each dataset and subtracted individually, so that the joint image no longer contained the variable source. An accurate model of the source is necessary for subtraction, since if the model contains minihalo emission, the resulting image will feature a negative bowl where it was removed. Conversely, if the model underestimates the source flux, the minihalo emission will be artificially peaked and the total flux density will be boosted.

To ensure that the subtraction model does not include the minihalo, it is desirable to remove

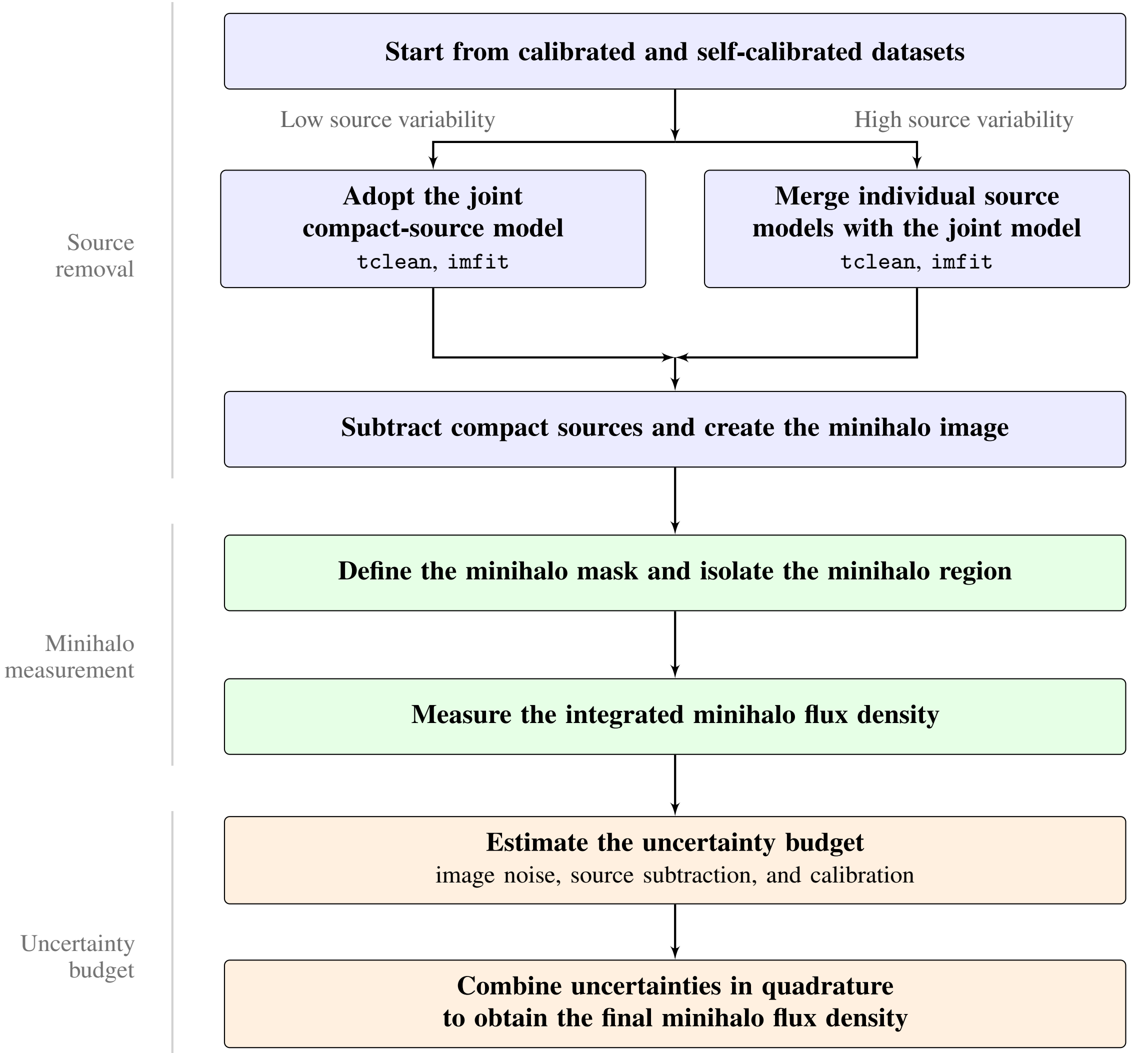


Figure 4.1: Overview of the minihalo analysis pipeline.

flux corresponding to large-scale emission. Since baseline length corresponds to spatial scale in the image, removing short baselines will cause the large scale minihalo emission to be strongly suppressed in the image. This was achieved by specifying a `uvrange` in `tclean`, which allows specification of a baseline cutoff. The weighting of these images was set to uniform, or `robust` $= -2$, to maximise the sidelobe suppression for imaging the compact sources. To determine an optimal *uv* value to perform the cutoff, testing was performed on the observations of RXJ1720+2638. The spatial region of this minihalo is contaminated with the bright BCG, while a variable AGN to the southwest caused negative radial spikes in proximity to the minihalo emission (see Figure 4.2).

To evaluate an appropriate cutoff, the lower *uv*-bound was varied and the BCG was fitted as a Gaussian point source using the `imfit` task for each dataset. As this region contains the sum of the minihalo and BCG fluxes, there exists a cutoff value for which the BCG emission alone remains, and the peak flux will converge to a constant value. The peak flux was used instead of the integrated flux as the BCG represents an unresolved point source, though these values

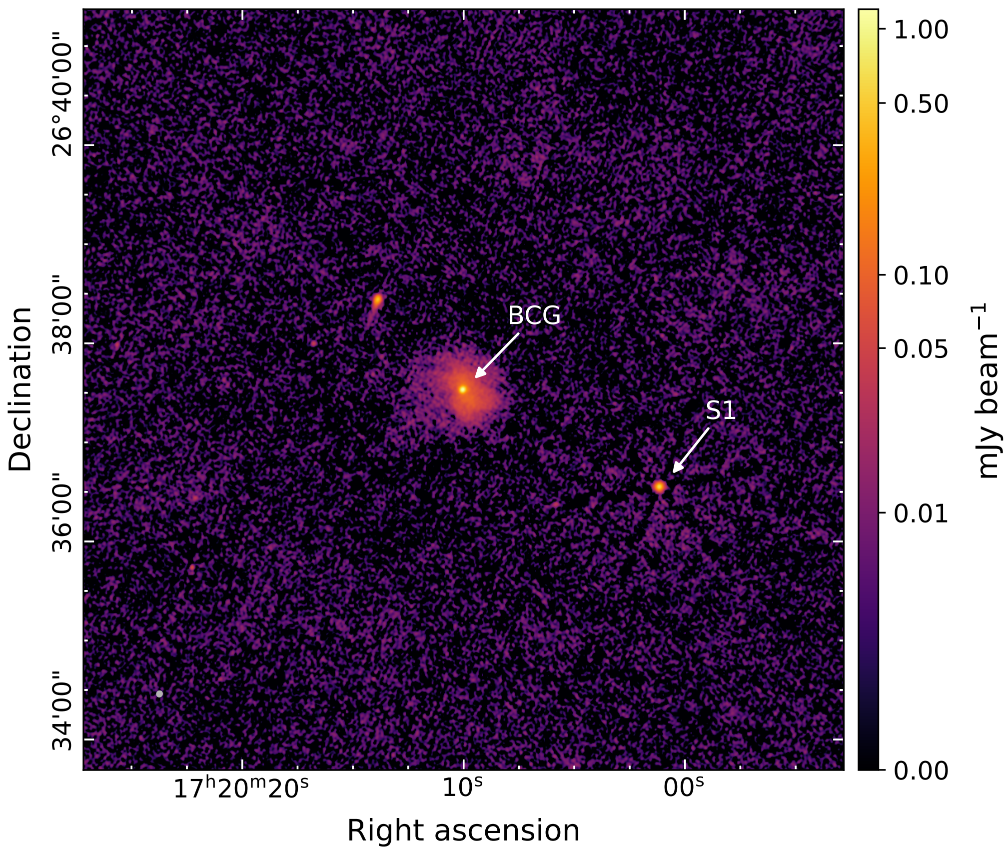


Figure 4.2: Calibrated image of RXJ1720+2638, showing minihalo contamination from the central unresolved BCG and radial PSF artefacts from a variable AGN (S1). The image has a restoring beam of 2.79”×2.51” at a position angle of -25.6° (shown in grey at bottom left). The noise rms is 3.49 μJy beam^{-1} and the image was cleaned to a threshold of 3σ.

will also converge as extended emission is removed. Figure 4.3 shows the results of the testing, with convergence at baselines > 10 kλ. Associated images at a selection of these *uv*-cutoffs are shown in Figure 4.4 to illustrate the removal of the diffuse emission. The peak values for the D-configuration datasets are inconsistent at low cutoffs due to the greater minihalo contamination (from higher extended brightness sensitivity). This caused the fitted Gaussian to exhibit a low peak and wide spread, boosting the integrated flux over the extended region and suppressing the peak flux. A minimum cutoff was chosen at 12 kλ to ensure no minihalo emission remained, while minimising the effect on the signal-to-noise ratio.

A clear difference of approximately 10% in the measured AGN flux exists between the D-configuration observations. This may be due to calibration error or true variability. This important distinction is discussed in more detail in Section 4.3. The models for AGN subtraction used a 12 kλ minimum *uv*-cut, and all models were subtracted on an individual dataset basis to prevent under- or over-subtraction due to time variability.

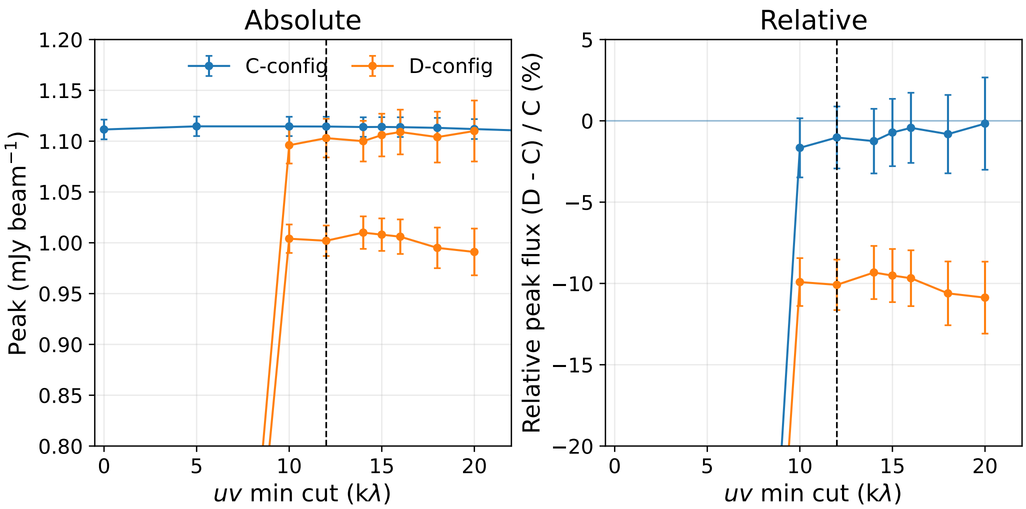


Figure 4.3: Flux density fit for RXJ1720+2638 BCG as a function of minimum uv-range used during imaging. The black vertical line indicates the adopted minimum uv-cut. *Left Panel:* Absolute peak flux with 1σ error for each dataset. *Right Panel:* Relative peak flux density of D-configuration to C-configuration.

4.2 Minihalo spatial extent

To estimate the total minihalo flux density, the individual fluxes within the spatial extent of the minihalo must be integrated. For valid comparisons with previous studies, the ideal approach would use an identical spatial mask. However, these masks are often not available or lack detailed information. For these clusters, the spatial extent is instead estimated from pixels with flux $> 3\sigma$, where the noise σ was determined from the rms per beam across the residual image. The initial binary selection mask was segmented into separate emission structures. The minihalo region was identified and confirmed to be isolated from other discrete sources prior to final summation. The choice of mask is noted where relevant within the results.

4.3 Flux density uncertainty

There are three primary sources of error for estimations in the total flux density of the minihalo emission: the spatial image noise σ_{noise}, the source subtraction σ_{sub}, and the calibration error σ_{cal}. The final uncertainty is then the total of these components added in quadrature,

$$\sigma = \sqrt{\sigma_{sub}^2 + \sigma_{noise}^2 + \sigma_{cal}^2} \tag{4.1}$$

The image noise from the spatial sum over the minihalo region is calculated as the residual rms per beam of the minihalo image, weighted by the beams in the spatial region. The number of beams is given by $N_{beam} = A_{MH}/A_{beam}$, where A_{MH} is the area of the minihalo and A_{beam} is the restoring beam area. This gives the total image noise as $\sigma_{noise}^2 = (\sigma_{rms}\sqrt{N_{beam}})^2$.

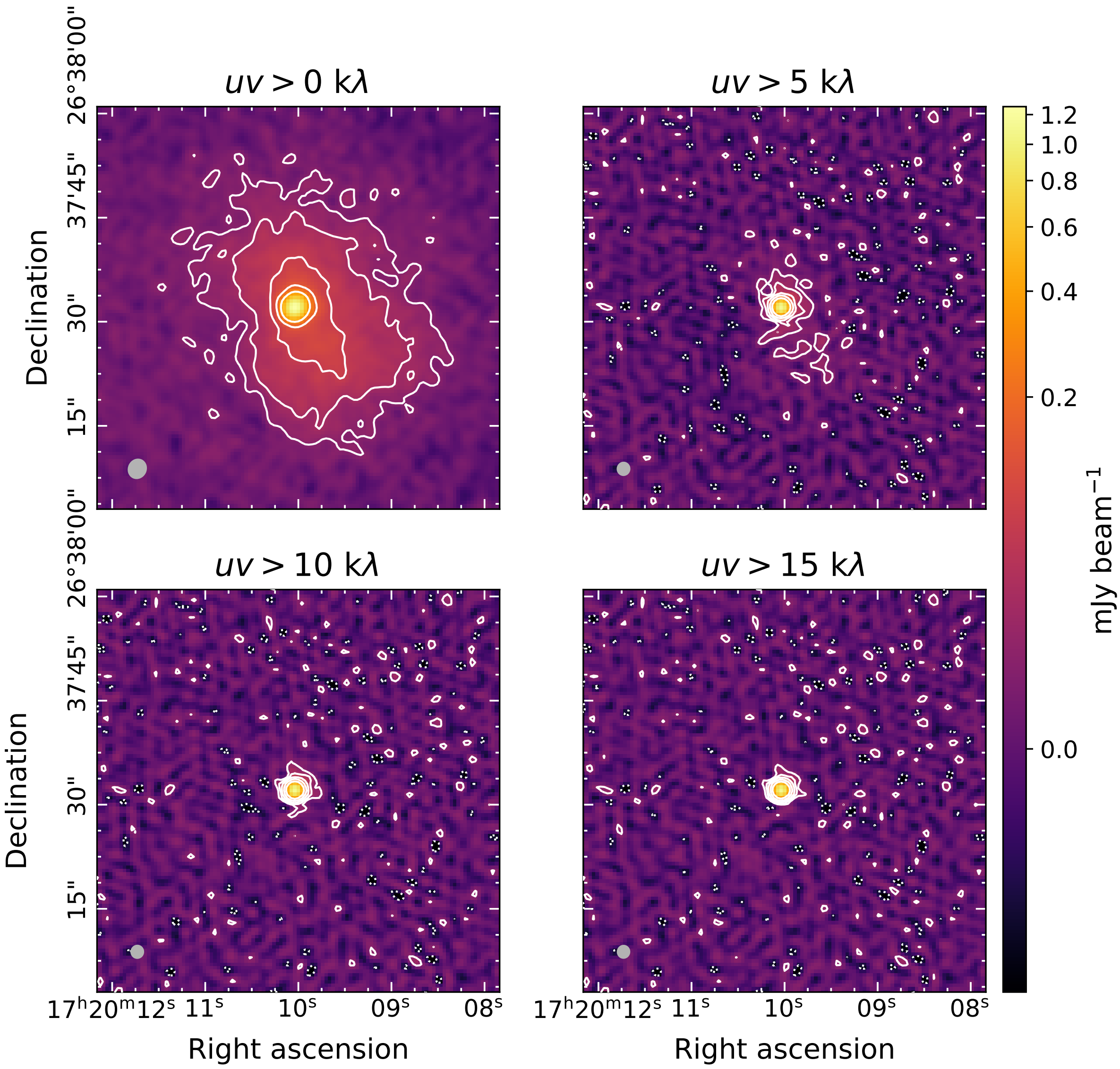


Figure 4.4: Imaging of the central BCG of RXJ1720+2638 with progressively increasing minimum uv-cut. The top-left panel shows no minimum cutoff, while subsequent panels apply thresholds of 5, 10, and 15 kλ. Top-left image is zoom in of Figure 4.2. The remaining images have beam sizes of approximately 1.85” $\times$ 1.75” arcsec at position angles $\approx$-7.5° (shown in grey, bottom left) and rms $\approx$6.5 μJy. White contours are spaced by factors of two, starting from 20 μJy, with negative contours at -20 μJy shown as white dotted lines.

To estimate the subtraction error, the identical methodology is adopted as to previous studies. Giacintucci et al. (2014a) defined this error as,

$$\sigma_{sub}^2 = \left(I_{MH,s} \times A_s\right)^2 \tag{4.2}$$

where $I_{MH,s}$ is the average surface brightness of the minihalo close to the source, and A_s is the area occupied by the subtraction. This is a conservative estimate as it assumes that the area is entirely contaminated and potentially erroneous. As the embedded sources are primarily located near the brightest region of the minihalo, this estimate can dominate the overall uncertainty for bright minihalos.

An alternative method would be to consider the subtraction error as arising from uncertainties in the point source imaging. This would correspond to $\sigma_{noise,s}$ of the image used for source subtraction over the subtraction area A_s. To robustly use this metric, source subtraction would require robust testing and evaluation to ensure that the source is accurately separated from the minihalo emission. The benefit would be a considerably lower uncertainty as the image noise is naturally below that of the minihalo. Such subtraction testing was not performed over the wider cluster sample, and so is not implemented in this thesis.

Secondary calibrators and compact AGN used for amplitude and phase calibration can show flux density fluctuations between 5–10% at month- to year-timescales (Kurinsky et al., 2013). This is in addition to the 3% intrinsic uncertainty at 8–12 GHz in the primary calibrators (Perley et al., 2017). Any variation in a target source is, therefore, a combination of calibrator variability and residual calibration errors. This motivates the inclusion of a systematic calibration uncertainty to account for fluctuations in the primary and secondary calibrators. To estimate this effect, the modelled power-law spectra of the secondary calibrators was compared between consecutive month observations to constrain residual errors. Such month-to-month variations should be minimal, broadband, and comparable to the intrinsic primary calibrator uncertainty. In contrast, highly variable sources are expected to exhibit changes in the spectral shape in addition to large amplitude variations. A purely broadband change can be modelled as $S_{obs1} = C\, S_{obs2}$, where C is a constant scaling factor, and was fit using χ^2 minimisation. Deviations from the linear relation indicate changes in the spectral shape, resulting in large residuals and leading to the exclusion of such cases from further analysis. The error-weighted absolute variation over all clusters is measured at 2.14%, with only a single cluster exceeding 5% deviation (Figure 4.5). This level of variation is consistent with the expected time variability of calibration sources, and all larger deviations correspond to changes in spectral shape.

The observed variation could be entirely real (for perfect calibration accuracy) or entirely due to uncertainty (for no source variability), or more likely, to be some combination of the two. Since there is no robust method to disentangle the relative proportions, a conservative estimate maximises the calibration error. As such, the calibration uncertainty was adopted as 5%.

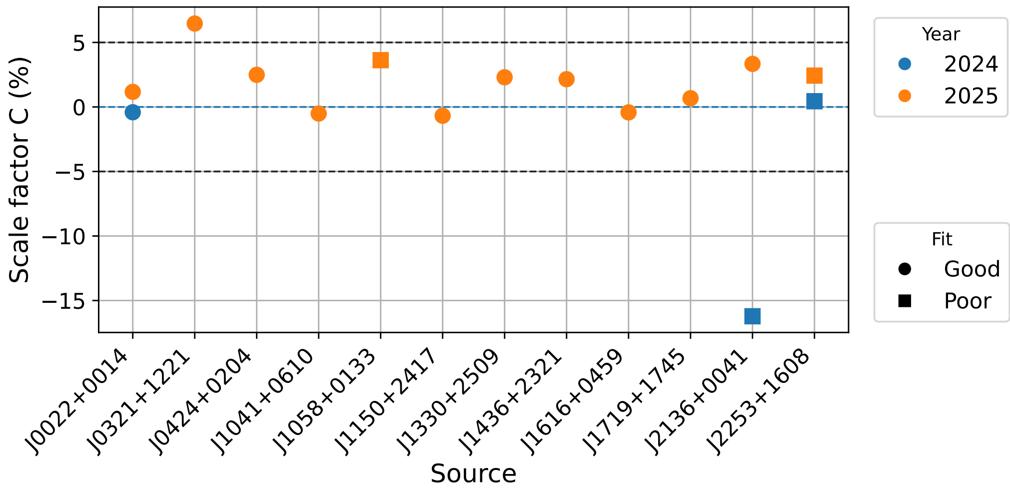


Figure 4.5: Month-to-month broadband variation for all secondary calibrators. The year of measured variation is denoted by colour. A good fit (circles) corresponds to highly linear broadband scaling, while a poor fit (squares) indicates changes in spectral shape. The adopted calibration uncertainty (5%) is marked for comparison.

Cluster	Calibration	Self-calibration	Minihalo flux
2A0335+096	Done	Done	No
A1413	Done	Done	No
A1795	Done	No	No
A2204	Done	Done	No
A2626	Done	No	No
A478	Done	Done	Done
ACTJ0022-0036	Done	Done	No
MS1455+2232	Done	Done	No
RXCJ1115+0129	Done	Done	Done
RXJ1720+2638	Done	Done	Done
RXJ2129+0005	Done	Done	Done
Z3146	Done	No	No

Table 4.1: Processing completeness for each cluster, showing the status of calibration, self-calibration, and final minihalo flux measurement.

4.4 Sample completion

The completeness of analysis for the observed clusters is provided in Table 4.1. All calibration has been completed, alongside the majority of self-calibration. Estimates of the minihalo flux are obtained for four clusters. The corresponding results are presented in the following chapter, alongside the status of the remainder of clusters. Incompleteness is primarily attributed to the time constraint of a Master's thesis, and challenges in the modelling of more complex cluster morphologies.

Chapter 5

Results

The following sections present background information and summaries of previous minihalo studies for each cluster. Any cluster-specific deviations to the analysis procedures outlined in the previous chapter are noted for clarity. The results from this work are presented here, with their broader implications discussed in the following chapter. These clusters are presented in order of highest completion of imaging and analysis, to the lowest. The minihalo flux densities and measured source fluxes are summarised in Tables 5.2 and 5.3, respectively.

5.1 RXJ1720+2638

RXJ1720+2638 (also known as RXJ1720.1+2638) is a cool core galaxy cluster ($M_{500} = 5.9 \times 10^{14}$ $M_\odot$, $R_{500} = 1.39$ Mpc (Planck Collaboration et al., 2016))[1] at $z = 0.160$ (Owers et al., 2011). Early X-ray observations detected the presence of two cold fronts, a sharp edge to the southeast and an extended plateau to the northwest (Mazzotta et al., 2001b). This suggested the presence of a dense, cool gas core ($T \approx 4$ keV) embedded within a more diffuse and hot ICM ($T \approx 10$ keV), with the cold fronts likely arising via a minor merger. Weak gravitational lensing studies (Okabe et al., 2010), together with galaxy density mapping (Owers et al., 2011), revealed the presence of a substructure approximately 550 kpc north of the cluster core. Owers et al. (2011) suggested that a northward passage of this substructure, passing east of the cluster core, provided the perturbation required to induce large-scale gas sloshing. RXJ1720+2638 was one of the first two clusters where cold fronts and minihalos were found to exhibit a spatial correlation, with the minihalo emission bounded by the cold fronts (Mazzotta et al., 2008).

A detailed spectral study was performed by Giacintucci et al. (2014a), who analysed observations at six frequencies from 317 MHz to 8.4 GHz taken by the GMRT and VLA. To perform

[1]This mass was calculated for a redshift of 0.164. The value of R_{500} for this cluster, and those for the other clusters, are calculated from the spherical mass overdensity relation $M_\Delta = \frac{4}{3}\pi R_\Delta^3 \Delta\rho(z)$ and matching cosmology used to find the mass.

accurate comparisons between these frequencies, they limited all baseline ranges to 1–50 kλ to enforce a unified sampling of the uv-plane. They noted three primary radio components in the core region, the central radio galaxy (BCG, $\alpha = 0.87 \pm 0.03$), a head-tail radio galaxy to the northeast with the tail to the south ($\alpha = 1.05 \pm 0.04$, steepening down the tail), and diffuse minihalo emission. The BCG was noted to be unresolved with no jets or lobes detected at scales larger than 1.4 kpc. Higher angular resolution observations by the extended Multi-Element Radio Linked Interferometer Network (e-MERLIN) at 4.76 GHz found the BCG to exhibit sub-kpc jets (Perrott et al., 2023). These jets extend approximately 0.8 kpc to both the east and west, although uncertainties in the recovered flux suggested an additional undetected extent.

Giacintucci et al. (2014a) separated the minihalo emission into a central region with a diameter of 160 kpc and a tail extending approximately 200 kpc to the south. These were found to have spectral indices of $\alpha_c = 1.0 \pm 0.1$ and $\alpha_t = 1.5 \pm 0.1$, respectively. The core region exhibited a relatively constant spectral index across its extent, while clear spectral steepening was observed along the tail. Low-frequency LOFAR images at 144 MHz are consistent with the constant spectral index in the core, with $\alpha_{144}^{612} = 0.93 \pm 0.10$, and pronounced steepening across the cold fronts (Savini et al., 2019; Biava et al., 2021). Biava et al. (2021) also imaged at 54 MHz with the LOFAR Low Band Antenna (LBA), though they noted the resultant spectral indices for the minihalo and AGN were steeper than expected. They argued that this is due to the presence of a larger scale steep-spectrum diffuse component super imposed over the minihalo emission at lower frequencies.

Although the spectral index for the core has been found to be consistent up to 4.86 GHz, the 8.44 GHz measurement was lower than predicted by the single power-law model. While this could indicate possible steepening, Giacintucci et al. (2014b) noted that the 4.86 GHz and 8.44 GHz observations suffered from significantly poorer uv sampling, likely leading to flux losses through the filtering of large-scale emission. Perrott et al. (2021) performed a high-frequency measurement of the central minihalo region at 15.5 GHz, using the Arcminute Microkelvin Imager (AMI). By constructing a model of the minihalo directly in the uv-plane to account for spatial filtering, they found that a single power-law with $\alpha = 1.0$ remains consistent up to 15.5 GHz. They similarly reprocessed the 8.44 GHz data using the same method of uv-plane modelling, finding that the resultant best fitting model closely matched the single power-law spectrum. This demonstrated the previously observed steepening of Giacintucci et al. (2014b) was caused by spatial filtering, rather than an intrinsic physical process.

5.1.1 10 GHz observation

During imaging, two contaminating sources were identified prior to minihalo flux estimation (see Figure 5.1). These comprised the BCG (S1) embedded within the minihalo emission, and a variable AGN (S3) to the southwest. The head-tail source (S2) was well cleaned, with its tail

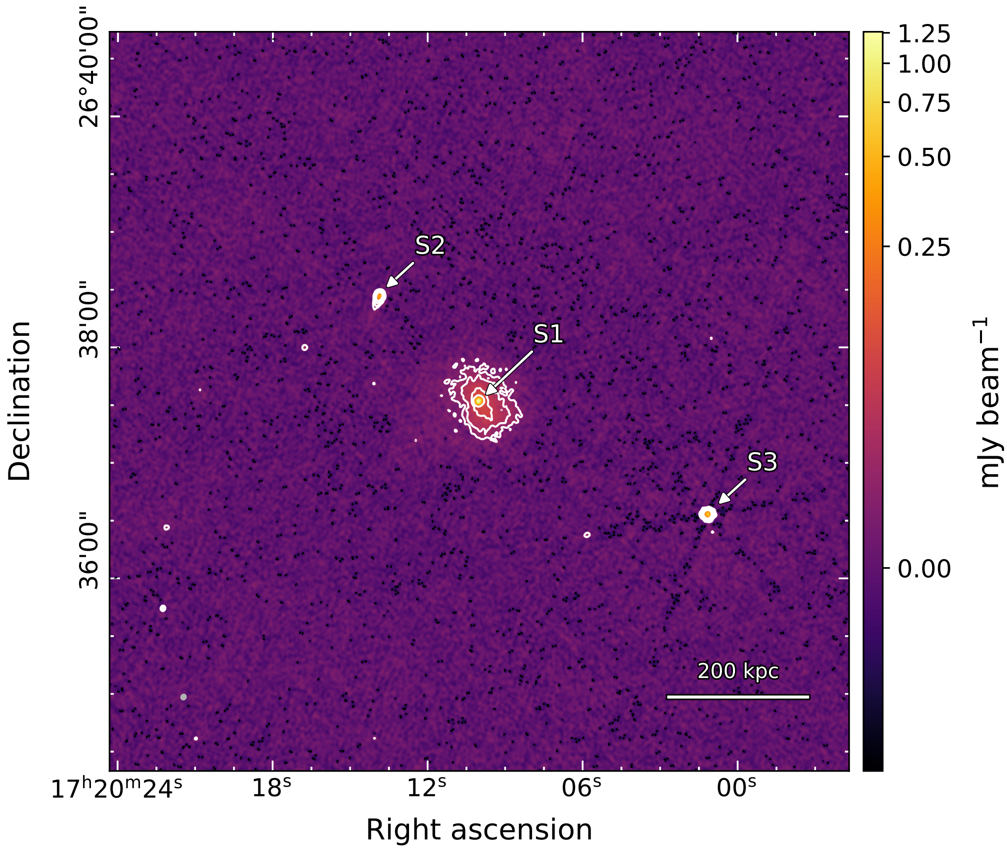


Figure 5.1: RXJ1720+2638. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.79" × 2.51" at a position angle of -25.6° (shown in grey, bottom left) and an rms noise of 3.48 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

emission isolated from that of the minihalo. Sources S1 and S3 were modelled individually per dataset with a *uv*-cut of 12 kλ and subsequently subtracted.

After subtraction, the minihalo emission was imaged using a `robust` of 1. The multi-scale `tclean` scales were set to $[0, 4, 8, 16, 32]$ with a `smallscalebias` of -0.5. The wideband primary beam correction was applied to the subtracted image, correcting for loss of flux in the tail region. The resulting image is shown in Figure 5.2. As Giacintucci et al. (2014a) provide details of the masking region, including the separation of centre and tail regions, the same masking was applied for flux estimation. A *uv* constraint of 1–50 kλ was also applied to ensure matched sampling. This allowed an accurate reproduction of these regions for analysis and comparison to previous studies.

The total minihalo flux density is measured as 8.61 ± 0.56 mJy, with the majority 7.60 ± 0.52 mJy attributed to the central region and only 1.01 ± 0.35 mJy arising from the tail. The measured flux densities for the relevant sources are provided in Table 5.3. Spectral indices are fitted for the total minihalo emission, the central region, the tail, and the BCG using flux densities from this work combined with measurements from Giacintucci et al. (2014a) (excluding 8.46 GHz). The resulting spectral energy distributions are shown in Figure 5.3. The spectral index for the total minihalo emission is $\alpha = 1.1 \pm 0.06$, with the flatter central region at $\alpha_c = 1.08 \pm 0.06$, and the steep tail region at $\alpha_t = 1.41 \pm 0.09$.

Extrapolating the spectral index of the total minihalo emission found by Giacintucci et al. (2014a), a flux density of $S_p = 9.1 \pm 1.4$ mJy is predicted, in agreement with the actual 10 GHz measurement. The extrapolation for the central region finds an expected flux density of $S_{p,c} = 8.7 \pm 1.1$ mJy, agreeing within certainty to the measured 10 GHz flux density $S_c = 7.60 \pm 0.52$ mJy. The measured emission also closely agrees with the constraints found by Perrott et al. (2021) at 15.5 GHz for the central region (blue shading in Figure 5.3). In contrast, the expected tail flux density $S_{p,t} = 0.34 \pm 0.13$ mJy is found to be three times lower than measured 1.01 ± 0.35 mJy.

The BCG was measured to vary ~20% over one year (between C- and D-configuration runs) and ~10% over one month (subsequent D-configurations). This degree of variation could suggest a systematic calibration error between the D-configuration observations. However, a systematic offset of only 0.7% was observed in the flux density models of the secondary calibrator for these observations. Other point sources in the image were also measured and no systematic offsets were observed. Given the lack of systematic offsets in other sources, the presence of active jets from the BCG, and a slight change in the spectral shape, the observed variability is likely real.

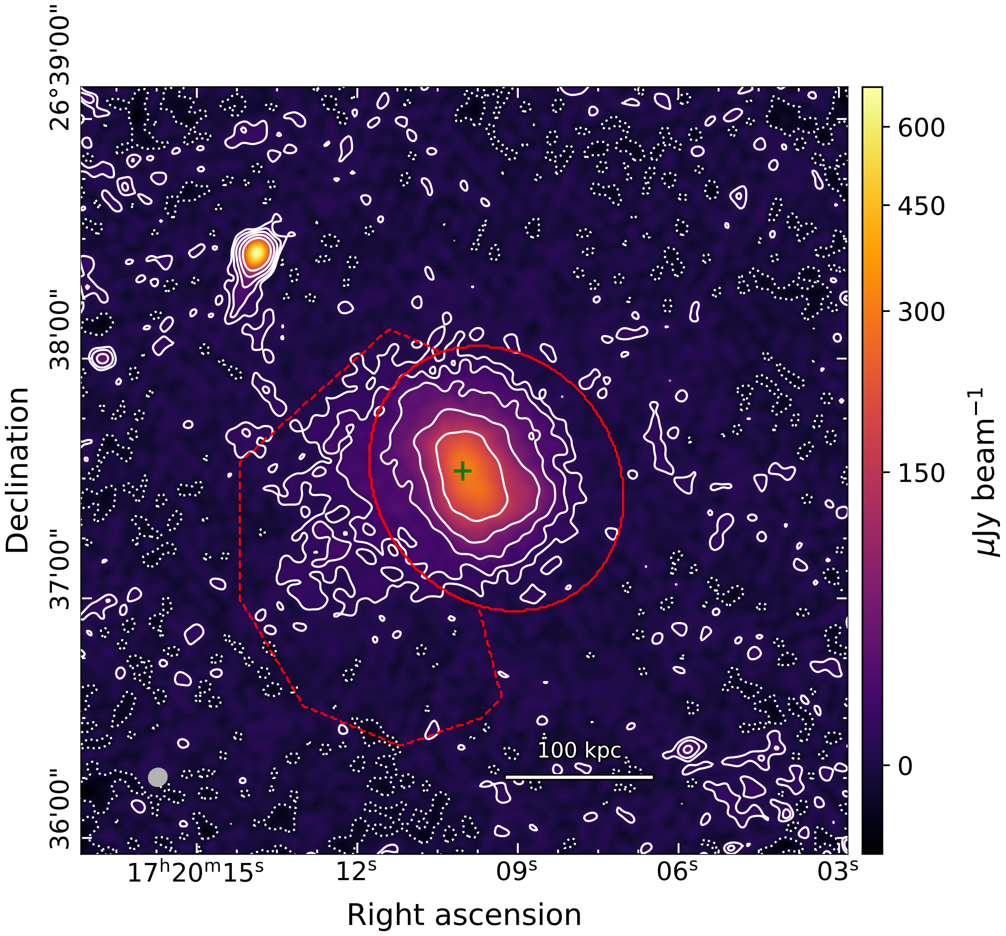


Figure 5.2: RXJ1720+2638. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 4.65”×4.38” at a position angle of -55.7° (shown in grey, bottom left) and an rms noise of 3.38 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The masks used for flux estimation are shown in red, marking the central (solid) and tail (dashed) regions. The green cross marks the location of the subtracted BCG. Note that for primary beam corrected images, the rms is valid at the centre, while the edges have a boosted noise due to the correction. This causes the exaggerated positive and negative contours near the border of the image.

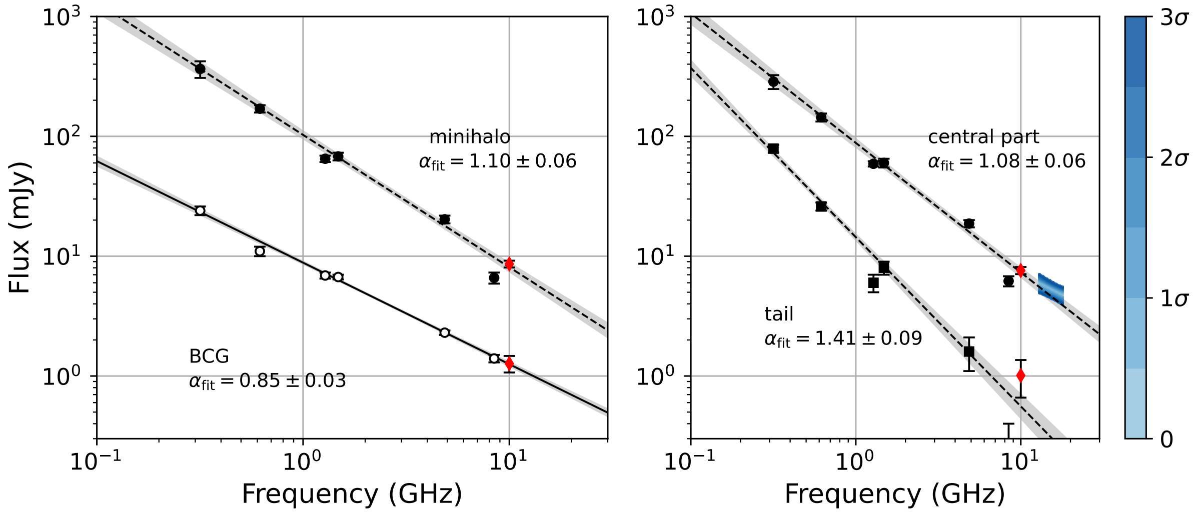


Figure 5.3: RXJ1720+2638 radio spectra between 317 MHz and 10 GHz. *Left Panel:* Spectrum of the total minihalo (filled circles) and the BCG (empty circles). The red diamond marks the 10 GHz measurement from this work. The dashed line represents the best-fitting power-law fit for the minihalo emission, while the solid line indicates the BCG power-law spectrum. Both fits exclude the 8.44 GHz data point. Grey shaded regions indicate the 1σ confidence interval, with fitted spectral indices and uncertainties annotated in the panel. *Right Panel:* Spectra of the central (circles) and tail (squares) minihalo components. Dashed lines show the corresponding power-law fits over same frequency range as the left panel, with grey shading indicating the 1σ confidence intervals. The shaded region in blue exhibits the constraints found from the *uv*-modelling by Perrott et al. (2021). Fitted spectral indices are reported in the panel.

5.2 A478

A478 is a nearby cool core galaxy cluster ($M_{500} = 6.95 \times 10^{14}$, $R_{500} = 1.31$ Mpc (Planck Collaboration et al., 2016)) at a redshift of $z = 0.088$ (Liu et al., 2024). *Chandra* and *X-ray Multi-Mirror Mission* (XMM-Newton) observations revealed a relaxed cluster morphology with a slightly elliptical ICM structure and a central BCG hosting a double-lobed AGN (Sun et al., 2003; Bourdin et al., 2008). The core temperature was measured as $T \approx 1.54$ keV, increasing to $T \approx 7.6$ keV at a radius of 260 kpc (Ng et al., 2024). Two small (~ 10 kpc) X-ray cavities were found to the northeast and southwest and were spatially aligned with the lobes of the BCG, indicating interaction between the BCG and the ICM (Sun et al., 2003). A cold front was detected approximately 60 kpc to the southwest of the cluster core (Markevitch et al., 2003).

In the initial detection of minihalo emission, Giacintucci et al. (2014b) identified several nearby radio sources to the minihalo. At 1.4 GHz, these included the central BCG (S1); a head-tail to the east with a tail to the northwest (S4); a second head-tail embedded in the tail of S4 and extending to the northeast (S2); and an unresolved source (S3) to the north of S2 and S4. They noted that the latter source, S3, was likely a background source due to the absence of an optical counterpart. Giacintucci et al. (2014b) found the minihalo to be slightly more extended to the northeast ($R_{max} \sim 150$ kpc) compared to the other directions ($R_{min} \sim 120$ kpc). This region blended with the tail of S4, resulting in a large uncertainty in the minihalo flux estimate due to difficulty in separating the tail emission from the diffuse minihalo. They found the minihalo emission to be largely confined by the southwest cold front, with a total flux density of 16.6 ± 3 mJy. LOFAR imaging at 144 MHz did not detect the diffuse minihalo emission, placing an upper limit on the spectral index of $\alpha < 1$, implying a flatter spectrum than most observed minihalos (Savini et al., 2019). Using archival GMRT data, Savini et al. (2019) measured S1 and S4 to have a spectral indices of $\alpha \approx 1.1$ and $\alpha \approx 0.6$, respectively, between 144 and 610 MHz.

5.2.1 10 GHz observation

The minihalo detected at 10 GHz is significantly smaller than that observed at 1.4 GHz, measuring approximately 80 kpc north-south and 45 kpc east-west. This resulted in only two contaminating sources, S1 and S4 (see Figure 5.4). The S4 AGN doubled in brightness in the month between the D-configuration observations, causing severe PSF artefacts in the joint image. As such, this source was modelled on a per-dataset basis for the final self-calibration rounds. No comparable variation was observed in other compact sources, suggesting this behaviour is consistent with an AGN flare. The BCG exhibited variability consistent with the estimated calibration uncertainty over all observations. The sources S2 and S3 are detected, but outside the radius of the observed minihalo. Three faint compact sources are located between S2 and the minihalo, but are spatially separate from the minihalo emission at the 3σ level. Point subtraction was therefore limited to

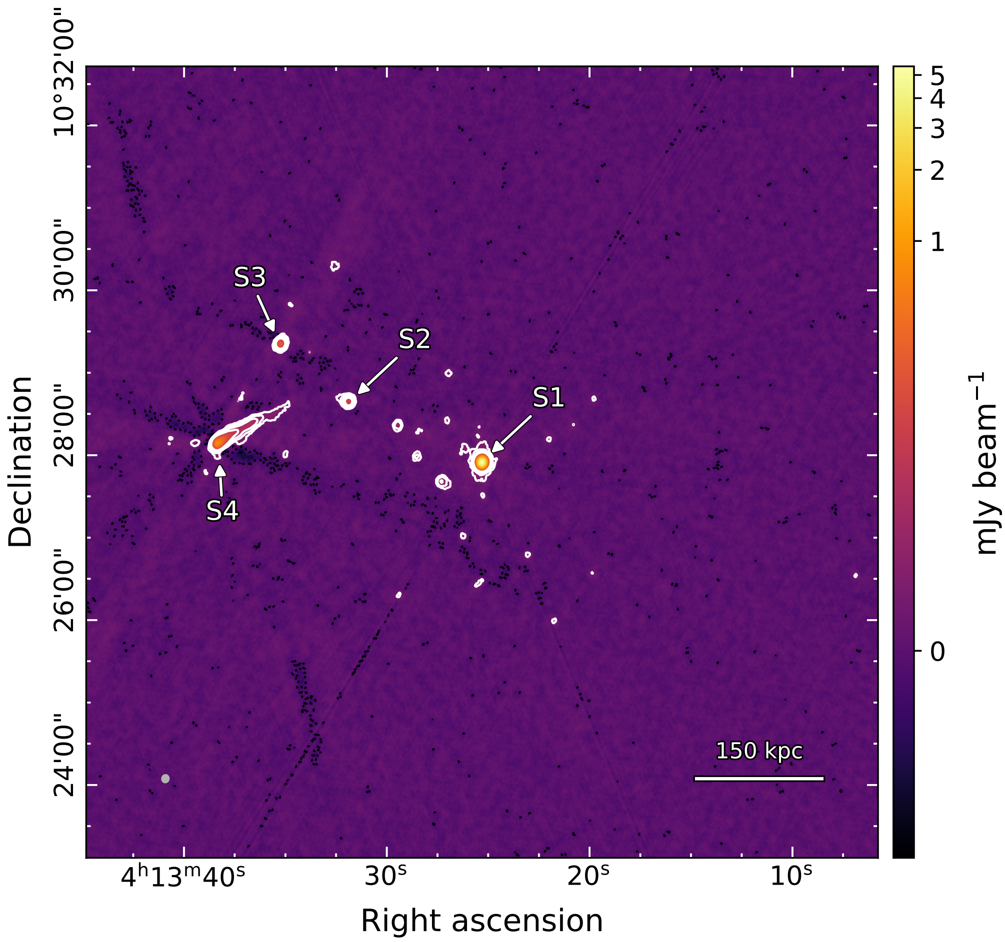


Figure 5.4: A478. Non-subtracted (robust = 0.5) 10 GHz image with a restoring beam of 2.79” × 2.51” at a position angle of -25.6° (shown in grey, bottom left) and an rms noise of 3.48 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

S1 and S4 (including the tail), using the standard 12 kλ uv-cut. This uv-cut resulted in some residual extended emission from S4, however, it was spatially distinct and the artefacts were successfully suppressed.

Following subtraction, the minihalo was modelled with scales $[0, 2, 4, 8, 16]$, a `smallscalebias` of -0.5, and a `robust` of 1. This was followed by primary beam correction. The masking region was not explicitly defined in Giacintucci et al. (2014b), so flux estimation mask was determined using the 3σ region as discussed in the previous chapter.

The total minihalo flux is measured as 0.31 ± 0.03 mJy. Associated compact sources are listed in Table 5.3. Combining this measurement with the 1.4 GHz measurement by Giacintucci et al. (2014b), the resulting spectral index is $\alpha = 2.02 \pm 0.10$. This is exceptionally steep for a minihalo and is inconsistent with the spectral index upper limit reported by Savini et al. (2019). To investigate further, the 8–12 GHz observation was split into four 1 GHz sub-bands. Each sub-band was imaged independently, and the total minihalo flux was computed using the spatial

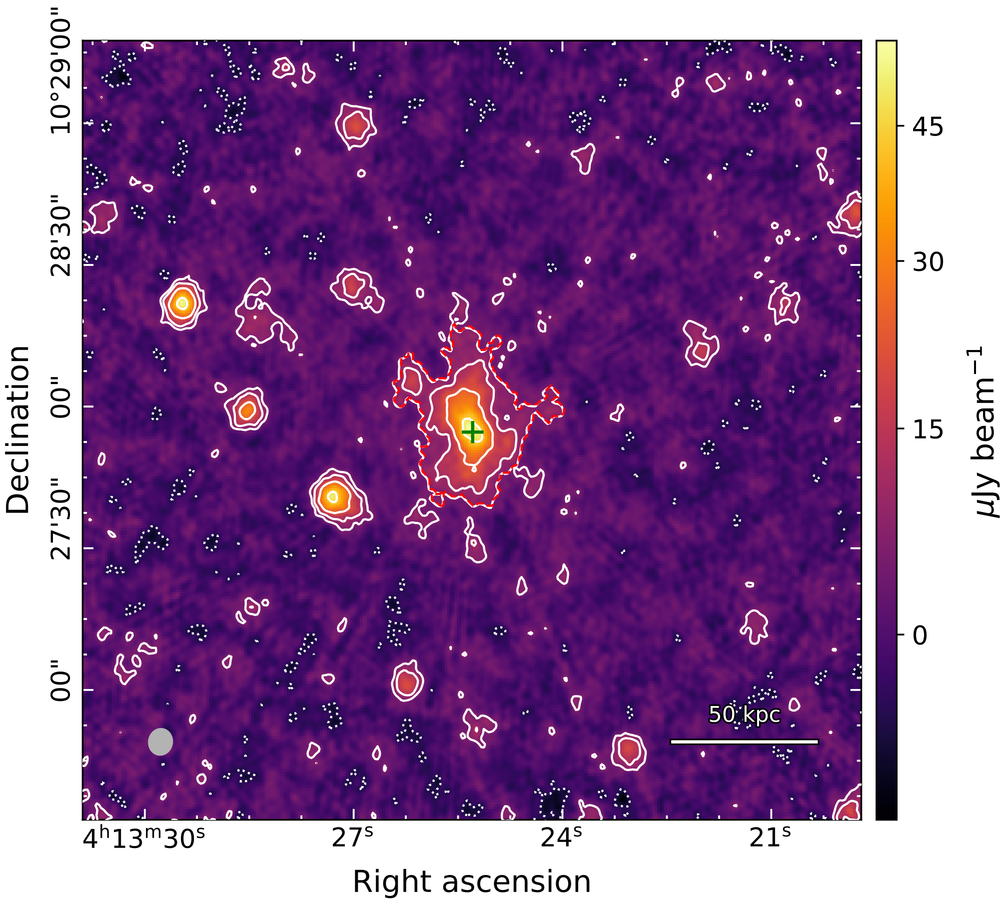


Figure 5.5: A478. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 4.65"×4.38" at a position angle of -55.7° (shown in grey, bottom left) and an rms noise of 3.38 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The mask used for flux estimation is shown by the dashed red line. The green cross marks the location of the subtracted BCG.

mask from the joint image. The resulting flux densities are listed in Table 5.1. Over the 8–12 GHz range the spectral index is even steeper at $\alpha = 3.34 \pm 0.18$. The BCG spectral index is 0.78 ± 0.02, close to the 0.88 ± 0.03 measured over the full 1.4–10 GHz range. The sub-band fits and the total frequency fit are shown in Figure 5.6. It is important to note that there is a significant decrease in short *uv* baselines for the 11–12 GHz sub-band due to the high amount of RFI. This could artificially steepen the spectral index due to the instrumental spatial filtering. A comparison of the *uv* sampling is shown in Figure 5.7. The implications of this unusually steep minihalo spectrum are discussed in the following chapter.

Freq band (GHz)	8–9	9–10	10–11	11–12
Minihalo flux (mJy)	0.52 ± 0.06	0.35 ± 0.03	0.26 ± 0.03	0.17 ± 0.08
BCG flux (mJy)	6.03 ± 0.02	5.53 ± 0.02	5.15 ± 0.02	4.75 ± 0.02

Table 5.1: A478 total minihalo flux and BCG flux measured across 1 GHz sub-bands.

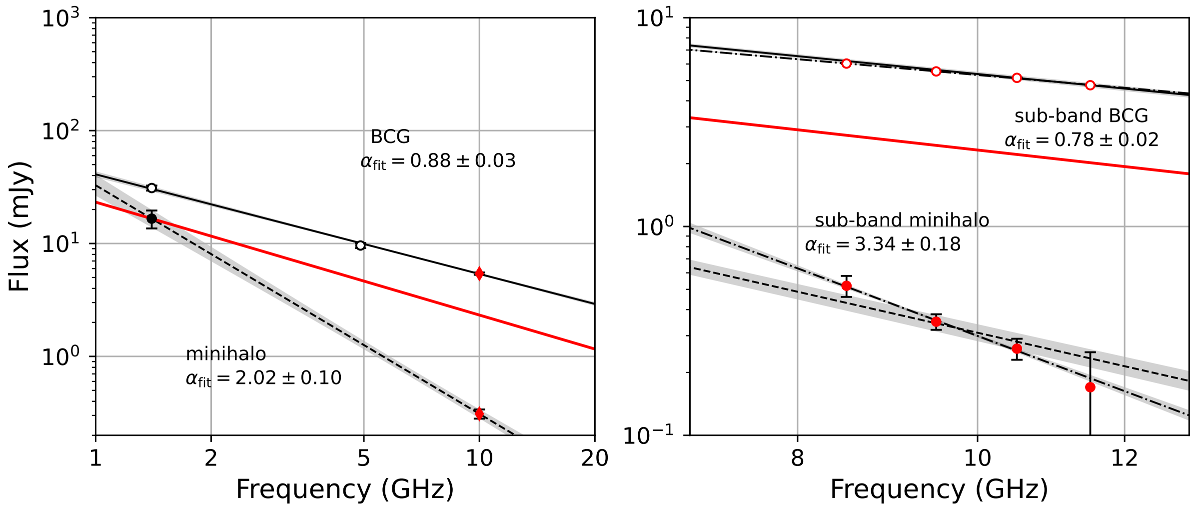


Figure 5.6: A478 radio spectra between 317 MHz and 10 GHz. *Left Panel:* Spectrum of the total minihalo (filled circles) and the BCG (empty circles). The red diamond marks the 10 GHz measurement from this work. The dashed line represents the best-fitting power-law fit for the minihalo emission, while the solid line indicates the BCG power-law spectrum. The red solid line represents the $\alpha = 1$ upper limit found by Savini et al. (2019). Grey shaded regions indicate the 1σ confidence interval, with fitted spectral indices and uncertainties annotated in the panel. *Right Panel:* Sub-band spectra for the total minihalo (filled red circles) and the BCG (empty red circles) between 8–12 GHz. The dashed and solid black lines, along with the red solid line, denote the spectra from the left panel. The dash-dotted lines represent the best-fitting power-law for the sub-band spectrum for the minihalo and BCG emission, with the spectral indices annotated in panel.

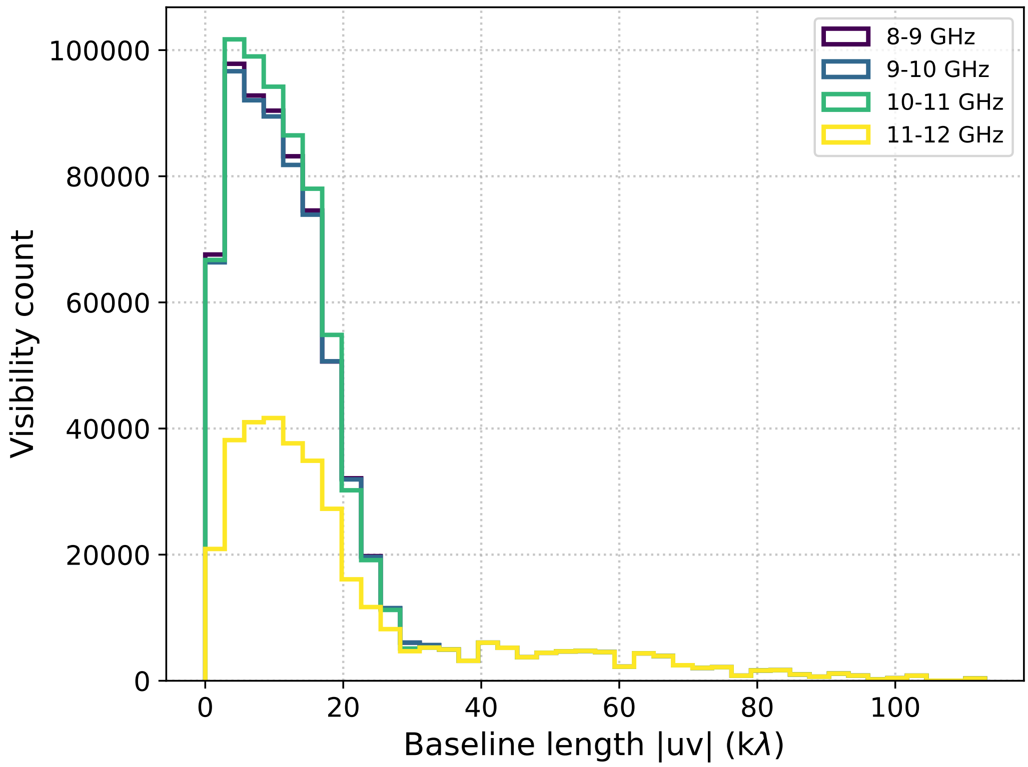


Figure 5.7: Baseline-length distributions for the 1 GHz sub-bands derived from the A478 observations.

5.3 RXCJ1115+0129

RXCJ1115+0129 (also known as MACS J1115.8+0129) is a large cool core cluster (M_{500} = 6.86×10^{14} $M_\odot$, R_{500} = 1.19 Mpc (Planck Collaboration et al., 2016)) at $z = 0.340$ (Wen et al., 2015) that has recently been confirmed to host a minihalo (Trehaeven et al., 2023). This cluster exhibits a close spatial alignment between the X-ray peak and the central BCG, and relatively regular elliptical morphology (Ebeling et al., 2010; Pandey-Pommier et al., 2016). Giovannini et al. (2020) reported a 7 mJy discrepancy between the flux density measured in the *Faint Images of the Radio Sky at Twenty-Centimeters* (FIRST) survey[2] and a deeper 1.4 GHz VLA observation. They suggested that the additional flux recovered in the pointed VLA data, which is more sensitive to the extended emission, corresponds to the brightest region of diffuse minihalo emission surrounding the central source. Trehaeven et al. (2023) subsequently confirmed the presence of the minihalo using MeerKAT archival data at 1.28 GHz. They found that the minihalo is orientated towards the northwest and southeast, aligned with the radio lobes of the central BCG. The minihalo extends 412 kpc along its longest axis, with an average diameter of 372 kpc. The minihalo flux density was measured as $S_{1.28\,\mathrm{GHz}} = 7.91 \pm 2.59$ mJy, consistent with the discrepancy noted by Giovannini et al. (2020). Using a 1–1.5 GHz sub-band comparison, they derived a spectral index of $\alpha = 1.0 \pm 1.1$. The large uncertainty is due to the subtraction uncertainty of the strong BCG within the minihalo. They also measured a BCG flux density of $S_{1.28\,\mathrm{GHz}} = 8.73 \pm 0.88$ mJy and spectral index $\alpha = 0.78 \pm 0.26$. They noted an unusual spatial variation in the spectral index across the minihalo, with significantly flatter values ($\alpha \sim 0.5$) to the east and west, compared to steeper values ($\alpha \sim 1.5$) toward the north and south.

5.3.1 10 GHz observation

Initial examination of the 10 GHz image revealed four bright sources, including the BCG (S1). The remaining sources (S2-S4) are compact and located outside an 800 kpc radius of the central region (see the `robust` = 0.5 image in Figure 5.8). The BCG increased in flux density by ∼25% between the C- and D-configuration observations. As joint imaging was not significantly affected by this variation, no separate flux modelling was applied during self-calibration. In addition to S1, sources S2 and S3 were subtracted to suppress PSF artefacts resulting from their variability. After subtraction, the minihalo was imaged with `robust` = 1, scales [0, 4, 8], followed by primary beam correction. Masking for flux estimation was performed within the 3σ contour.

The total minihalo flux is measured as 0.97 ± 0.15 mJy. As with Trehaeven et al. (2023), the uncertainty is dominated by the subtraction contribution $\sigma_{sub} = 0.14$ mJy. The BCG has a measured flux density of 2.27 ± 0.21 mJy. Combining these measurements with those of Trehaeven et al. (2023), the derived spectral index for the minihalo emission is $\alpha = 1.03 \pm 0.18$,

[2] A 20 centimetre wavelength corresponds to a 1.4 GHz frequency.

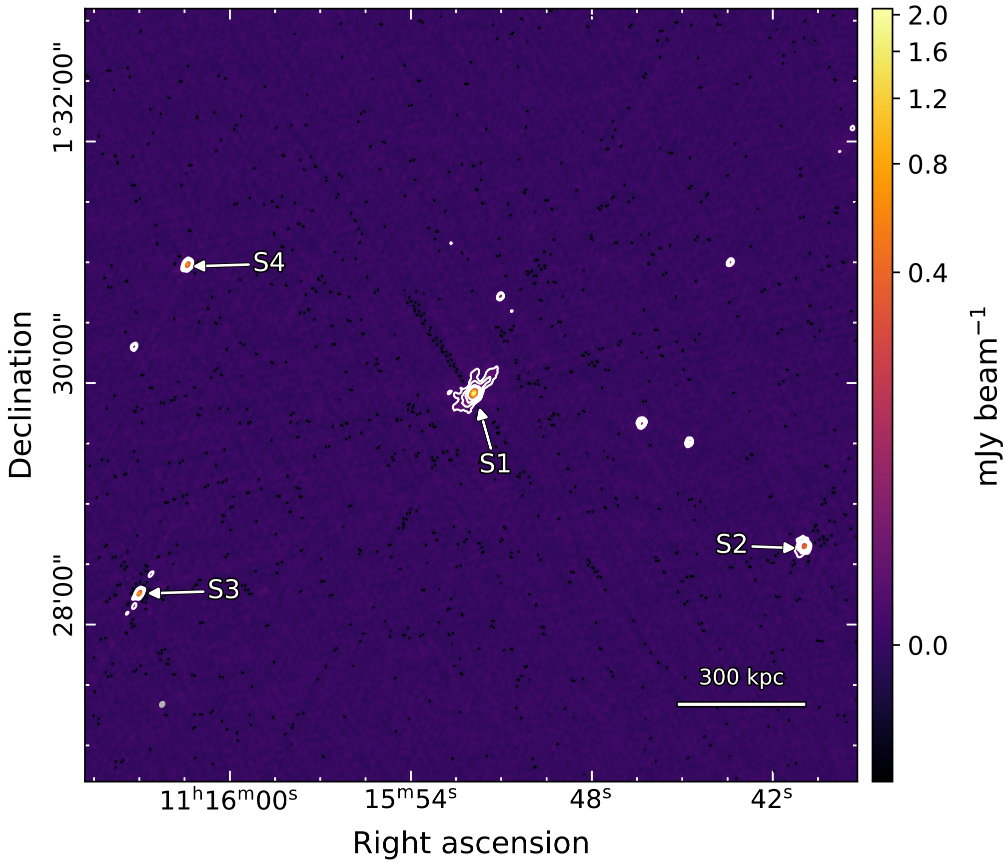


Figure 5.8: RXCJ1115+0129. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.89” × 2.21” at a position angle of -31.9° (shown in grey, bottom left) and an rms noise of 2.66 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

while the BCG spectral index is 0.66 ± 0.07. These findings are consistent with Trehaeven et al. (2023), and offer a lower uncertainty by a factor of 6 over the broader frequency range. Figure 5.9 displays the `robust` = 1 minihalo image, which reveals two mirrored decreases of flux in the region of S1. These features are likely due to imperfect source subtraction, however, their contribution is incorporated into the subtraction uncertainty.

The spatial extent of the minihalo is ~ 160 kpc along the northwest-southeast axis, and ~ 100 kpc on the perpendicular axis. This is orientated similarly to that found by Trehaeven et al. (2023), though narrower in the perpendicular. This may reflect the steepening observed towards the north and south tapering the emission at these higher frequencies.

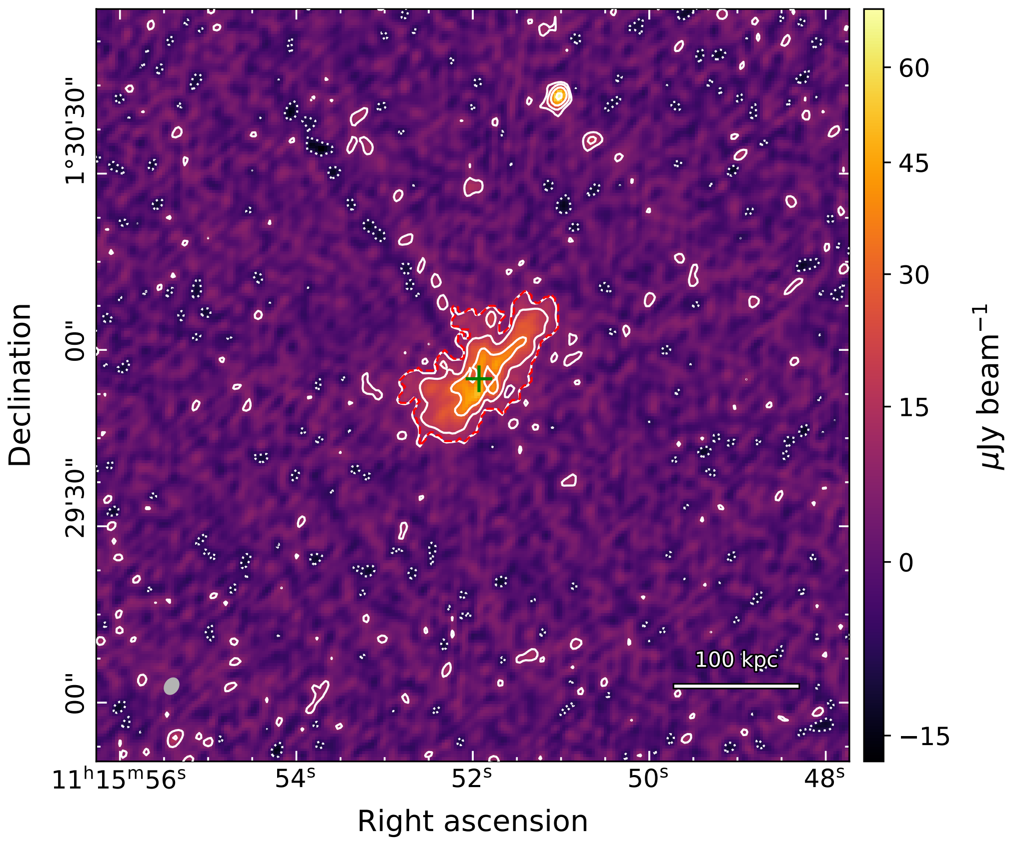


Figure 5.9: RXCJ1115+0129. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 2.89”×2.20” at a position angle of -31.9° (shown in grey, bottom left) and an rms noise of 2.63 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The mask used for flux estimation is shown by the dashed red line. The green cross marks the location of the subtracted BCG.

5.4 RXJ2129+0005

RXJ2129+0005 is a relaxed cool core cluster ($M_{500} = 4.33 \times 10^{14}$ M$_{\odot}$, $R_{500} = 1.06$ Mpc (Planck Collaboration et al., 2016)) at redshift $z = 0.234$ (Sanders et al., 2011). *Chandra* X-ray observations measured a central core temperature of $T = 4.5 \pm 0.5$ keV, rising to $T = 7.4 \pm 1.2$ keV at a radius of ~ 200 kpc (Cavagnolo et al., 2009). The X-ray emission was elongated along the northeast-southwest direction, with a cold front identified ~ 65 kpc to the southwest (Jiménez-Teja et al., 2024). Weaker cold fronts were evidenced to the southeast and northwest, consistent with widespread gas sloshing in the core.

The BCG is closely aligned with the cluster centre. GMRT observations at 235 and 610 MHz measured flux densities of 87 ± 5 mJy and 48.9 ± 2.5, respectively (Kale et al., 2015). VLA observations at 8.46 GHz revealed a compact core and a small (~ 4 kpc) jet extending southwards, with some evidence of a much fainter counter-jet to the north (Giacintucci et al., 2019). Across this frequency range, the BCG has a spectral index of $\alpha = 0.77 \pm 0.02$.

Kale et al. (2015) also reported the discovery of a minihalo surrounding the central BCG using the aforementioned GMRT observations at 235 and 610 MHz. The minihalo extends 225 kpc along the east-west axis and 110 kpc along the north-south axis. They measured total minihalo flux densities of $S_{235\,\mathrm{MHz}} = 21 \pm 1.6$ mJy and $S_{610\,\mathrm{MHz}} = 8 \pm 0.7$ mJy. An additional estimate at 1.4 GHz of 2.4 mJy was obtained by subtracting the compact source flux measured by FIRST from that measured in the *NRAO VLA Sky Survey* (NVSS). Giacintucci et al. (2019) detected the minihalo in a GMRT observation at 1.3 GHz, measuring the total emission as 2.4 ± 0.2 mJy, comparable with the previous estimate. They derived a spectral index of $\alpha = 1.2 \pm 0.1$ for the minihalo between 235 MHz and 1.4 GHz.

5.4.1 10 GHz observation

Self-calibration was successful for three of the four datasets, with phase solutions for one C-configuration observation failing to converge. The phase solutions were inconsistent across the frequency band, varying between -50° and 50°. This observation (ID: `24A-411.sb45228321.eb45252099.60350.80962270833`) also included two other science targets, A2626 and ACTJ0022-0036, which did not exhibit similar phase calibration failures. As such, this observations was discarded for the images presented here.

Two bright sources are present in the central region: the BCG (S1) and a second radio galaxy to the south (S2). Source S2 is not explicitly mentioned by Giacintucci et al., 2019, potentially indicating a non-detection as matching the coordinates reveals no explicit source in their image. The BCG varied by $\sim 20\%$ between the remaining C- and D-configuration observations, producing radial PSF artefacts in the `robust` = 0.5 image (Figure 5.10). Only S1 was subtracted, as S2 was not modelled in the 12 kλ uv-cut image. Following subtraction, the minihalo was imaged with `robust` = 1 and multiscales of $[0, 3, 8, 16, 32]$, before primary beam correction. The mask used for the calculation of the flux density was estimated based on that shown by (Giacintucci et al., 2019) (see Figure 3a of paper). To account for the contribution of the unsubtracted source S2, the region within 5” was excluded from the mask. The resulting image and mask is shown in Figure 5.11.

The total minihalo and BCG flux densities are measured as 0.41 ± 0.10 mJy and 5.36 ± 0.25 mJy, respectively. The uncertainty in the measured minihalo flux is dominated by subtraction uncertainty, $\sigma_{sub} = 0.099$ mJy. Combining these measurements with previous results, the spectral indices of the minihalo and BCG were calculated as $\alpha = 1.18 \pm 0.09$ and $\alpha = 0.78 \pm 0.02$, respectively. The spectra are shown in Figure 5.12. Extrapolating from the previously measured minihalo spectral index yields an expected flux density of 0.21 ± 0.05. This is slightly inconsistent with the measured value at the $\approx 2\sigma$ level. The derived BCG spectral index is consistent with the previous measurements.

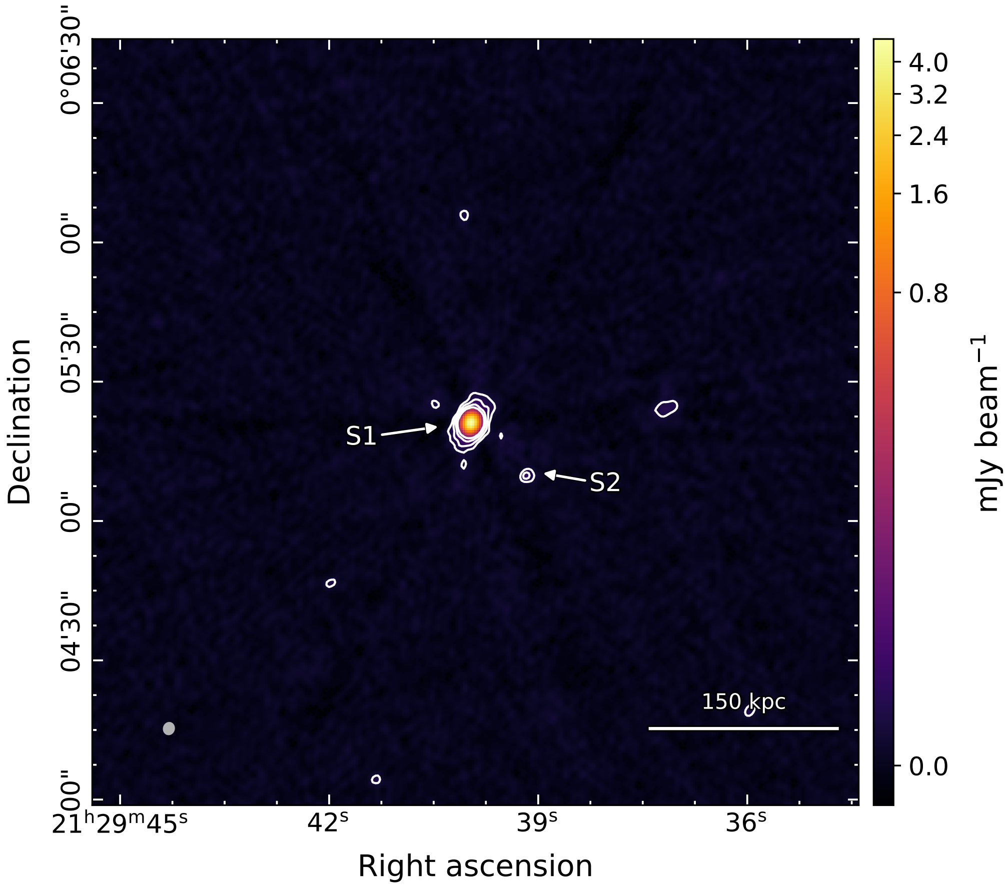


Figure 5.10: RXJ2129+0005. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.53"×2.23" at a position angle of -15.4° and an rms noise of 2.24 μJy beam^{-1} (shown in grey, bottom left). White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

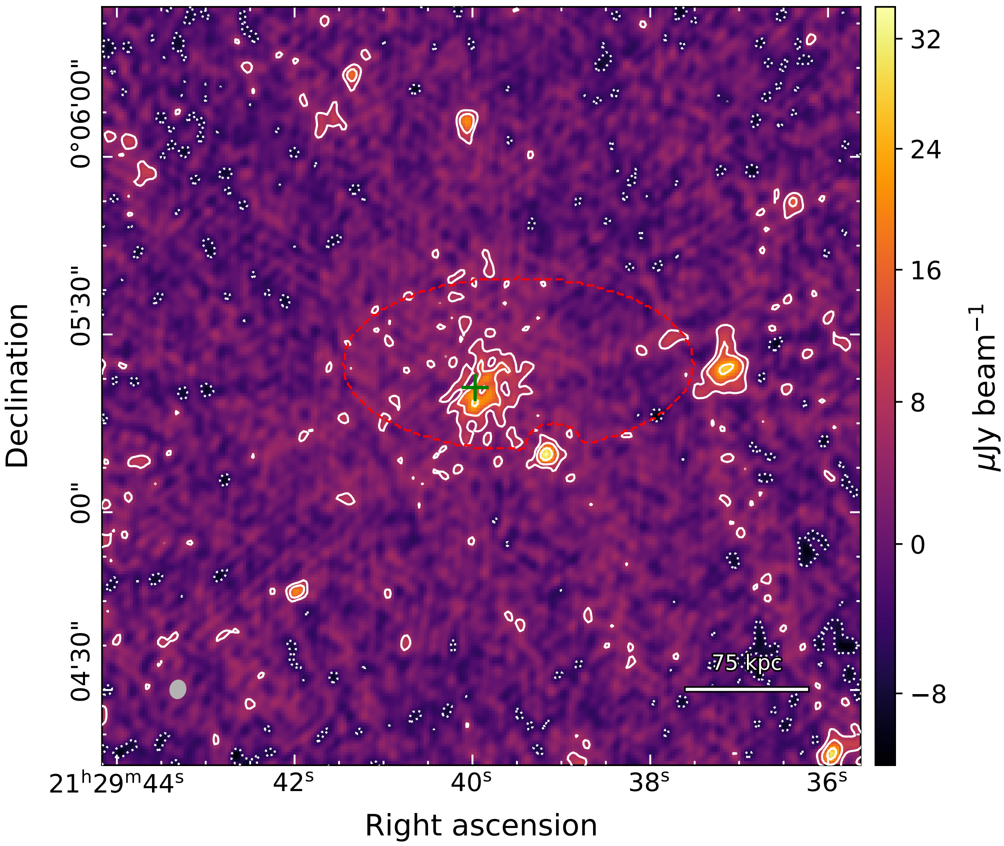


Figure 5.11: RXJ2129+0005. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 3.67”×2.79” at a position angle of -32.4° (shown in grey, bottom left) and an rms noise of 2.03 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The mask used for flux estimation is shown by the dashed red line. The green cross marks the location of the subtracted sources.

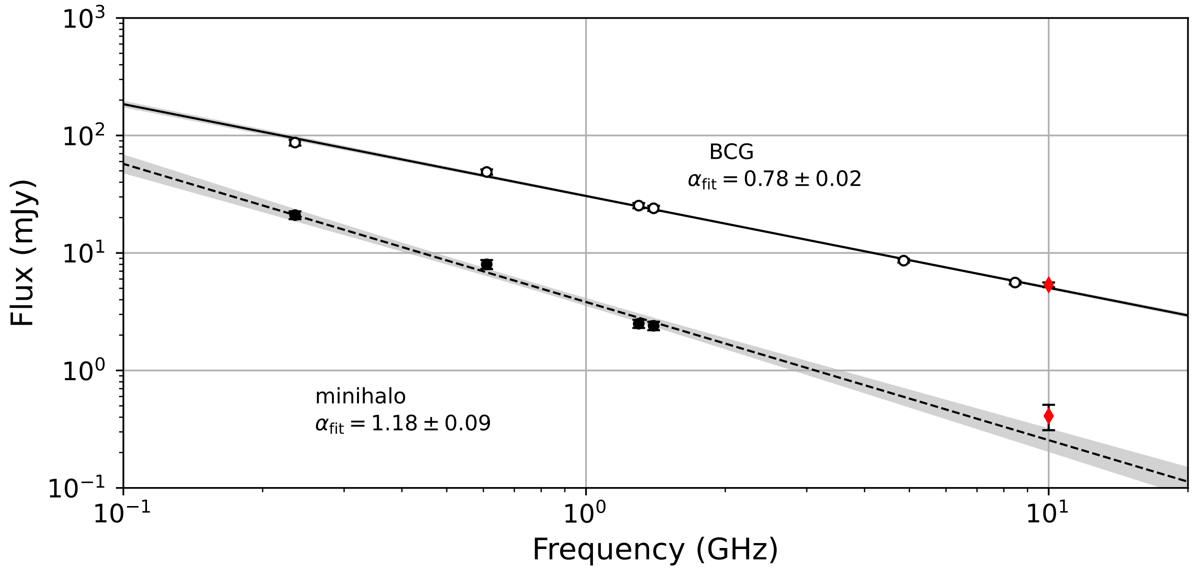


Figure 5.12: RXJ2129 radio spectra between 235 MHz and 10 GHz. Spectrum of the total minihalo (filled circles) and the BCG (empty circles). The red diamond marks the 10 GHz measurement from this work. The dashed line represents the best-fitting power-law fit for the minihalo emission, while the solid line indicates the BCG power-law spectrum. Grey shaded regions indicate the 1σ confidence interval, with fitted spectral indices and uncertainties annotated in the panel.

5.5 2A0335+096

2A0335+096 is a nearby, low-mass cluster ($M_{500} = 2.27 \times 10^{14}$ M$_\odot$, $R_{500} = 0.92$ Mpc (Planck Collaboration et al., 2016))[3] at $z = 0.036$ (Sanders et al., 2011), the lowest redshift of this sample. *Chandra* X-ray observations revealed a cool core with an average temperature of $T \leq 3.6$ keV within R$_{2500}$ (Cavagnolo et al., 2009; Giacintucci et al., 2017). Two X-ray cavities were detected to the northwest and southeast of the central BCG, alongside several bright X-ray structures within the central 20 kpc (Mazzotta et al., 2003). A cold front was observed approximately 40 kpc south of the cluster centre. These features are indicative of gas sloshing in the core, while the outer regions ($r > 40$ kpc) exhibited a symmetric, slightly elliptical X-ray surface brightness distribution (Mazzotta et al., 2003).

Early VLA radio observations at 1.5 GHz discovered two fossil radio lobes around the BCG (S1) and a separate relic lobe located ~ 18 kpc northwest of the core (Sarazin et al., 1995). The fossil lobes are the remnants of past AGN outbursts from the central BCG, featuring an aged spectrum. 144 MHz LOFAR observations confirmed that the fossil lobes are spatially aligned with the X-ray cavities (Bîrzan et al., 2020; Ignesti et al., 2022). The relic lobe was interpreted as aged plasma from an even older AGN outburst (Giacintucci et al., 2019). The central BCG has a nearby (~ 5 kpc) compact source, connected with a tail of emission, that was identified as a companion galaxy (S2) (Sarazin et al., 1995; Giacintucci et al., 2019). The BCG exhibited two active lobes aligned northeast and southwest, rotated clockwise with respect to the fossil lobes. Giacintucci et al. (2019) measured the flux density of the BCG at 1.4 GHz as $S_{BCG,1.4\,\mathrm{GHz}} = 16 \pm 1$ mJy, and the companion galaxy as $S_{S2,1.4\,\mathrm{GHz}} = 1 \pm 0.1$ mJy. The fossil lobes were also measured: the southeast lobe with flux density 4.1 ± 0.2 mJy, the northwest lobe 2.7 ± 0.1 mJy, and the outer northwest lobe at 3.1 ± 0.2 mJy. Ignesti et al. (2022) detected the fossil lobes at 144 MHz and in 610 MHz GMRT observations, and reported an estimated spectral index of $\alpha \approx 1.5 \pm 0.1$.

A head-tail galaxy is located ~ 330 kpc to the northwest, with a long tail extending 1 Mpc to the south and passing to the west of the minihalo (Ignesti et al., 2022). As with minihalos, this spatial extent exceeds that expected from spatial diffusion over the CRe lifetime. Ignesti et al. (2022) suggested that interaction between the tail and the ICM may generate localised turbulence, which re-energises the tail CRe in a process termed ‘gentle reacceleration’.

Sarazin et al. (1995) also detected larger diffuse emission corresponding to the minihalo, which extends over 70 kpc. They measured a flux of 22.6 mJy, though they did not state an uncertainty. Giacintucci et al. (2019) subtracted the contributions from S1, S2, and the fossil lobes, and measured a total minihalo flux of 21.1 ± 2.1 mJy at 1.4 GHz. Ignesti et al. (2022) measured the minihalo emission at 144 MHz to be 387 ± 53.9 mJy, which they combined with

[3]Retrieved as PSZ2 G176.27-35.04, which has a slight offset to the default coordinates of 2A0335+096. The value for R$_{500}$ was inferred from M$_{500}$.

the previous measurement to estimate a spectral index of $\alpha = 1.2 \pm 0.1$. They also analysed a 610 MHz GMRT observation, but were unable to estimate the minihalo flux due to insufficient sensitivity of the data. As the minihalo exhibits a flatter spectrum than the fossil lobes, this implies it originates from comparatively younger or more energetic CRe. Ignesti et al. (2022) suggested that the CRe forming the minihalo are not injected from the lobes and subsequently diffused spatially, and must instead be reaccelerated. Their results supported the hadronic and turbulent reacceleration scenarios, with no clear discriminating evidence.

5.5.1 10 GHz observation

During self-calibration, several bright sources were located in the central region. These include the BCG, companion galaxy, and the two active lobes of the BCG. The self-calibration of the core was challenged by the presence of significantly negative model points ($\gtrsim -50$ mJy) in the vicinity of the BCG. Separate imaging of the individual datasets revealed that this issue was localised to the C-configuration data. Robust values were varied over $[-1, -0.5, 0.5, 1]$ in an attempt to minimise this occurrence. However, a fully satisfactory self-calibration was not achieved. The resulting image with `robust` = 0.5 is shown in Figure 5.13. The point S3 may correspond to a peak of the southeast fossil lobe, though these are not clearly visible in the image. To further investigate the diffuse emission of the minihalo and fossil lobes, the active lobes, S1, and S2 were subtracted using models derived from a 12 kλ uv-cut image.

Following this subtraction, the minihalo was re-imaged with `robust` = 1 and multiscales of $[0, 4, 8, 16, 32, 64]$, followed by primary beam correction. The resulting image is displayed in Figure 5.14. The diffuse emission of the southeast fossil lobe is difficult to disentangle from the minihalo emission in this region. The peak S3 is likely associated with this lobe, based on comparisons to the subtracted image of Giacintucci et al. (2019). The northwest fossil lobe is not reliably detected at the 3σ level. The measured flux densities for the BCG and S2 are 3.01 ± 0.18 mJy and 0.27 ± 0.07 mJy, respectively. The shape of the BCG lobes is still observable in the subtracted image, evidencing that the subtraction is imperfect. Due to this, and the residual fossil lobe emission, no reliable measurement of the minihalo flux was performed.

The expected fossil lobe flux can be estimated by extrapolating the 1.4 GHz flux values measured by Giacintucci et al. (2019) using the spectral index $\alpha \approx 1.5 \pm 0.1$ reported by Ignesti et al. (2022). These predicted fluxes are 0.21 ± 0.04 mJy for the southeast lobe, 0.14 ± 0.03 mJy for the northwest lobe, and 0.16 ± 0.03 mJy for the relic lobe. As there is evidence for the detection of the fossil lobes, the regions specified by Giacintucci et al. (2019) were used to estimate the combined lobe and minihalo flux at 10 GHz. These are found as 0.23 ± 0.02 mJy for the southeast region, and 0.13 ± 0.01 mJy for the northwest region. Note that there is minimal subtraction uncertainty for these regions, although Figure 5.14 shows potential contamination from a positive PSF artefact within the northwest region. These estimates are close to the

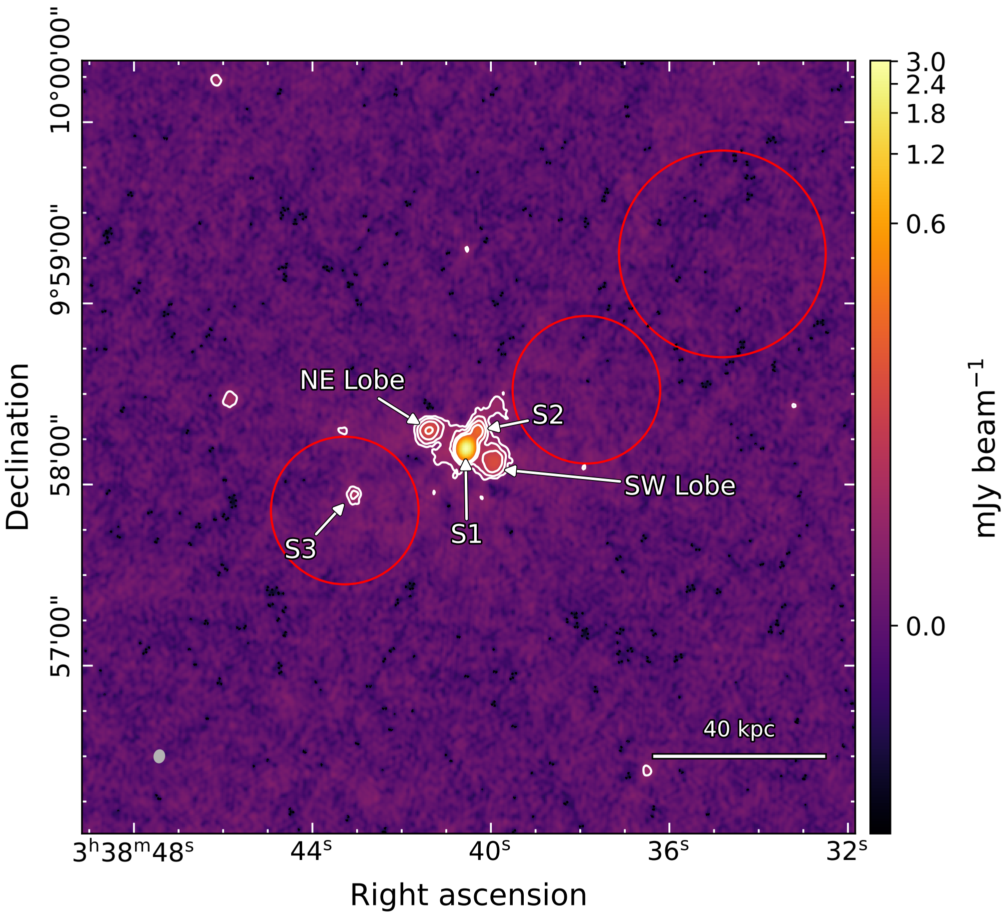


Figure 5.13: 2A0335+096. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 4.14" × 3.32" at a position angle of -9.9° (shown in grey, bottom left) and an rms noise of 2.52 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The +3σ contour has been omitted for visual clarity in highlighting the compact sources. The BCG lobes and sources are annotated. The red circles correspond to the locations of the fossil lobes, with the larger northwest circle being the relic lobe, estimated from Giacintucci et al. (2019).

predicted measurements, illustrating the difficulty in disentangling the minihalo and fossil lobe emission over these regions.

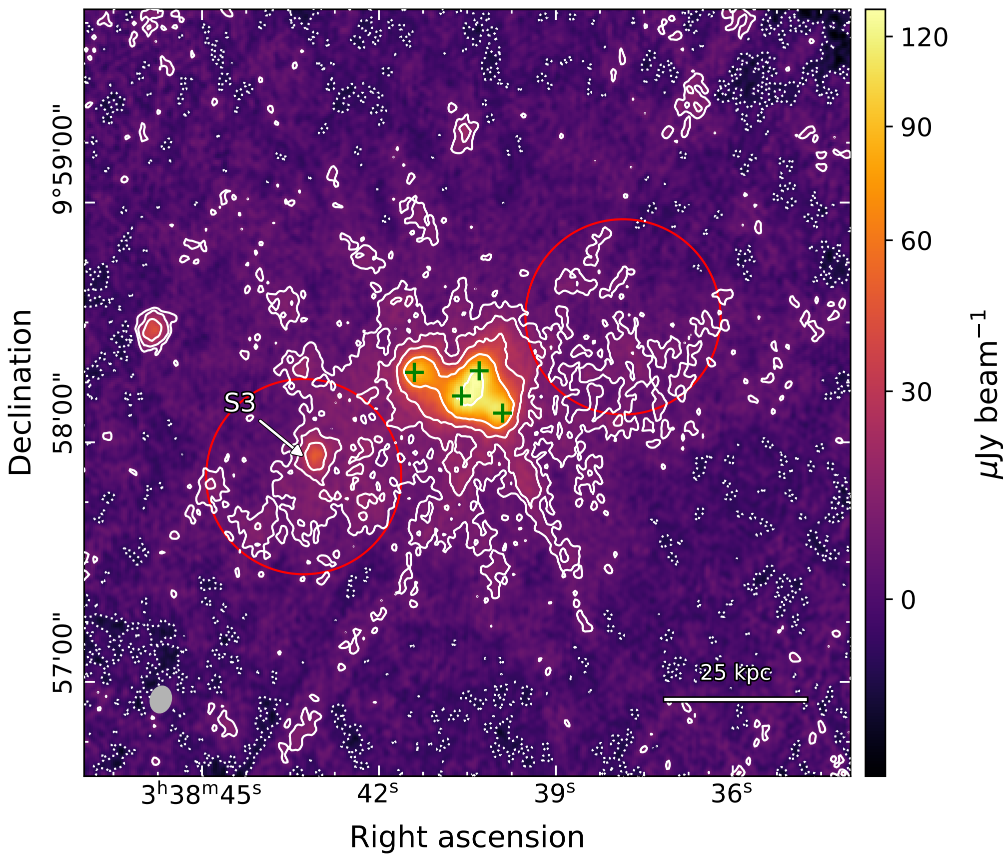


Figure 5.14: 2A0335+096. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 6.68"×5.14" at a position angle of -14.2° (shown in grey, bottom left) and an rms noise of 2.43 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The green cross marks the locations of the subtracted BCG+lobes and S2. The red circles correspond to the locations of the fossil lobes, estimated from Giacintucci et al. (2019).

5.6 A2204

A2204 is a low-mass cool core cluster ($M_{500} = 1.84 \times 10^{14}$ M$_\odot$, $R_{500} = 0.82$ Mpc (Planck Collaboration et al., 2016)) at redshift $z = 0.152$ (Sanders et al., 2011). Detailed *Chandra* X-ray observations revealed a complex core morphology, characterised by a prominent clockwise spiral pattern (Sanders et al., 2005). The central temperature was measured as $T \approx 3.7$ keV, rising to $T \approx 8$ keV at a radius of 100 kpc. Within this region, two cold fronts were identified to the east ($\sim$28 kpc) and west ($\sim$55 kpc), delineating the spiral structure. This is similar to the simulations of ZuHone et al. (2013) and the morphology of RXJ1720+2638. Numerous $\sim$4 kpc depressions in the X-ray surface brightness of the inner core ($r \leq 20$ kpc) may correspond to cavities. However, a lack of spatially aligned radio emission meant that these could not be confirmed as radio bubbles (Sanders et al., 2009).

The central BCG consists of a compact nucleus (S1) and a companion galaxy (S2) located $\sim$25 kpc to the north (Sanders et al., 2005; Sanders et al., 2009). VLA observations at 8.4 and

1.4 GHz revealed weak components north and south of the BCG, potentially corresponding to AGN lobes. Giacintucci et al. (2014b) reprocessed the 1.4 GHz observations and obtained flux densities of 58.9 ± 2.9 mJy and 1.6 ± 0.1 mJy for S1 and S2, respectively. After subtracting these components, they identified a minihalo with a radius of 50 kpc and a total flux density of 8.6 ± 0.9 mJy. The spatial extent matches that of the cold fronts, suggesting that the emission is confined by the sloshing fronts.

5.6.1 10 GHz observation

Three sources were identified during self-calibration: the BCG (S1), its companion galaxy (S2), and a northwest compact source (S3). Imaging with `robust` = 0.5 reveals a northward protrusion from the BCG, with no corresponding southern component at this frequency (Figure 5.15). Source S2 is blended with the emission of S1 at the angular resolution of the D-configuration observations. To minimise contribution from S2 for flux estimation, a fitting radius of 4" was adopted for the Gaussian modelling. This resulted in $\sim 20\%$ variability between the C- and D-configuration measurements, likely reflecting intrinsic variability. Phase calibration was performed using a joint model to aid in disentangling the emission from S1 and S2 in the D-configuration datasets. Sources S1 and S2 were subtracted from each dataset using their respective sky models from a 12 kλ uv-cut image.

Following this subtraction, the minihalo was imaged with `robust` = 1 and multiscales $[0, 2, 4, 8, 16]$, before primary beam correction. The resulting image features significant negative radial residuals, particularly in the northeast-southwest axis, indicating imperfect subtraction of S1 (Figure 5.16). An attenuation of the minihalo emission is evident along the northwest-southeast axis, forming a 'butterfly-like' pattern. Independent modelling of each subtracted dataset showed that these artefacts originate from the C-configuration data. Several attempts were made to improve the subtraction for this dataset by varying the modelling parameters of the uv-cut image, however, no satisfactory solution was achieved. As these artefacts affect emission beyond the region of the subtraction, a reliable uncertainty estimate cannot be obtained and, therefore, no estimate of the minihalo flux is reported.

The BCG flux was measured as 11.1 ± 1.0 mJy, which combined with the previous 1.4 GHz measurement, yields a spectral index of $\alpha = 0.85 \pm 0.05$. The flux of S2, from the resolved C-configuration data only, was 0.15 ± 0.02 mJy. Combined with the 1.4 GHz measurement, this provides a spectral index of $\alpha = 1.2 \pm 0.1$, indicative of a steep-spectrum radio galaxy.

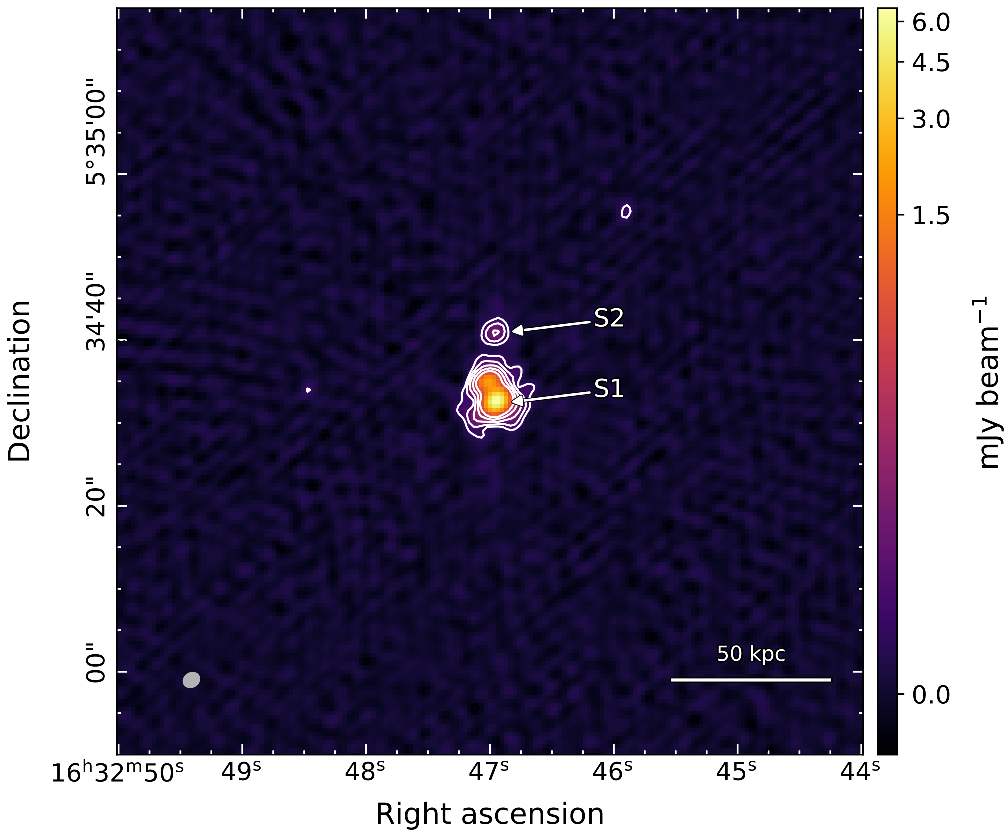


Figure 5.15: A2204. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.02" × 1.69" at a position angle of -58.0° (shown in grey, bottom left) and an rms noise of 6.04 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

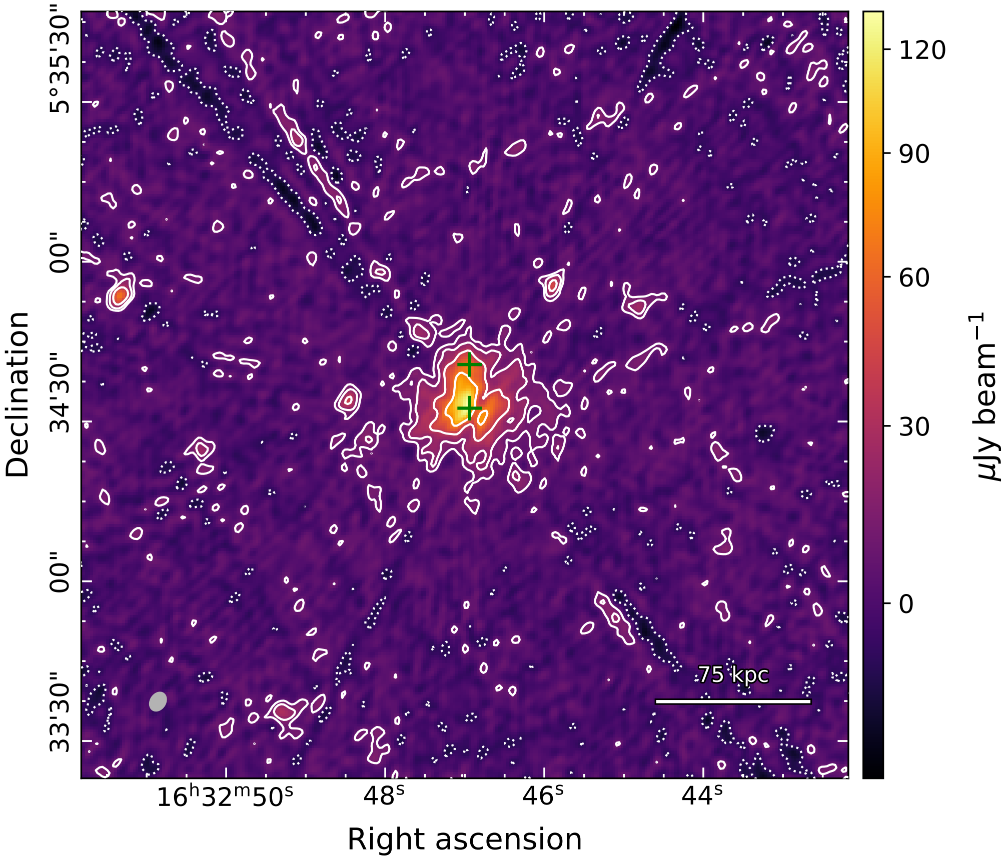


Figure 5.16: A2204. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 3.67”$\times$2.79” at a position angle of -32.4° (shown in grey, bottom left) and an rms noise of 2.99 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The green cross marks the locations of the subtracted sources.

5.7 MS1455+2232

MS1455+2232 (also referred to as Z7160 and ZwCl 1454.8+2233) is a low-mass cluster ($M_{500} = 3.5 \times 10^{14}$ M$_\odot$, $R_{500} = 1.02$ Mpc (Giacintucci et al., 2017))[4] at redshift 0.258 (Wen et al., 2015). The central temperature was measured as 4 ± 0.5 keV, rising to 5.5 ± 0.5 keV at a radius of $\sim$200 kpc (Mazzotta et al., 2001a). Two cold fronts have been identified, one $\sim$140 kpc to the north and the other $\sim$40 kpc to the south of the centre (Mazzotta et al., 2008). These features form a clear spiral structure, indicative of gas sloshing (Mazzotta et al., 2008). Riseley et al. (2022) performed a detailed analysis of this cluster using *Chandra* X-ray, 144 MHz LOFAR, and 1300 MHz MeerKAT observations. They found that the sloshing spiral is slightly asymmetrical and elongated towards the southwest, covering a region 254×232 kpc. Given similarities to RXJ1720+2638 and numerical simulations, they suggested that this sloshing may be the result of a minor merger, though no studies have specifically searched for substructure in the galaxy distribution.

The minihalo was first reported by Venturi et al. (2008) in 610 MHz GMRT observations, surrounding a central source. They measured the longest extent as 360 kpc, with a total flux density of 43.6 ± 2.2 mJy. VLA observations at 1.4 GHz only partially resolved the minihalo, with a maximum extent of $\sim$230 kpc and total flux density of 8.5 ± 1.1 mJy (Giacintucci et al., 2019). The sensitive LOFAR and MeerKAT imaging by Riseley et al. (2022) found a largest linear size of 586 kpc at 1300 MHz. They measured the total flux densities as $S_{145\,\mathrm{MHz}} = 154.5 \pm 15.6$ mJy and $S_{1300\,\mathrm{MHz}} = 18.7 \pm 0.9$ mJy, yielding a spectral index of the minihalo as $\alpha = 1.07 \pm 0.1$.

The size of the minihalo notably exceeds the radius of the sloshing spiral core, implying that the emission is not contained by the cold fronts. Riseley et al. (2022) calculated the average spectral index within the spiral core and outside, finding that the differences were within 1σ. They also noted the minihalo exhibits multiple radio components, with a sharp exponential taper in radio brightness to the north at a 100 kpc radius, while a second component appears to the south at the same radius, exhibiting a less steep exponential decrease. These features were insufficient to determine whether hadronic production or turbulent reacceleration is favoured, with a combination of the two providing the best explanation.

The BCG was investigated by Riseley et al. (2022), who found a two-component power-law fit, with a break frequency of 3.1 ± 1.4 GHz. Below this frequency the spectrum was consistent with a relatively flat power law of $\alpha_{low} = 0.45 \pm 0.05$, while at higher frequencies it steepens to $\alpha_{high} = 0.81 \pm 0.18$. There was a bright western component and much fainter eastern component associated with the BCG, of uncertain origin.

[4] Giacintucci et al. (2017) derive this mass from the average cluster temperature T_X (excluding the core) using the M_{500}–T_X relation found by Vikhlinin et al. (2009).

5.7.1 10 GHz observation

Several bright sources were identified during self-calibration, including the central BCG (S1), a southern compact source (S2), and an eastern head-tail (S3). The source S2 is a highly variable quasar[5] that induced severe artefacts during self-calibration. This source was individually modelled for each dataset in an attempt to mitigate these issues, however, this was only partially successful. The joint `robust` = 0.5 image is shown in Figure 5.17, depicting this issue. As these residuals directly interfere with the emission of S3, additional artefacts from this source are introduced.

This becomes particularly evident following the subtraction of S1 and S2. Imperfect modelling and subtraction of S2 result in artefacts from S3 that contaminate the diffuse emission of the minihalo. An image of the minihalo produced using `robust` = 1, scales $[0, 2, 4, 8, 16, 32]$, and `smallscalebias` = -0.5 is presented in Figure 5.18. This image is not primary beam corrected to highlight the sources in the wider field. Negative and positive correlations blend with the central minihalo emission, preventing reliable separation of diffuse structure at the 3σ contour. Further work is required to achieve a satisfactory subtraction and allow an unbiased estimate of the minihalo flux density. The flux densities of the point sources are reported in Table 5.3.

[5] A quasar is an extremely luminous AGN, where the emission is caused by accretion of matter onto a supermassive black hole.

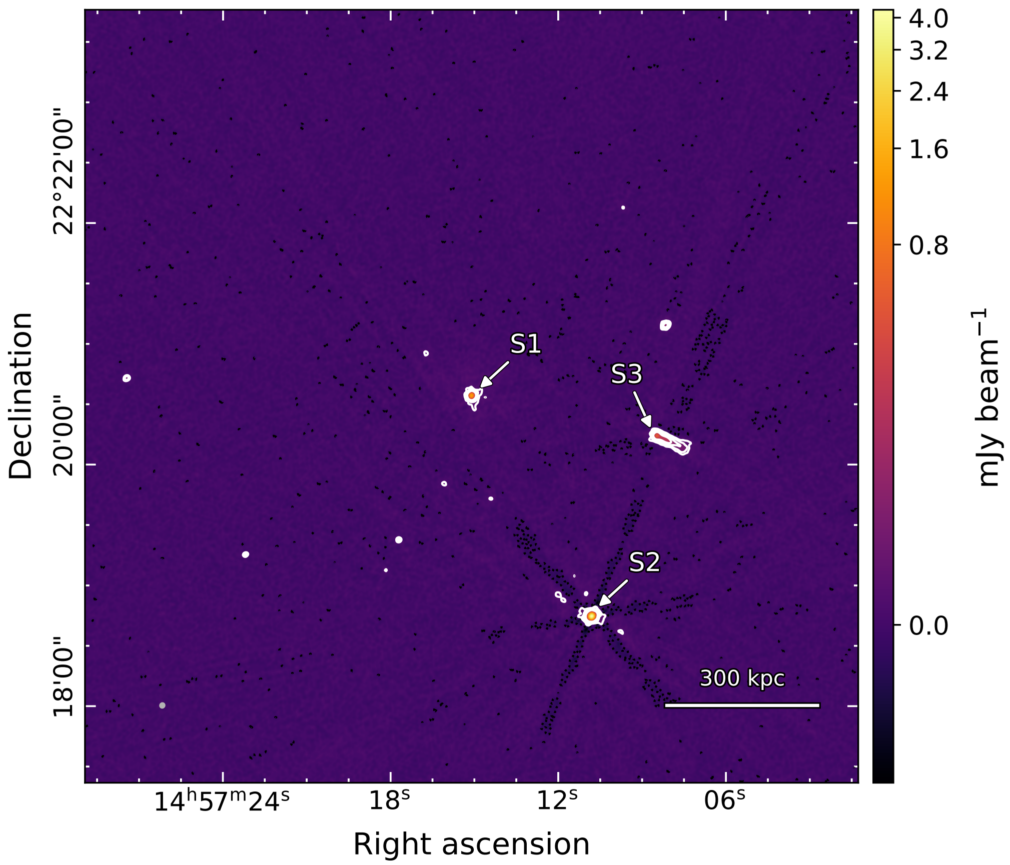


Figure 5.17: MS1455+2232. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.42"$\times$2.27" at a position angle of -27.5° (shown in grey, bottom left) and an rms noise of 2.99 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

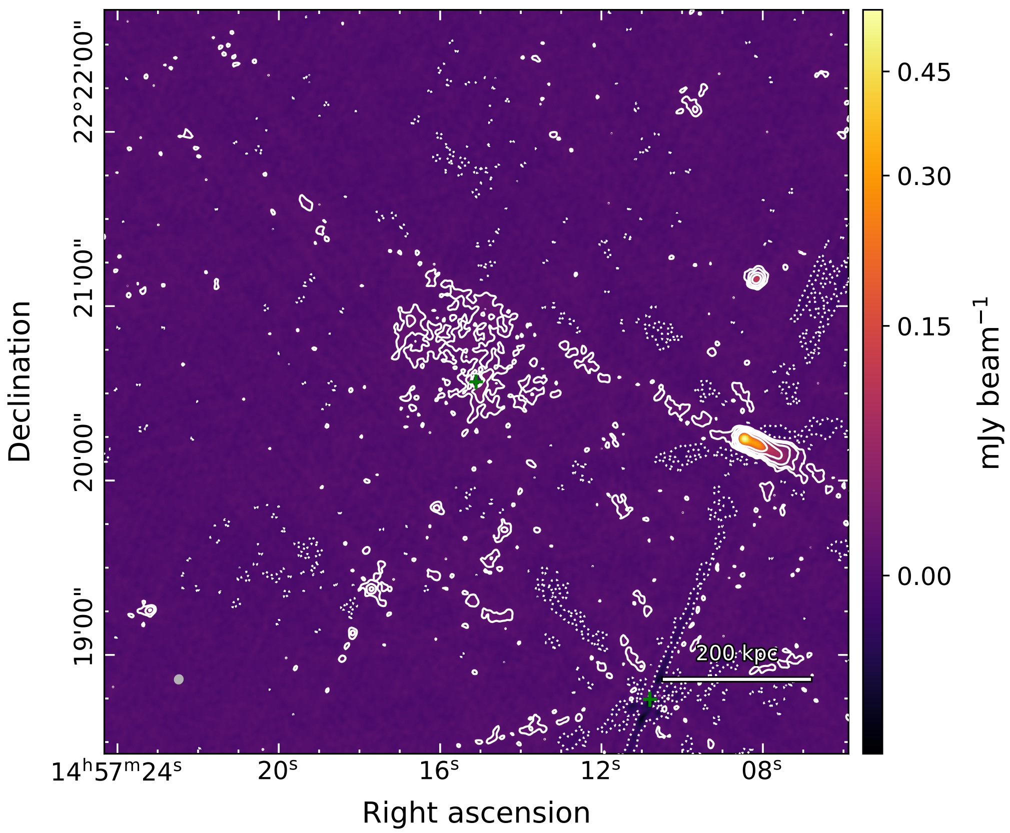


Figure 5.18: MS1455+2232. Point source subtracted (`robust` = 1) 10 GHz image with a restoring beam 3.02”$\times$2.80” at a position angle of -25.1° (shown in grey, bottom left) and an rms noise of 2.85 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The green cross marks the locations of the subtracted sources.

5.8 A1413

A1413 is an non-cool core galaxy cluster ($M_{500} = 5.95 \times 10^{14}$ M$_{\odot}$, $R_{500} = 1.22$ Mpc (Planck Collaboration et al., 2016)) at redshift $z = 0.139$ (Wen et al., 2015). It is currently the only known non-cool core cluster to host a minihalo. The dynamical state of this cluster was initially classified as undisturbed, due to the highly symmetric elliptical morphology at large scales not indicative of recent merger activity (Govoni et al., 2009). However, recent *Chandra* and XMM-Newton observations noted the absence of a sharp temperature decline in the core, which is uncharacteristic of a cool core system (Botteon et al., 2018; Lusetti et al., 2024). This borderline relaxed state has been interpreted as evidence for a weak cool core system, potentially stabilising after a past merger. The current consensus suggests A1413 exhibits neither a fully relaxed nor a highly disturbed morphology (Riseley et al., 2023; Trehaeven et al., 2023; Lusetti et al., 2024).

Diffuse minihalo emission was reported by Govoni et al. (2009) using 1.4 GHz VLA observations. They measured a total flux density of 1.9 ± 0.7 mJy over a spatial scale of ~ 220 kpc. Savini et al. (2019) confirmed this detection with 144 MHz LOFAR observations and reported a size of ~ 210 kpc, total flux density of 40 ± 7 mJy, and spectral index of $\alpha \approx 1.3$. A similar spatial extent was found by Trehaeven et al. (2023) using 1.3 GHz MeerKAT observations, who found a total flux density of 2.05 ± 0.27 mJy. They also measured an in-band spectral index of $\alpha = 1.5 \pm 0.46$.

Deeper observations with 144 MHz LOFAR and 1.3 GHz MeerKAT discovered a much larger halo ~ 500 kpc (Riseley et al., 2023). The total flux densities at 144 MHz and 1.3 GHz over the larger extent were measured as 29.5 ± 3.4 mJy and 3.23 ± 0.17 mJy, respectively. The spectral index over the larger halo had a median value of $\langle\alpha\rangle = 1.18 \pm 0.11$. Tentative evidence of steepening towards the edges of the minihalo was reported, with $\langle\alpha\rangle_{outer} = 1.12 \pm 0.13$ compared to $\langle\alpha\rangle_{inner} = 0.97 \pm 0.07$.

Several sources were identified within the central minihalo, including the BCG, an eastern head-tail radio galaxy, and a third galaxy to the southwest. The spectral index of the BCG was measured as $\alpha = 1.13 \pm 0.07$ (Riseley et al., 2023). They suggested that this steep AGN spectrum was due to radio emission being dominated by lobes rather than the core.

5.8.1 10 GHz observations

The sources observed during self-calibration did not exhibit significant variability. After several rounds of self-calibration, the solutions did not fully converge, remaining within -10° and 10°. The prominent sources in the image are the central BCG (S3), its companion head-tail galaxy (S2), two nearby radio galaxies to the southwest (S1, S4), and a more distant head-tail galaxy to the west (S5). These are imaged using `robust` = 0.5 and shown in Figure 5.19. The minihalo is difficult to discern due to the prominence of the embedded sources. Sources S1–S4 were

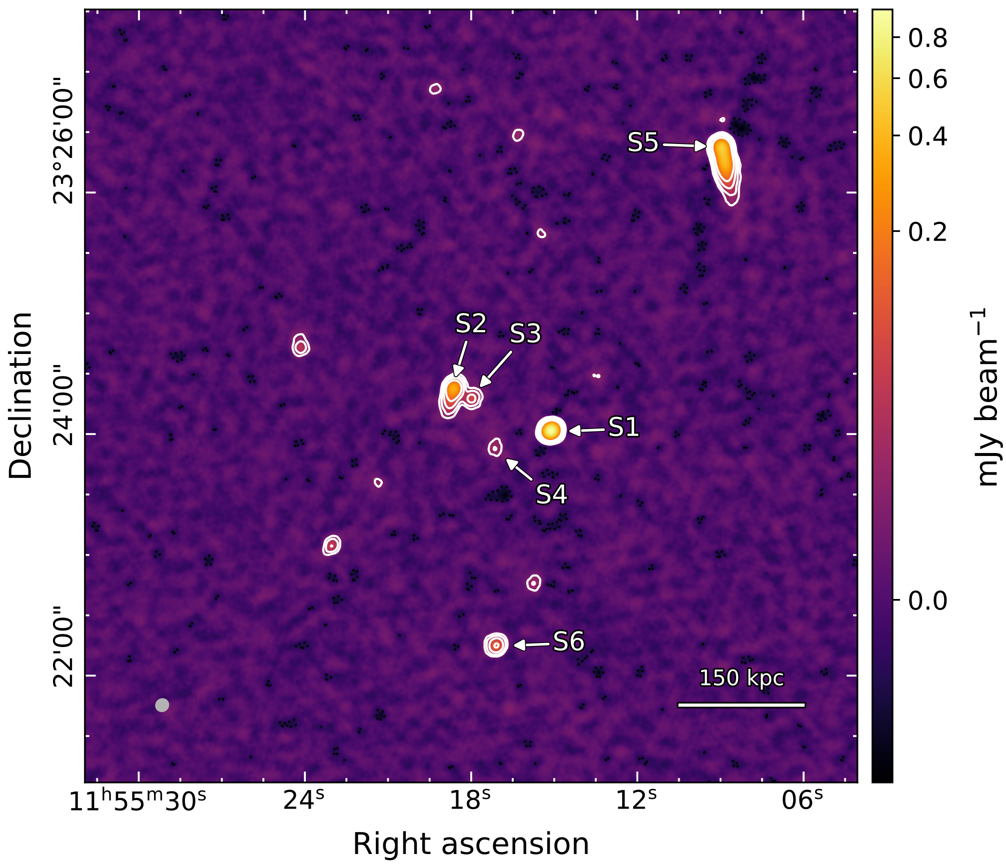


Figure 5.19: A1413. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 6.35”×5.97” at a position angle of -49.9° (shown in grey, bottom left) and an rms noise of 2.60 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

subtracted using the model from a 12 kλ *uv*-cut image.

The subtracted datasets were imaged using `robust` = 1 and multiscales $[0, 4, 8, 16, 32]$, followed by primary beam correction (Figure 5.20). Residual emission from S2 and S4 remains, complicating the estimation of minihalo flux. This is complicated by the large synthesised beam, likely due to the lack of remaining C-configuration data compared to the other clusters. Further work is required to robustly disentangle the complex source emission from that of the diffuse minihalo, in order to produce a minihalo flux estimation.

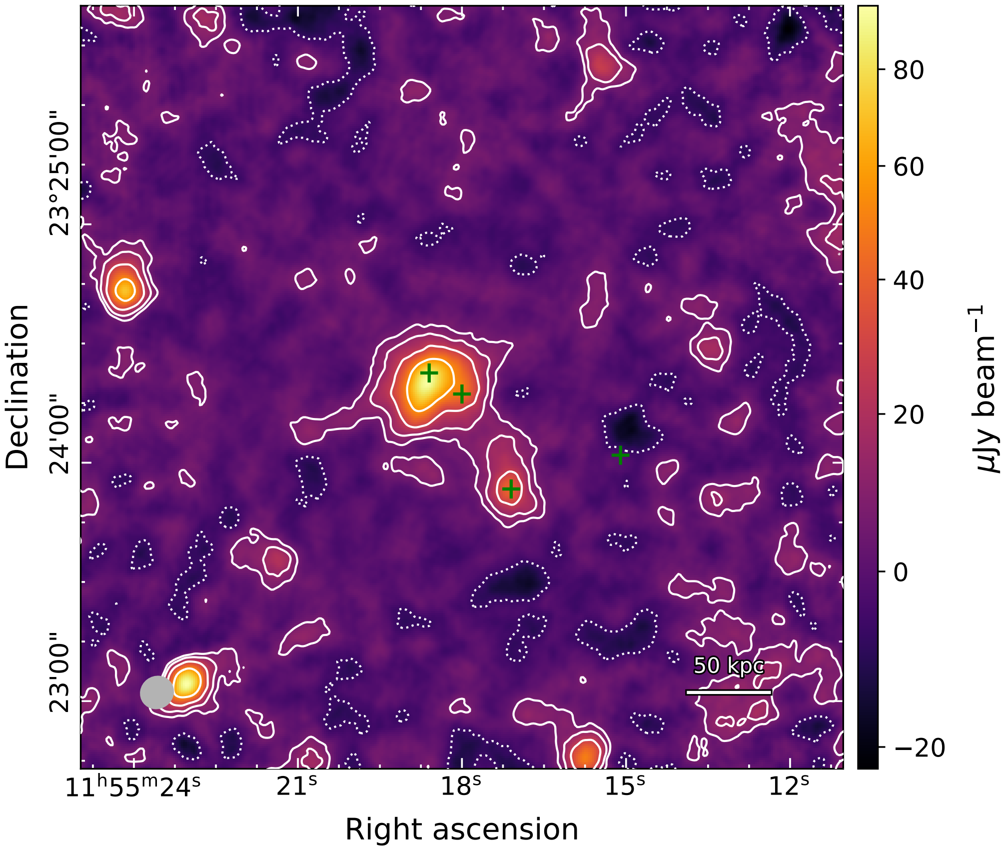


Figure 5.20: A1413. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 8.32"×7.85" at a position angle of -52.6° (shown in grey, bottom left) and an rms noise of 2.27 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The green cross marks the locations of the subtracted sources.

5.9 ACTJ0022-0036

ACTJ0022-0036 is a galaxy cluster ($M_{500} = 5.5 \times 10^{14}$ M$_\odot$, $R_{500} = 0.92$ Mpc (Hasselfield et al., 2013))[6] at redshift $z = 0.805$ (Menanteau et al., 2013). It is the most distant cluster confirmed to host a minihalo, as the detection at $z = 1.7$ is only a candidate (HLavacek-Larrondo et al., 2025). Only limited information is available regarding the cluster's dynamical state, as no dedicated studies have been conducted. Knowles et al. (2019) fitted temperature models to *Chandra* X-ray data, with the best-fitting model having a temperature drop towards the core and a central temperature $T = 3.2^{+1.7}_{-0.7}$ keV. Although this is an indication of a cool core, this was only significant at the 2σ level, lending tentative evidence.

Diffuse emission surrounding the BCG was reported by Knowles et al. (2019) using 610 MHz GMRT observations and subsequently categorised as a minihalo. To disentangle the diffuse component from the BCG emission, they fitted a two-component model to the radio surface brightness. This modelled the BCG as a 2D Gaussian source and the minihalo as a 2D exponential.

[6]Derived using a Universal Pressure Profile and SZ measurements.

Their best-fitting model revealed a strong preference for the minihalo component. They estimated the radius from the model as ~ 125 kpc and integrated the model surface brightness within this radius to derive a total flux density of 6.4 ± 1.8 mJy. The corresponding model for the BCG resulted in a flux density of 46.9 ± 0.2 mJy at 610 MHz. This was combined with a FIRST measurement of 19.02 ± 0.93 mJy at 1.4 GHz, yielding the BCG spectral index as $\alpha = 0.9 \pm 0.1$.

5.9.1 10 GHz observation

The self-calibration indicated four sources of minimal time-variability. These include the BCG (S1), a bright source to the east (S2), a compact source to the west (S3), and a bright source to the north (S4). The observations were imaged with `robust` = 0.5 and shown in Figure 5.21. Knowles et al. (2019) identified S3 as a cluster member. S2 and S4 were not identified in their analysis, potentially due to their location beyond R_{500}. In both the 610 MHz GMRT and the 10 GHz observation, S4 exhibited mild extension. In these higher angular resolution VLA images, there is some evidence of a protrusion to the southeast and to the north from a compact core. To image the minihalo, S1 was individually subtracted from each dataset using the model from a 12 kλ uv-cut image. After subtraction, the datasets were imaged using `robust` = 1 and scales $[0, 2, 4, 8, 16]$, followed by primary beam correction. The resulting image is shown in Figure 5.22.

The subtracted image has clear evidence of over-subtraction of the AGN, with negative residuals at the centre. However, the $+3\sigma$ contours in the surrounding region are potential indicators of diffuse emission. With more careful modelling and subtraction of the BCG, the minihalo emission may be recovered at 10 GHz. The subtracted model likely contains the BCG and minihalo flux, with the measured BCG flux density in the image used to generate the model as 3.89 ± 0.04 mJy. Extrapolating from the prior estimate of the spectral index, the expected flux density for the BCG is 2.25 ± 0.37 mJy. This is significantly lower than the measured value, providing further evidence for the contamination of minihalo emission in the BCG model.

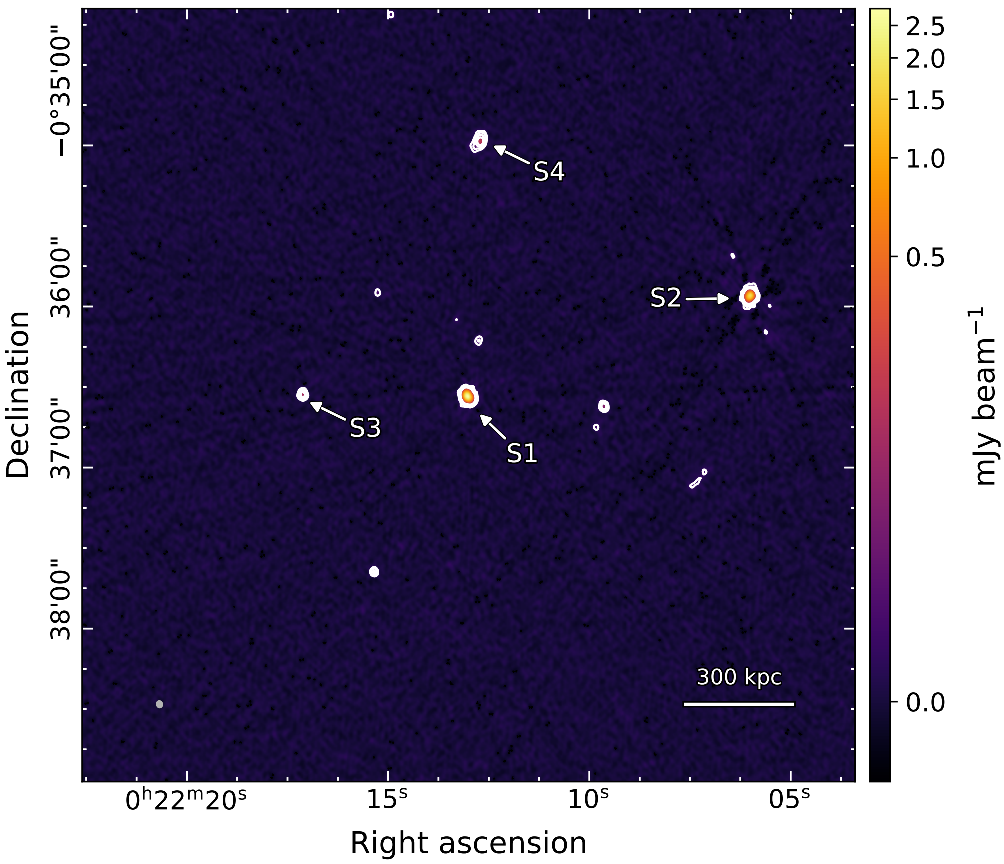


Figure 5.21: ACTJ0022-0036. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.47" × 2.10" at a position angle of -13.2° (shown in grey, bottom left) and an rms noise of 2.10 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources.

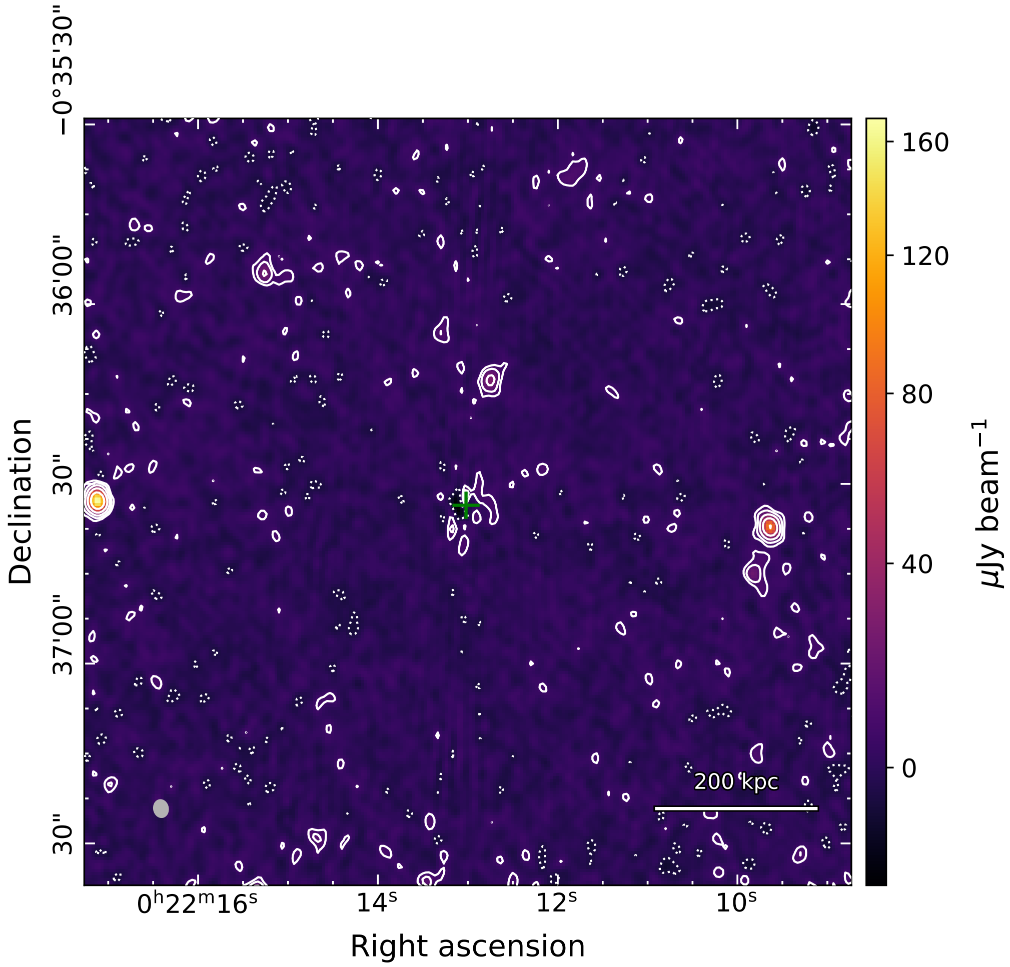


Figure 5.22: ACTJ0022-0036. Point source subtracted and primary beam corrected (`robust` = 1) 10 GHz image with a restoring beam 2.90”$\times$2.36” at a position angle of -10.7° (shown in grey, bottom left) and an rms noise of 1.94 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, with negative contours at -3σ shown as white dotted lines. The green cross marks the location of the subtracted source.

5.10 Z3146

Z3146 (also known as ZwCl 1021+0426) is a cool core cluster ($M_{500} = 6.53 \times 10^{14}$ $M_{\odot}$, $R_{500} = 1.2$ Mpc (Planck Collaboration et al., 2016)) at redshift $z = 0.289$ (Wen et al., 2015). *Chandra* X-ray observations identified three cold fronts, two located ~ 13 and ~ 35 kpc north of the core, and a third front lying ~ 150 kpc to the southeast (Forman et al., 2002). XMM-Newton observations reported a core temperature of ~ 5.5 keV, rising to ~ 7 keV at a radius of ~ 150 kpc (Romero et al., 2020). X-ray surface brightness fluctuations detected by XMM-Newton depict variations in the density and pressure, reflecting sloshing gas in the core (Romero et al., 2023).

VLA observations at 4.9 GHz and 8.5 GHz revealed a minihalo with a radius of 90 kpc (Giacintucci et al., 2014b). Two point sources corresponding to the BCG and a background galaxy at $z = 0.34$ are embedded in the diffuse emission. These were measured to have spectral indices of $\alpha = 0.66 \pm 0.14$ and $\alpha = -0.33 \pm 0.16$, respectively, in this frequency range. After subtraction of these sources, Giacintucci et al. (2014b) reported a minihalo flux density of 1.7 ± 0.2 mJy at 4.9 GHz. As the minihalo was not reliably detected at 8.5 GHz, the spectral indices of S1 and S2 were used to extrapolate flux densities at 1.4 GHz. These were then subtracted from a 1.4 GHz NVSS measurement to yield an estimated minihalo flux of ~ 5.2 mJy. This corresponds to a spectral index $\alpha \approx 1$ for the minihalo. Kale et al. (2015) image the minihalo using 610 MHz GMRT observations. They report a largest linear extent of 190 kpc, similar to that found at 4.9 GHz. They measured a total flux density of 11.5 mJy, with no stated uncertainty, which is consistent with $\alpha \approx 1$ over 610 MHz to 4.9 GHz.

5.10.1 10 GHz observation

Several prominent sources were identified during self-calibration. These include the BCG (S1), the background galaxy (S2), an eastern two-component system (S3, S4), a northeastern double-lobe galaxy (S5), and a second double-lobe source to the south (S6). The sources S3 and S4 are the brightest objects in the image and exhibit significant variability between the observations. These sources were modelled independently for each dataset to aid self-calibration. However, negative point fluxes were continually present in the models of these sources. This caused severe positive and negative sidelobes across the image. Several robusts between -1 and 1 were trialled alongside various scales, but a satisfactory clean model was not obtained. Figure 5.23 shows these observations imaged with `robust` = 1 and scales of $[0, 2, 4]$. A sidelobe artefact passes directly across the minihalo emission visible around S1, contaminating the flux values. As such, no estimate for the total minihalo flux can be derived. Further work is required to sufficiently model and subtract the contaminating effects from the minihalo emission.

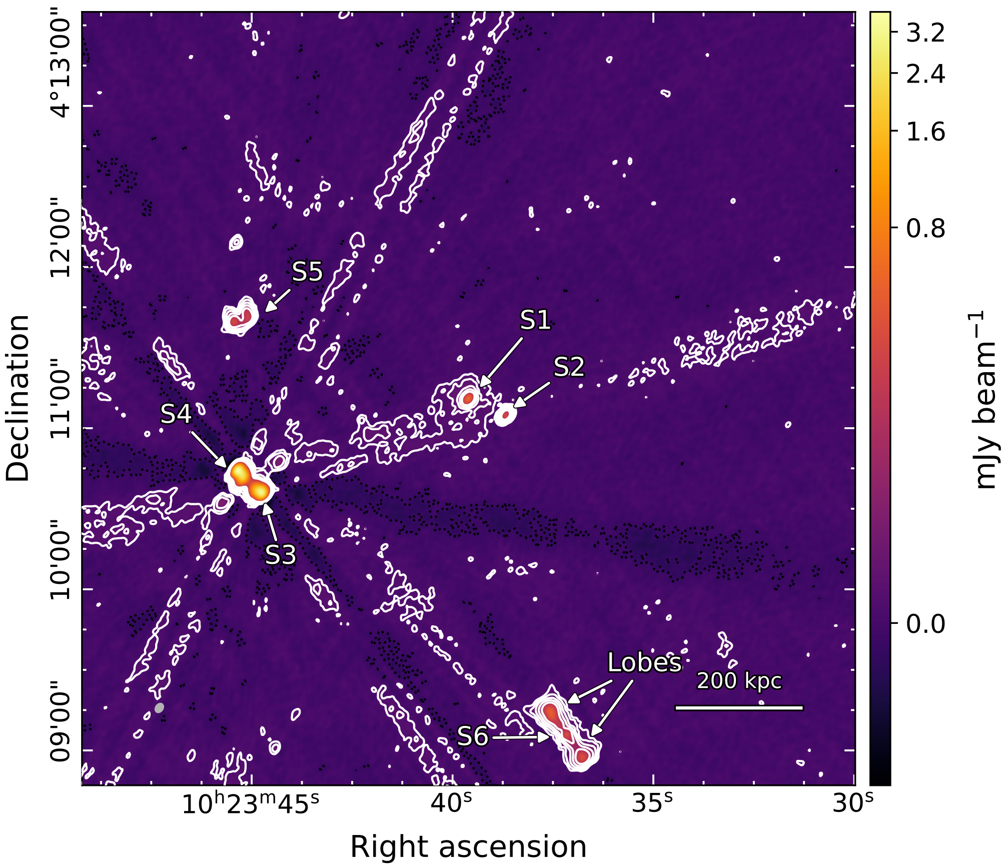


Figure 5.23: Z3146. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 3.56”×2.62” at a position angle of -35.0° (shown in grey, bottom left) and an rms noise of 3.19 μJy beam^{-1}. White contours are spaced by factors of two, starting from 6σ, while negative contours at -3σ are shown in black. The $+3\sigma$ contour has been omitted for visual clarity in highlighting the compact sources and imaging artefacts.

5.11 A1795

A1795 is a cool core cluster ($M_{500} = 4.47 \times 10^{14}$ $M_\odot$, R_{500} = 1.15 Mpc (Planck Collaboration et al., 2016)) at redshift $z = 0.062$ (Sanders et al., 2011). *Chandra* X-ray observations yielded a central temperature of 3.36 ± 0.12 keV, rising to 6.49 ± 0.43 keV at a radius of 160 kpc (Ettori et al., 2002). The morphology beyond the cool core is reported to be relaxed, with a slight elongation to the north. A cold front was also detected ~ 78 kpc to the south, suggesting gas sloshing in the core (Markevitch et al., 2001).

VLA observations at 1.4 GHz revealed a minihalo surrounding an extremely bright central BCG (Giacintucci et al., 2014b). Arcsecond resolution imaging of the BCG exhibited two prominent lobes extending to the north and south, before bending to the east and west, respectively. The length of the lobes was found to be ~ 10 kpc. The flux density of the BCG was measured as 917 ± 46 mJy from a FIRST image at 1.4 GHz. A second source is located to the west of the BCG, and was associated with a background galaxy at $z = 0.57$ (Giacintucci et al., 2014b). The radius of the minihalo was estimated at ~ 100 kpc, and the total flux density at 1.4 GHz was reported as 85 ± 5 mJy. Kale et al. (2015) attempted to image the minihalo using GMRT observations at 1.3 GHz. However, strong artefacts centred on the BCG prevented usable image quality and the cluster was excluded in their further analysis.

5.11.1 10 GHz observation

Self-calibration was complicated by the presence of the bent lobes of the BCG (S1). Modelling tended towards overestimating the lobe flux and subsequently compensating by inserting significant (≥ 50 μJy) negative point model. These resulted in severe sidelobe artefacts across the image. Small scales aided in modelling at high thresholds ($> 20\sigma$). However, deeper clean thresholds ($\leq 20\sigma$) reintroduced the negative model points. As such, a self-calibrated image was not obtained. Two other prominent sources are present in the image, the aforementioned background galaxy S2, and a head-tail galaxy to the southeast. Flux values are not provided for these sources due to uncertainty in the self-calibration. A joint image using `robust` = 0.5 and scales $[0, 2, 4]$ is shown in Figure 5.24 to illustrate these artefacts. Figure 5.25 depicts a closer view of the BCG, showing the lobe structure.

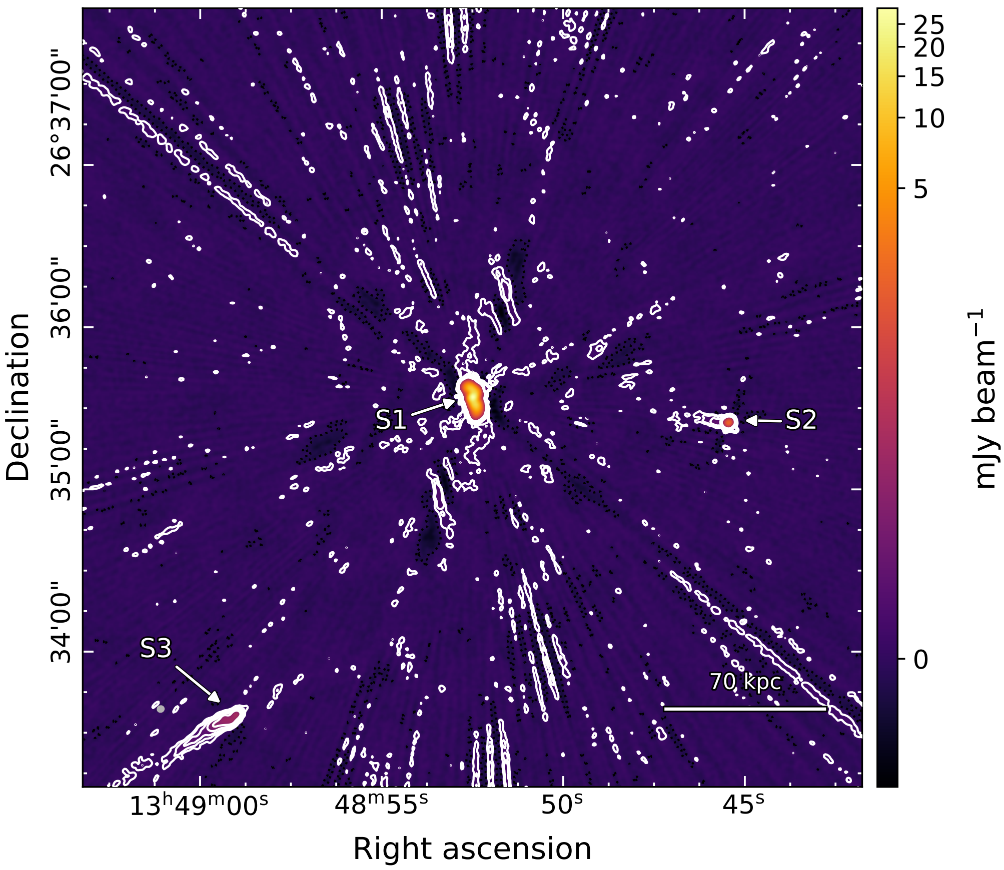


Figure 5.24: A1795. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.37”×2.17” at a position angle of -35.0° (shown in grey, bottom left) and an rms noise of 3.54 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, while negative contours at -3σ are shown in black.

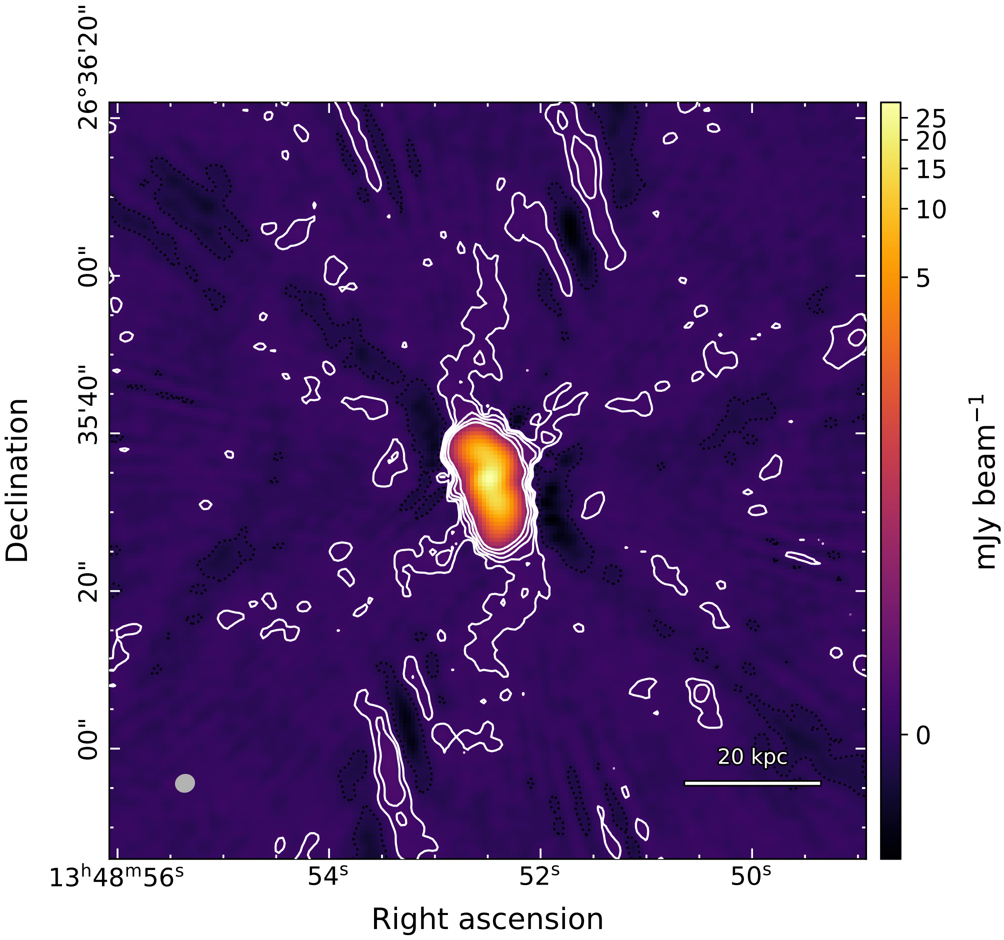


Figure 5.25: A1795. Zoom of the image shown in Figure 5.24, to illustrate the lobe structure of the central BCG. Details of the image are given in the full image caption.

5.12 A2626

A2626 is a nearby low-mass galaxy cluster ($M_{500} = 1.84 \times 10^{14}$ $M_\odot$, $R_{500} = 0.85$ Mpc (Planck Collaboration et al., 2016)) at redshift $z = 0.056$ (Wen et al., 2015). XMM-Newton X-ray observations revealed temperature variations of only ~ 1 keV out to a radius of 600 kpc (Wong et al., 2008). The central temperature was measured as ~ 2.5 keV, rising to ~ 3.5 keV at a radius of 230 kpc, before declining at larger radii. Two extended X-ray regions were identified ~ 450 kpc to the northeast and southwest from the centre, attributed to smaller merging subclusters (Mohr et al., 1996; Wong et al., 2008). The cluster morphology is therefore complex, but not highly disturbed.

Several unusual arc-like radio sources have been detected in the core of A2626. Four arcs were observed in a symmetric diamond pattern around the central BCG, convex with respect to the centre (Gitti, 2013; Ignesti et al., 2017; Kale et al., 2017). The arcs were detected by VLA observations at 1.4 GHz, but not at 5 GHz, implying a steep spectrum (Gitti, 2013). GMRT observations at 610 MHz confirmed a steep spectrum with reported spectral indices in the range 3.2–3.6 (Kale et al., 2017). They also measured the linear arc extents in the range 79–152 kpc. Within the region defined by the arcs, Gitti et al. (2004) and Gitti (2013) report diffuse emission proposed to be a minihalo with an extent of 60 kpc. Ignesti et al. (2017) measured the spectral index of the inner diffuse emission separately from the arcs using VLA observations at 3 and 5 GHz They found a relatively constant average spectral index of $\alpha = 0.7 \pm 0.1$ in this region, significantly different from the arcs. This minihalo candidate remains to be confirmed as the smaller scale may allow it to be a result of AGN activity.

5.12.1 10 GHz observation

The central BCG was unable to be cleanly modelled during self-calibration. Significant negative point models were introduced at even high thresholds ($\geq 20\sigma$). These introduced radial sidelobe artefacts across the full image. Robusts were trialled between -2 and 1, alongside various scales, without success. A self-calibrated image was not obtained for this cluster. An image using `robust` $= 0.5$ and scales $[0, 2, 4, 8]$ is presented in Figure 5.26. The central BCG (S1) is visible, alongside a double-lobe compact source (S2) to the west. The lobes protrude north and south for ~ 10 kpc and slightly bend towards the west. No flux densities are estimated for these sources due to uncertainty in image quality.

5.12. A2626

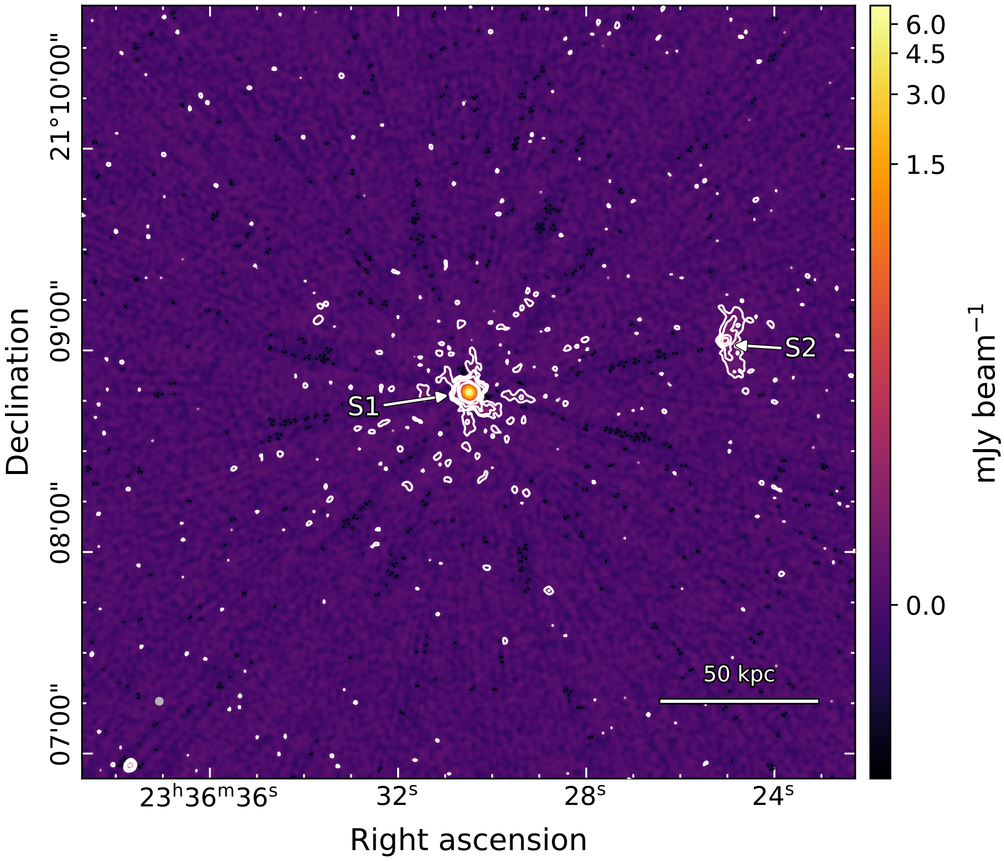


Figure 5.26: A2626. Non-subtracted (`robust` = 0.5) 10 GHz image with a restoring beam of 2.15”×2.11” at a position angle of -48.4° (shown in grey, bottom left) and an rms noise of 3.04 μJy beam^{-1}. White contours are spaced by factors of two, starting from 3σ, while negative contours at -3σ are shown in black.

Table 5.2: Flux densities of minihalos.

Cluster	Flux$_{10\,\mathrm{GHz}}$ (mJy)	α
RXJ1720+2638	8.61 ± 0.56	1.10 ± 0.05
A478	0.31 ± 0.03	2.02 ± 0.10
RXCJ1115+0129	0.97 ± 0.15	1.03 ± 0.18
RXJ2129+0005	0.41 ± 0.10	1.18 ± 0.09

Notes. Minihalo fluxes are derived from primary beam-corrected images. The reported flux for RXJ1720+2638 is the total across the tail and centre regions.

Table 5.3: Flux densities of discrete radio sources.

Cluster	Source	R.A.$_{J2000}$ (h, m, s)	Decl$_{J2000}$ (°, ’, ”)	Flux$_{10\,\mathrm{GHz}}$ (mJy)	α	Notes
RXJ1720+2638	S1	17 20 10.0	+26 37 31.9	1.27 ± 0.07	0.85 ± 0.03	BCG
	S2	17 20 13.9	+26 38 26.0	0.64 ± 0.04		
	S3	17 20 01.2	+26 36 33.3	1.02 ± 0.19		Subtr.
A478	S1	04 13 25.3	+10 27 54.6	5.41 ± 0.03	0.88 ± 0.03	BCG
	S2	04 13 31.9	+10 28 38.9	0.18 ± 0.02		
	S3	04 13 35.3	+10 29 21.2	0.30 ± 0.03		
	S4	04 13 38.4	+10 28 08.5	0.65 ± 0.21		Subtr.
RXCJ1115+0129	S1	11 15 51.9	+01 29 55.1	2.27 ± 0.21	0.67 ± 0.07	BCG
	S2	11 15 45.0	+01 28 38.8	0.71 ± 0.12		Subtr.
	S3	11 16 03.1	+01 28 15.7	0.68 ± 0.10		Subtr.
	S4	11 16 01.4	+01 30 58.7	0.82 ± 0.04		
RXJ2129+0005	S1	21 29 40.0	+00 05 21.1	5.36 ± 0.25	0.78 ± 0.02	BCG
	S2	21 29 39.2	+00 05 09.5	–		
2A0335+096*	S1	03 38 40.6	+09 58 11.8	3.01 ± 0.18	0.85 ± 0.04	BCG
	S2	03 38 40.3	+09 57 17.9	–		Subtr.
	S3	03 38 50.1	+09 57 16.5	0.26 ± 0.07		
A2204*	S1	16 32 46.9	+05 34 32.5	11.1 ± 1.29	0.85 ± 0.05	BCG
	S2	16 32 46.9	+05 34 40.7	0.15 ± 0.1	1.2 ± 0.1	Subtr.
MS1455+2232*	S1	14 57 15.1	+22 20 34.0	1.53 ± 0.07	0.81 ± 0.18**	BCG
	S2	14 57 10.8	+22 18 44.7	5.58 ± 0.51		Subtr.
	S3	14 57 08.2	+22 20 09.2	1.10 ± 0.20		
A1413*	S1	10 23 39.6	+04 11 10.8	0.99 ± 0.02		
	S2	10 23 38.7	+04 11 04.6	0.33 ± 0.03		
	S3	10 23 44.8	+04 10 35.9	0.54 ± 0.03	1.13 ± 0.07†	BCG
Z3146*	S1	10 23 39.6	+04 11 10.8	0.65 ± 0.06	0.75 ± 0.04	BCG
	S2	10 23 38.7	+04 11 04.6	0.35 ± 0.01		
	S3	10 23 44.8	+04 10 35.9	5.17 ± 0.47		
	S4	10 23 45.3	+04 10 44.2	6.58 ± 0.58		
	S5	10 23 45.2	+04 11 38.9	–		
	S6	10 23 37.2	+04 09 05.8	0.73 ± 0.10		
ACTJ0022-0036*	S1	00 22 13.0	-00 36 33.6	3.89 ± 0.04	0.89 ± 0.01	BCG
	S2	00 22 06.0	-00 35 56.1	2.61 ± 0.11		
	S3	00 22 17.1	-00 36 33.3	0.15 ± 0.02		
	S4	00 22 13.7	-00 34 58.5	0.21 ± 0.02		

Notes. Spectral index is given for BCG sources where available. The notes column denotes which sources correspond to a cluster BCG and subtracted sources. Source fluxes are derived from non-primary beam corrected images. Flux values are not reported for A1795 or A2626.
* Measured source fluxes for this cluster may be biased due to *uv*-cut biases.
** This source has a two-component power-law. The stated spectral index is accurate at 10 GHz.
† This spectral index is reported by Riseley et al. (2023) and was not fitted with the value measured in this thesis.

Chapter 6

Discussion

This sample of 12 clusters hosting minihalos was observed at 10 GHz to investigate the possibility of high-frequency spectral steepening. Of these, a robust measurement of the total minihalo flux density was obtained for four clusters. Combining these measurements with the results of previous studies, 75% of clusters exhibit a well-fitted single power-law spectrum. For a single cluster, A478, tentative evidence for steepening is observed. These results are summarised in the following section.

6.1 Summary of results

6.1.1 Clusters consistent with single power-law

The minihalo observed within RXJ1720+2638 is one of the most well-examined examples, though previous studies have stated conflicting interpretations. Notably, an 8.44 GHz measurement suggested tentative evidence for steepening relative to lower frequencies (Giacintucci et al., 2014a), while *uv*-plane modelling at 15.5 GHz was consistent with a single power-law spectrum (Perrott et al., 2021). The results of this thesis provide direct observational evidence in support of a single power-law spectrum up to at least 10 GHz. This is consistent with the conclusions of Perrott et al. (2021), who argued that incomplete *uv*-coverage at 8.44 GHz artificially reduced the measured flux density.

The flux density measured over the tail region of the RXJ1720+2638 minihalo is approximately three times higher than that predicted by extrapolation of previous measurements. The calculated spectral index reflects this, being flatter than previously reported. Giacintucci et al. (2014a) reported that spatial steepening was exhibited along the tail. As such, the averaged spectral index over the tail may differ at 10 GHz to the lower frequency studies. The detectable high-frequency emission is found to extend over only approximately half of the masked tail region. This reduction in spatial extent with increasing frequency is consistent with a steepening spectral index along the tail, although a spatially resolved study along the tail would be required to confirm this.

The findings for RXCJ1115+0129 are somewhat less constraining, primarily due to the limited number of flux density measurements for this minihalo. The previous study by Trehaeven et al. (2023) found a spectral index of $\alpha = 1.0 \pm 1.1$ between 1–1.5 GHz, with the uncertainty dominated by subtraction errors. The results of this thesis are consistent with their mean value and reduce the uncertainty by approximately a factor of six. Flux density uncertainty still remains dominated by source subtraction. The northern and southern spatial steepening reported in the previous study was not explored in depth for this thesis. However, the narrower observed spatial extent at 10 GHz is nearly aligned with the direction of their measured steepening. A spatially resolved point-to-point analysis combining the 1.3 GHz MeerKAT data with the 10 GHz observations presented here would provide a more robust test of this feature.

Several previous flux density measurements for the RXJ2129+0005 minihalo were available for comparison with the value measured for this thesis. The 10 GHz flux density is slightly higher than expected from that extrapolated from previous studies, although the 1σ uncertainties overlap. The excess flux may arise from two regions of emission near the masking boundary, including the unsubtracted source S2 and a diffuse region to the east (see Figure 5.11). These may contaminate the masked area and introduce flux not associated with the minihalo. The complex masking region defined by Giacintucci et al. (2014b) was approximated here by an ellipse, which further modifies the measured flux density. Notably, the flux density measured within the 3σ contour is less than half that of the larger mask ($S_{3\sigma} = 0.15 \pm 0.1$ mJy compared to $S_{mask} = 0.41 \pm 0.1$ mJy), which demonstrates the sensitivity of the integrated flux to the adopted mask.

6.1.2 Potential curvature for A478

The final cluster measured for this thesis, A478, presents a more intriguing case. Previous studies placed only an upper limit on the spectral index of $\alpha < 1$, whereas the flux density measured here shows a far greater spectral index of $\alpha = 2.02 \pm 0.10$. The in-band spectral index is even steeper at $\alpha = 3.34 \pm 0.18$, suggesting significant spectral curvature between 1.4 and 10 GHz. This result should be interpreted with caution, as it is based upon only two direct measurements and one non-detection. The adopted mask and *uv*-range likely differ from those used by Giacintucci et al. (2014b) and Savini et al. (2019). Spatial filtering at 10 GHz may result in missing flux, analogous to the 8.44 GHz measurement of RXJ1720+2638. Additional intermediate-frequency observations (e.g. 144 MHz to 5 GHz) are required to determine whether this apparent steepening reflects genuine particle ageing or residual systematic effects. Nevertheless, A478 remains an outlier compared to the other clusters in this sample.

6.1.3 Spectral implications for production mechanisms

For the turbulent reacceleration scenario, the radio spectrum is expected to steepen towards high frequencies. This is caused by the balancing of reacceleration efficiency and the CRe energy losses from synchrotron and inverse Compton radiation. Through simulation, ZuHone et al. (2013) predict that steepening from a single power-law occurs approximately beyond 1 GHz (see Figure 2.3). Their simulated minihalo also exhibited spectral indices in the range $\alpha \approx 1.5$–2. These results considered only a single initial injection of CRe. This is not entirely accurate if the minihalo has an embedded AGN, as is the case for many clusters in this sample (e.g. RXJ1720+2638, A1413, 2A0335+096, etc). The jets and lobes of AGN are a source of CRe injected into the core of the cluster. Depending on the CRe and energy injected, this may counteract steepening entirely.

In comparison, the hadronic scenario continually refreshes the CRe population. ZuHone et al. (2015) found this resulted in a single power-law spectrum to high frequencies, with a spectral index dependent on the energy spectrum of the CRe. Minor turbulence within the region had a negligible effect on the integrated spectrum, with the found spectral index as $\alpha \approx 1.1$.

The clusters RXJ1720+2638, RXCJ1115+0129, and RXJ2129+0005 exhibit total 10 GHz flux densities consistent, within uncertainties, with the extrapolated single power-law spectra derived from lower-frequency measurements. Their associated spectral indices are in the range $\alpha \approx 1$–1.1. These findings are more naturally explained by the hadronic scenario, than by the reacceleration mechanism for the production of the minihalo emission. In contrast, the minihalo within the cluster A478 appears more consistent with the reacceleration mechanism.

Given this evidence, the findings suggest both mechanisms may contribute to the overall minihalo emission and are dependent on the local cluster environment. This is reinforced by the presence of steepening along the tail of RXJ1720+2638. Evidence indicates the outer region is where turbulent acceleration begins to dominate the emission spectrum, outside of a hadronically dominated core. Both mechanisms being active within the minihalo at different scales is consistent with the discussions of previous studies (e.g. Perrott et al., 2021; Riseley et al., 2024).

Alternatively, turbulent acceleration could be the dominant mechanism in all clusters, with ongoing AGN activity offsetting the intrinsic spectral steepening. Under this interpretation, A478 would represent a cluster where the AGN-driven injections are comparatively weak or absent.

AGN activity can be approximated using the rest-frame 1.4 GHz radio luminosity of the BCG, $L_{1.4\,\mathrm{GHz}}$. This can be calculated from the redshift z,[1] the luminosity distance of the AGN D_L (which is based on the redshift and cosmology), the spectral index α, and the measured flux

[1]The cluster redshift is adopted as a proxy for the BCG redshift, which is sufficient for this comparative estimate.

density S_ν at the observing frequency ν via,

$$L_{\nu_{\rm rest}} = \frac{4\pi D_L^2}{(1+z)^{1-\alpha}} \left(\frac{\nu_{\rm rest}}{\nu}\right)^{-\alpha} S_\nu \tag{6.1}$$

The derived luminosity powers for the four clusters with minihalo flux measurements are presented in Table 6.1. The BCG of A478 exhibits radio power comparable to that of RXJ1720+2638, which hosts sub-parsec jets (Perrott et al., 2023). This suggests that the activity of this BCG is unlikely to be significantly less than systems where this steepening is not observed. Consequently, the lack of strong AGN activity cannot easily explain the apparent spectral steepening in A478. Further work in optical identifiers or high-resolution radio imaging would be required to investigate the level of AGN activity more definitively.

Table 6.1: Radio luminosity for the BCG at 1.4 GHz. Estimates from this thesis are presented in comparison to literature value derived from lower-frequency measurements.

Cluster	$L_{1.4\,\rm GHz}$ (this thesis) ($\times 10^{24}$ W Hz^{-1})	$L_{1.4\,\rm GHz}$ (Lit.) ($\times 10^{24}$ W Hz^{-1})	Reference
RXJ1720+2638	0.47 ± 0.18	0.50 ± 0.02	Giacintucci et al. (2014a)
A478	0.58 ± 0.21	0.60 ± 0.03	Giacintucci et al. (2014b)
RXCJ1115+0129	3.0 ± 1.8	3.2 ± 0.4	Trehaeven et al. (2023)
RXJ2129+0005	3.9 ± 1.2	3.1 ± 0.2	Kale et al. (2015)

6.2 Measurement Limitations

There are several fundamental observational limitations affecting the detectability of minihalos in this study. The primary constraints arise from the maximum recoverable scale, surface brightness sensitivity, and angular resolution of the interferometer configuration. Effects arising from these limitations are discussed here.

6.2.1 Maximum recoverable scale

The largest angular scale that the interferometer can recover is set by the minimum projected baseline length. For these VLA observations, the shortest baselines correspond to 1 kλ. Approximating the angular scale as $\theta \approx 1/\boldsymbol{u}$ yields 10^{-3} rad or ~ 3.44 arcminutes.

For a characteristic minihalo diameter of ~ 300 kpc, this angular scale corresponds to a minihalo located at redshift $z \sim 0.07$. Below this redshift, emission at scales comparable to or larger than this minihalo diameter will be partially or fully resolved out, leading to an underestimation of the total flux density.

This effect is most relevant for three clusters in this sample, 2A0335+096, A2626, and A1795. Of these clusters, only 2A0335+096 has a source-subtracted image. For this cluster

3.44′ corresponds to ≈ 150 kpc. Diffuse emission extending beyond this scale will not be fully recovered. The lowest redshift cluster for which a total 10 GHz flux density was successfully measured is RXJ1720+2638 at $z = 0.160$. This corresponds to a maximum physical scale of ~570 kpc, suggesting that spatial filtering is unlikely to significantly bias the integrated flux densities reported here.

Similarly, studies that have significantly different *uv*-coverage can be biased by the different fluxes they are able to recover. Matching the *uv*-coverage with prior studies, as performed for RXJ1720+2638, further ensures any spatial filtering is consistent between studies.

6.2.2 Surface brightness sensitivity

The surface brightness sensitivity is limited by image noise, which arises from residual RFI, calibration uncertainties, and instrumental thermal noise. For the 10 GHz observations presented here, the typical rms noise level is approximately 3 μJy beam^{-1}.

For steep-spectrum or intrinsically low-brightness minihalos, emission may fall below the adopted 3σ detection threshold. This can artificially reduce the apparent spatial extent of the emission relative to lower-frequency observations, where surface brightness is higher. In such cases, only lower limits on the spectral index can be derived. This would be relevant for a spatial analysis of the spectral index within the tail of the RXJ1720+2638 minihalo, where only half the region is greater than the detection threshold. As discussed previously, spatial masks defined in earlier studies were adopted where available to ensure consistency in flux density measurements.

The choice of adopted mask is of particular importance for A478, where the spatial extent of the minihalo observed at 10 GHz (approximately 80 kpc by 40 kpc) is far smaller than that of previous studies (approximately 150 kpc by 120 kpc (Giacintucci et al., 2014b)). As no mask was available, the final integrated area is likely far smaller, resulting in a systematic underestimation. Conversely, several compact sources are detected in this thesis that are not noted during the subtraction process at lower frequencies. This may instead bias those flux densities higher, resulting in a steeper observed spectrum.

6.2.3 Angular resolution

The angular resolution, defined by the restoring beam, has the greatest impact on the accuracy of the source subtraction. For the 10 GHz observations, the typical restoring beam is approximately 4”×3”. Higher angular resolution improves the modelling of compact sources and their associated lobes or jets, enabling more effective separation from the diffuse minihalo emission.

Many compact sources are unresolved at this beam size. Improved resolution allows subtraction to occur over smaller areas, thereby reducing subtraction uncertainties. Since the subtraction error scales with the area over which the model is removed, $\sigma_{sub} \propto A_{sub}$, improved resolution directly reduces residual contamination. As discussed previously, the inclusion of C-configuration data provides the necessary angular resolution with regards to the joint imaging process.

6.3 Incomplete clusters

For the majority of the remaining clusters in this sample a minihalo is detected at 10 GHz. However, difficulties encountered during the subtraction of contaminating emission prevented robust estimation of the minihalo flux in several cases. These difficulties can be categorised as arising from (i) variability of compact sources between observing epochs and (ii) residual diffuse emission not associated with the minihalo in the subtracted image.

Variability of compact sources between observations at different epochs presents a significant challenge for self-calibration and source subtraction. Point source modelling was performed independently for each dataset in order to mitigate this effect. However, this did not fully resolve the residual artefacts in the individual models nor the joint image. The C-configuration observations were typically the dominant contributor to subtraction errors in the combined dataset image, although the precise reason behind this remains unclear. Possible factors are the `tclean` parameters for imaging of the point sources and residual calibration errors.

The clusters where a minihalo was not detected at 10 GHz are A1795 and A2626. In both cases, the absence of a clear detection is attributed to imaging artefacts obscuring potential diffuse emission, rather than insufficient surface brightness sensitivity. These clusters contain bright central AGNs that complicate the self-calibration process. It is plausible that with additional refinement of the self-calibration process, a reliable image of the minihalo emission could be recovered.

The models used for source subtraction were constructed using a fixed *uv*-cut and imaging parameters derived from an analysis of a single cluster, RXJ1720+2638 (see Section 4.1). Although manual inspection of each resulting image was performed, these parameters likely require optimisation on a per-cluster basis. A key systematic limitation arises because the adopted *uv*-cut corresponds to a constant angular scale on the sky, rather than a constant physical scale.

The clusters in this sample span physical scales between 0.72 and 7.5 kpc arcsec^{-1}, increasing with redshift under the adopted cosmology. Consequently, the same angular *uv*-cut corresponds to substantially different physical scales across the sample. RXJ1720+2638 lies at an intermediate redshift within the cluster sample. For lower redshift clusters an identical *uv*-cut selects smaller physical scales. In such cases, diffuse emission from AGN lobes may not be fully included in the subtraction model, leading to residual contamination in the subtracted image (e.g. 2A0335+096).

Conversely, for higher redshift clusters, the same *uv*-cut corresponds to larger physical scales. This increases the risk that extended minihalo emission is partially incorporated into the subtraction model, resulting in overestimation of compact source flux densities, and subsequently, over-subtraction in the final image (e.g. ACTJ0022-0036). The impact of adopting a constant angular *uv*-cut is, therefore, dependent on the minihalo extent, cluster redshift, and the morphology of embedded radio sources.

6.4 SZ effect

The SZ effect represents a potential contaminant to the flux densities measured in this thesis. As it arises from IC-scattering throughout the cluster, it is naturally a cluster-scale phenomenon extending well beyond the spatial extent of the minihalo. Such a large-scale structure may be partially sampled at the shortest baselines of the 10 GHz observations. At 10 GHz this manifests as a negative signal, causing an underestimation of the minihalo flux. However, the relatively low frequency of these observations compared to the thermal SZ spectrum means that the expected amplitude of this decrement is small.

Perrott et al. (2021) estimated the maximum SZ signal for RXJ1720+2638 for the VLA observation at 8.44 GHz. They found a negative maximum amplitude of approximately 80 μJy at the shortest baselines. At the slightly higher frequency and enhanced *uv*-coverage of the shortest baselines for the 10 GHz observations, their modelling predicts an amplitude of approximately 144 μJy at the shortest baselines (~ 1 kλ). For RXJ1720+2638 this is negligible compared to the measured minihalo flux and lies well within the quoted uncertainty ($\sigma = 0.56$ mJy). Additionally, this particular flux density was measured from an image constructed using baselines > 15 kλ, which assists in filtering out and further reducing any SZ contribution.

The remaining clusters where minihalo fluxes are obtained (A478, RXCJ1115+0129, and RXJ2129+0005) have masses within -26% to $+18\%$ of RXJ1720+2638. As the SZ effect scales primarily with the integrated thermal pressure (and therefore cluster mass), a very conservative upper bound of < 0.18 mJy is considered for all clusters. This is of the same order as the measurement uncertainties for RXCJ1115+0129 ($\sigma = 0.10$ mJy) and RXJ2129+0005 ($\sigma = 0.15$ mJy). For A478, this represents approximately 50% of the measured flux and five times the stated uncertainty ($\sigma = 0.03$ mJy). However, even correcting by the maximum SZ estimate is insufficient to reconcile the order of magnitude discrepancy to the lower limit on the spectral index reported by Savini et al. (2019).

Any systematic under-estimation of the total minihalo flux would artificially steepen the spectral index. However, because the spectral index depends on the logarithm of the flux ratio between frequencies, moderate absolute flux offsets translate into comparatively small changes in α. Therefore, given that the SZ contribution is near the measurement uncertainties, it is unlikely to significantly affect the conclusions presented here.

6.5 Future work

Several future avenues arise from this thesis. The most significant is imaging and obtaining a total minihalo flux density for the remaining clusters in this sample. Expanding the analysed sample would strengthen the basis for the apparent absence of high-frequency spectral steepening in most minihalos. As discussed, achieving this primarily requires improvements in the self-calibration and source-subtraction procedures.

A major methodological improvement would be the adoption of a physically motivated *uv*-cut scaled to the expected angular extent of each minihalo. This would better isolate compact emission when constructing the subtraction model and reduce any redshift-dependent systematic bias. However, a simple *uv*-cut may not be sufficient for ACTJ0022-0036, where the apparent minihalo extent is small due to its high redshift. For this cluster, direct modelling of the diffuse emission using the surface brightness, as performed by Knowles et al. (2019), may be appropriate to provide a more robust approach.

A more rigorous treatment of systematic effects would involve explicit modelling of the SZ signal for each cluster using gas pressure profiles, followed by propagation through the interferometer response, as demonstrated by Perrott et al. (2021). This would enable direct correction for any SZ-induced bias in the measured minihalo flux densities, which is particularly important for fainter minihalos.

This thesis has focused primarily on the integrated spectral properties of the minihalo emission, with limited exploration of the spatially resolved structure. However, spatial morphology provides an additional diagnostic to analyse CRe production mechanisms. A point-to-point spectral comparison with previous low-frequency observations would test whether hadronic production and turbulent acceleration are operating over distinct spatial regions within individual minihalos. The smaller apparent extents observed at 10 GHz suggest that regions of spectral steepening, such as the tail of RXJ1720+2638, may be identified through spectral index mapping.

The signal-to-noise ratio of the diffuse emission ultimately limits the steepest spectral index that can be robustly measured. One potential method for enhancing low surface-brightness emission is spatial smoothing, typically through convolution with a Gaussian kernel. While such averaging increases sensitivity to diffuse emission, it reduces angular resolution. As this can complicate source-subtraction, it would be best investigated on the source-subtracted images. This approach was not explored in this thesis and remains a potential avenue for improving the detectability of extended emission in future analysis of the 10 GHz observations.

Chapter 7

Conclusion

The aim of this thesis was to investigate the production mechanisms behind the diffuse emission of minihalos. A key observational distinction between hadronic production and turbulent reacceleration lies in the high-frequency spectral steepening predicted by the reacceleration scenario. To investigate this, radio observations at 10 GHz for a sample of twelve galaxy cluster minihalos were calibrated and imaged. Within the time constraints of this thesis, four clusters were analysed in greater detail to obtain robust minihalo flux measurements. The results indicate that high-frequency spectral steepening < 10 GHz is not common among this subsample of minihalos.

All twelve clusters' observations were calibrated and jointly imaged to combine the higher angular resolution of C-configuration observations with the better sensitivity to extended emission of D-configuration data. A custom calibration stage was implemented to improve the removal of radio-frequency interference within the observations. The resulting images were source-subtracted to isolate the diffuse emission of the minihalo, and subsequently utilised to calculate the total flux density. This was successfully completed for four clusters, RXJ1720+2638, A478, RXCJ1115+0129, and RXJ2129+0005. The majority of the remaining clusters displayed evidence of a minihalo, however, imaging artefacts prevented a reliable measurement of the total minihalo emission.

The resulting flux densities were compared to previous studies and used to calculate the spectral indices for each minihalo. Three of these clusters (RXJ1720+2638, RXCJ1115+0129, RXJ2129+0005) are consistent with a single power-law up to 10 GHz, with spectral indices in the range $\alpha \approx 1$–1.1. One cluster (A478) exhibits potential steepening, with a derived spectral index of $\alpha = 2.02$, compared to the lower frequency upper bound of $\alpha \leq 1$. The single power-law spectrum is broadly consistent with the hadronic scenario, while the steepening is more consistent with the turbulent reacceleration method. These results suggest that both mechanisms can operate within the minihalo, with hadronic production potentially playing a dominant role in many systems.

The measured spectral indices can be biased by the defined spatial mask, *uv*-coverage, and source-subtraction between observations at different frequencies. Differences in masking strategies and *uv*-coverage between studies makes comparison to the results here more ambiguous, as spatial filtering from mismatched coverage can artificially reduce the measured flux. Furthermore, the sample size for measured flux densities limits a statistical inference on the general behaviour of minihalos. The final uncertainties for the flux densities measured for this thesis are dominated by the source-subtraction component, particularly for low surface-brightness minihalos.

A first step would be the completion of the self-calibration and source-subtraction for the calibrated datasets in this sample. This would enable a broader spectral comparison between the clusters. An exploration of the spatial dependence of the spectral index would also assist in distinguishing between the hadronic scenario and turbulent reacceleration. A spatial investigation comparing to lower frequency observations may reveal if these mechanisms operate over different minihalo regions, as previously observed for RXJ1720+2638.

The findings of this thesis extend high-frequency constraints on minihalo spectra. The absence of widespread spectral steepening challenges simple reacceleration-only models. These observations constitute the first high-frequency survey of radio minihalos, highlighting the importance of multi-frequency and spatially resolved studies.

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Appendix A

Example casa_pipescript.py file

Listing A.1: Example CASA pipeline script.

```
hifv_importdata(vis=["DATASET"])
hifv_flagdata(
    hm_tbuff='1.5int', fracspw=0.01, intents='*POINTING*,
    *FOCUS*,*ATMOSPHERE*,*SIDEBAND_RATIO*, *UNKNOWN*,
    *SYSTEM_CONFIGURATION*, *UNSPECIFIED#UNSPECIFIED*')
hifv_vlasetjy()
hifv_priorcals(show_tec_maps=False)
hifv_aoflagger(
    flag_target='PRIMARY', aoflagger_file='PRIMARY.lua')
hifv_testBPdcals()
hifv_semiFinalBPdcals()
hifv_aoflagger(
    flag_target='SECONDARY', aoflagger_file='SECONDARY.lua')
hifv_solint()
hifv_fluxboot()
hifv_aoflagger(
    flag_target='CALIBRATORS', aoflagger_file='CALIBRATORS.lua',
    use_corrected=True)
hifv_finalcals()
hifv_applycals()
hifv_aoflagger(
    flag_target='SCIENCE_TARGET', aoflagger_file='SCIENCE.lua',
    use_corrected=True)
hifv_statwt()
hifv_plotsummary()
```

```
hifv_exportdata()
h_save()
```